



US012403058B2

(12) **United States Patent**
Lim et al.

(10) **Patent No.:** **US 12,403,058 B2**

(45) **Date of Patent:** ***Sep. 2, 2025**

(54) **RECONFIGURABLE UPPER LEG SUPPORT
FOR A SURGICAL FRAME**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 146 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **18/208,670**

(22) Filed: **Jun. 12, 2023**

(65) **Prior Publication Data**
US 2023/0320915 A1 Oct. 12, 2023

Related U.S. Application Data

(63) Continuation of application No. 16/927,219, filed on
Jul. 13, 2020, now Pat. No. 11,672,718.
(Continued)

(51) **Int. Cl.**
A61G 13/00 (2006.01)
A61G 13/04 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **A61G 13/1245** (2013.01); **A61G 13/0036**
(2013.01); **A61G 13/04** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC A61H 1/0237; A61H 1/024; A61H 1/0255;
A61G 13/0036; A61G 13/0054;
(Continued)

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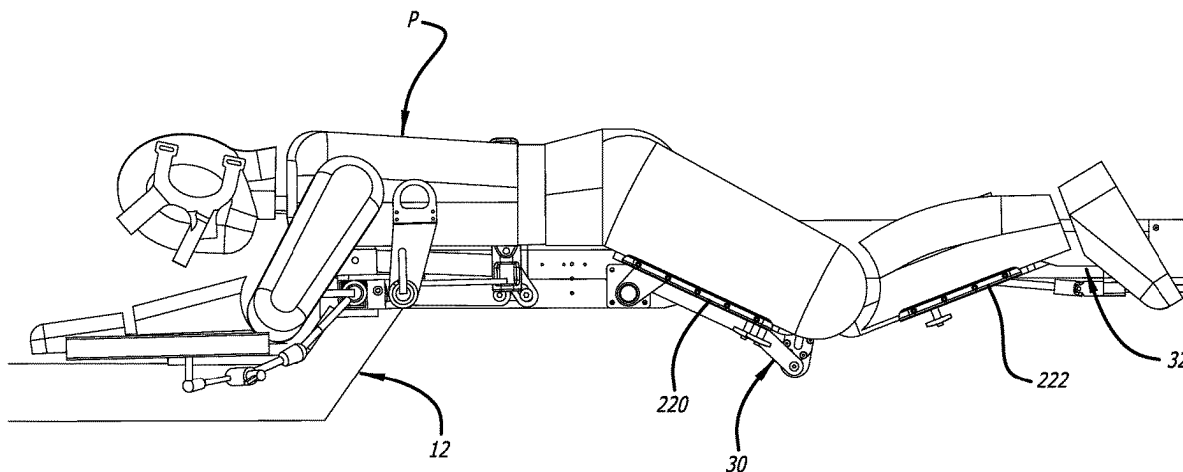
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Primary Examiner — Adam Baker

(57) **ABSTRACT**

A surgical frame and method for use thereof is provided. The surgical frame is capable of reconfiguration before, during, or after surgery. The surgical frame includes a main beam that can be rotated, raised/lowered, and tilted upwardly/downwardly to afford positioning and repositioning of a patient supported thereon. The surgical frame also includes a reconfigurable upper leg support for supporting portions of the upper legs, the hips, and the lower back of the patient to facilitate positioning and repositioning there during surgery. The upper leg support via reconfiguration thereof can accommodate patients of different sizes, can provide flexure of the patient's lumbar spine to facilitate surgical access thereto, and can prevent unwanted torsion of a patient's spine during such reconfiguration.

20 Claims, 52 Drawing Sheets



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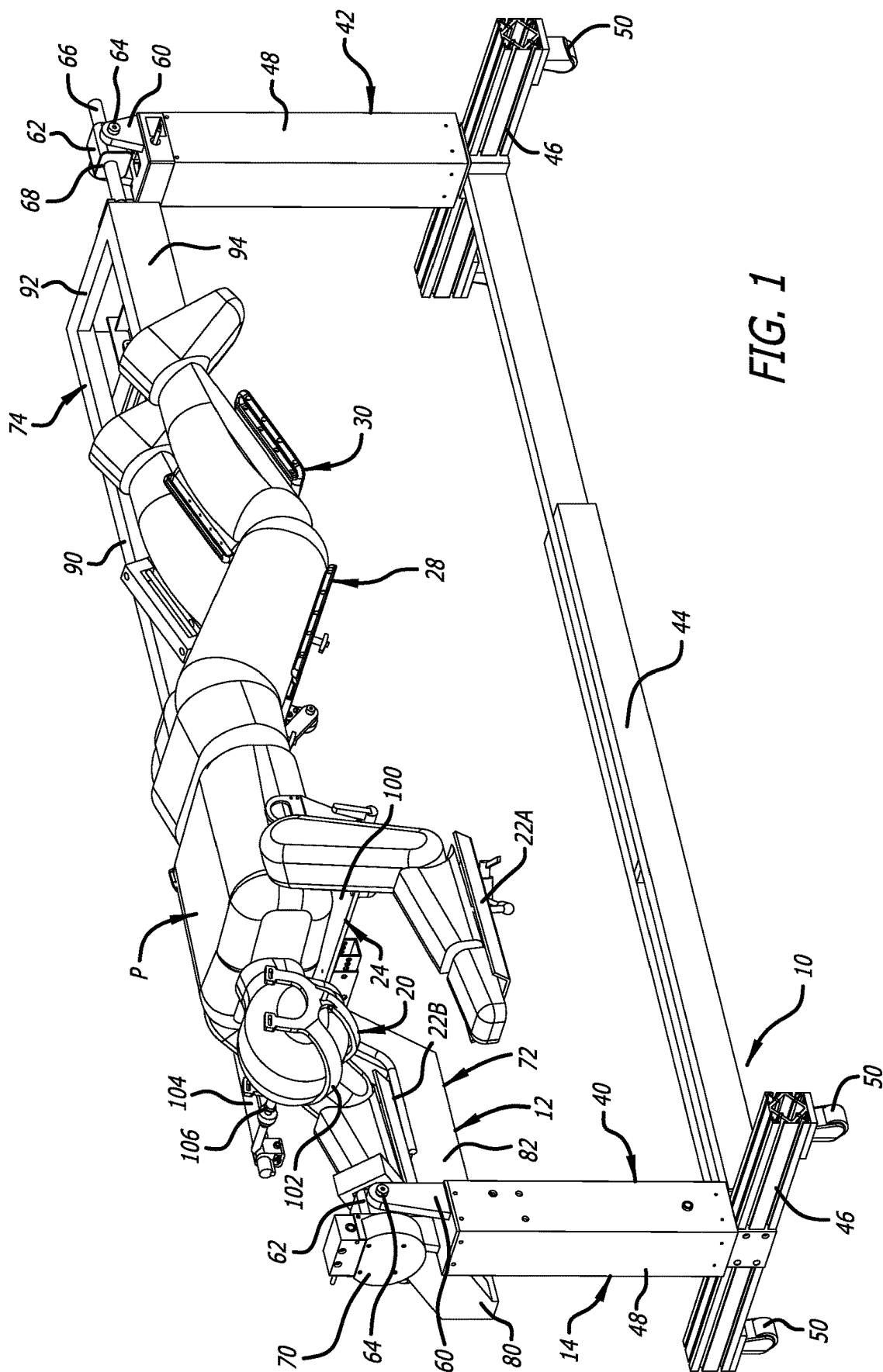
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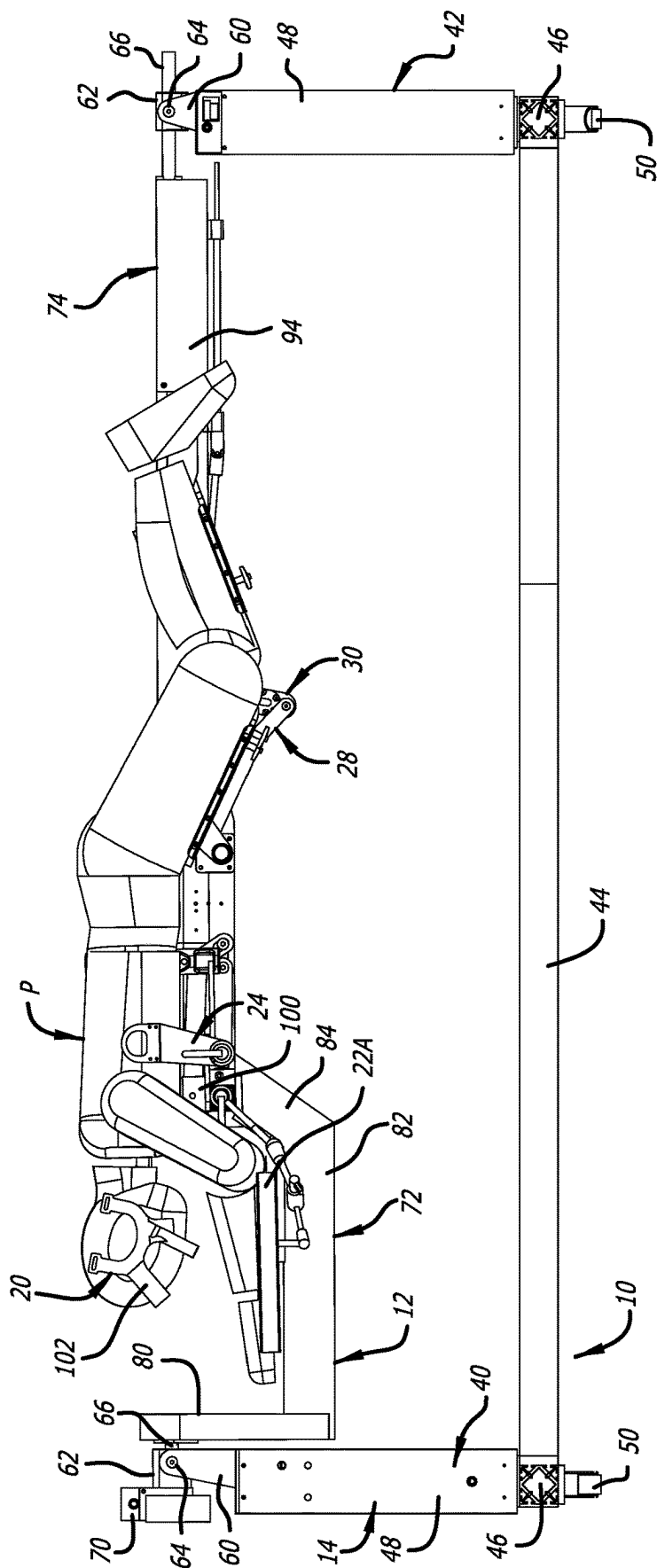


FIG. 2

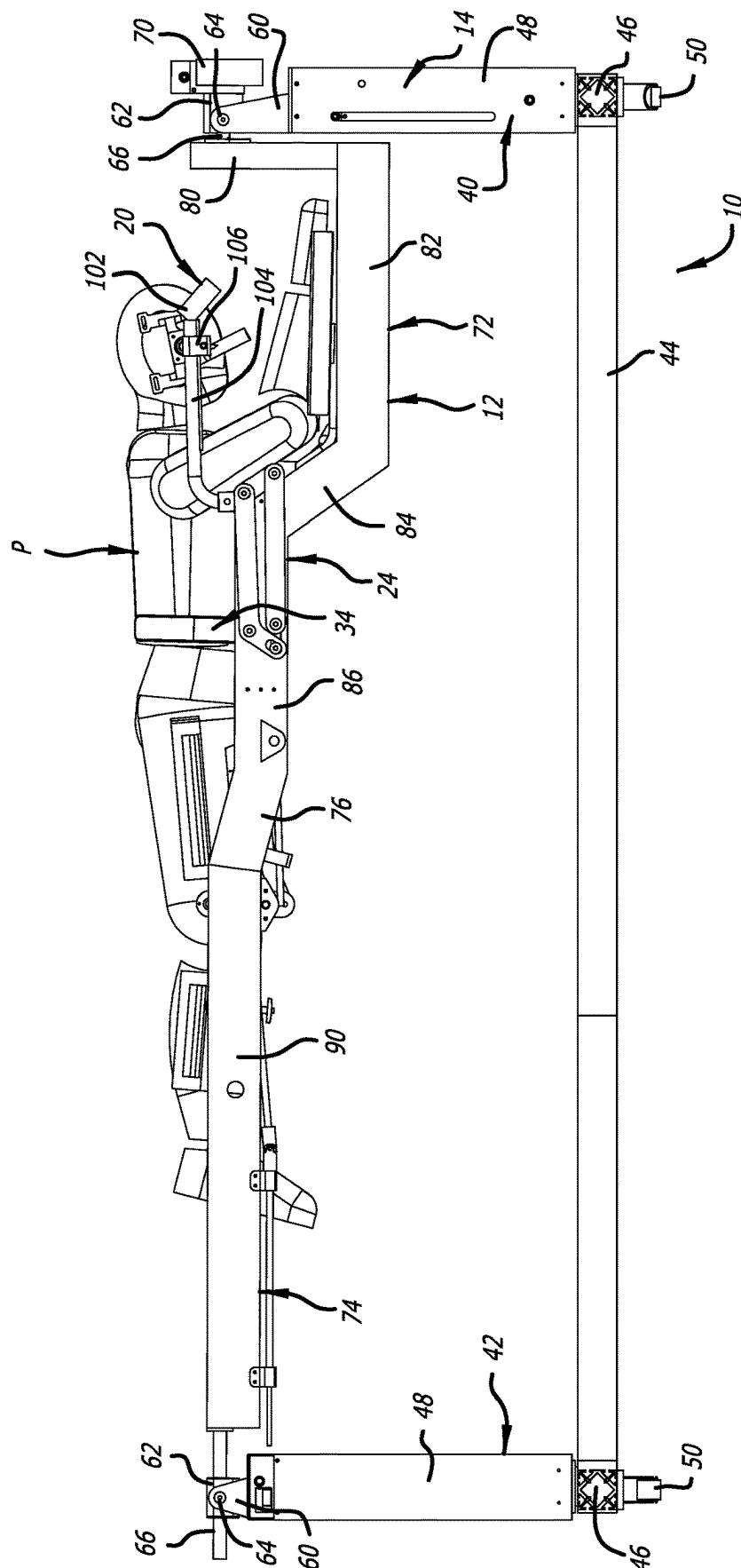
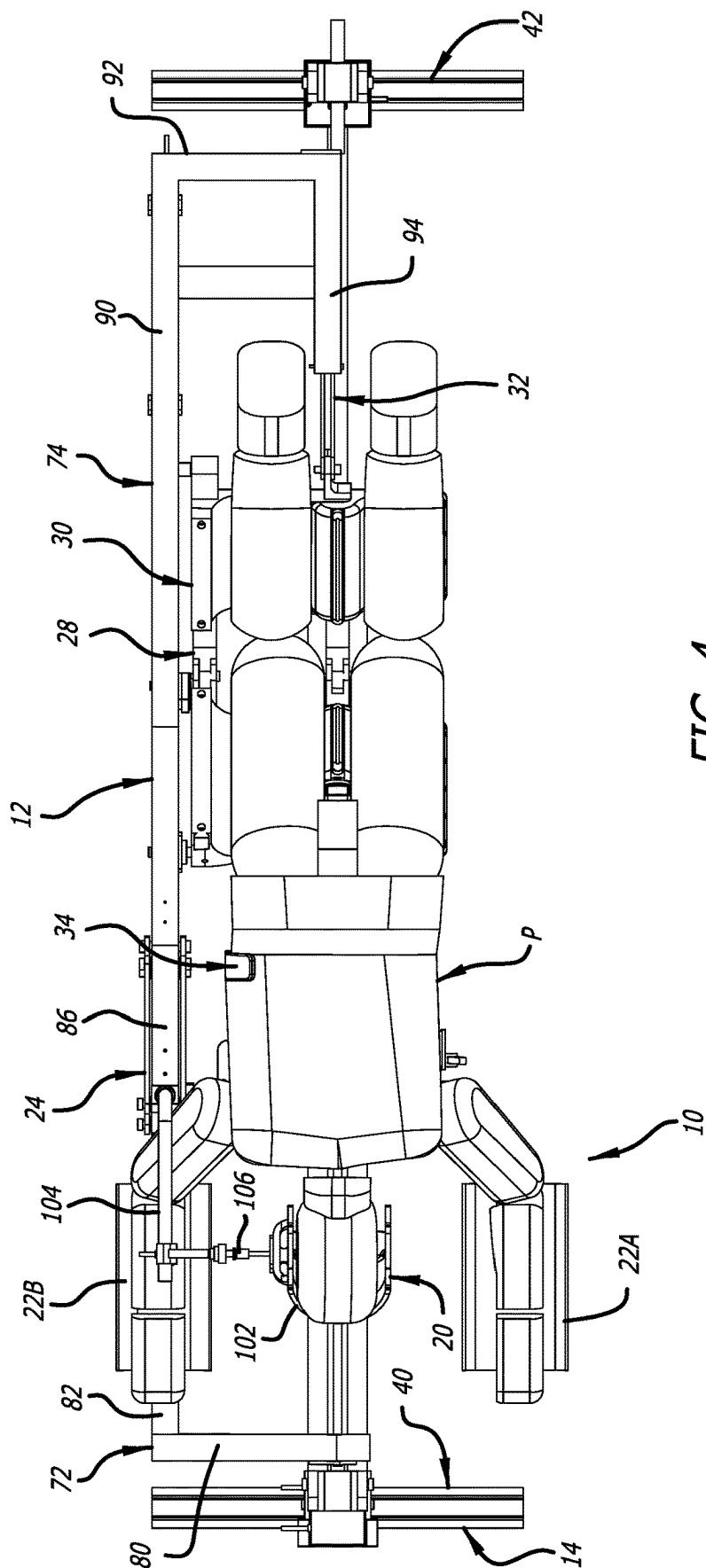


FIG. 3



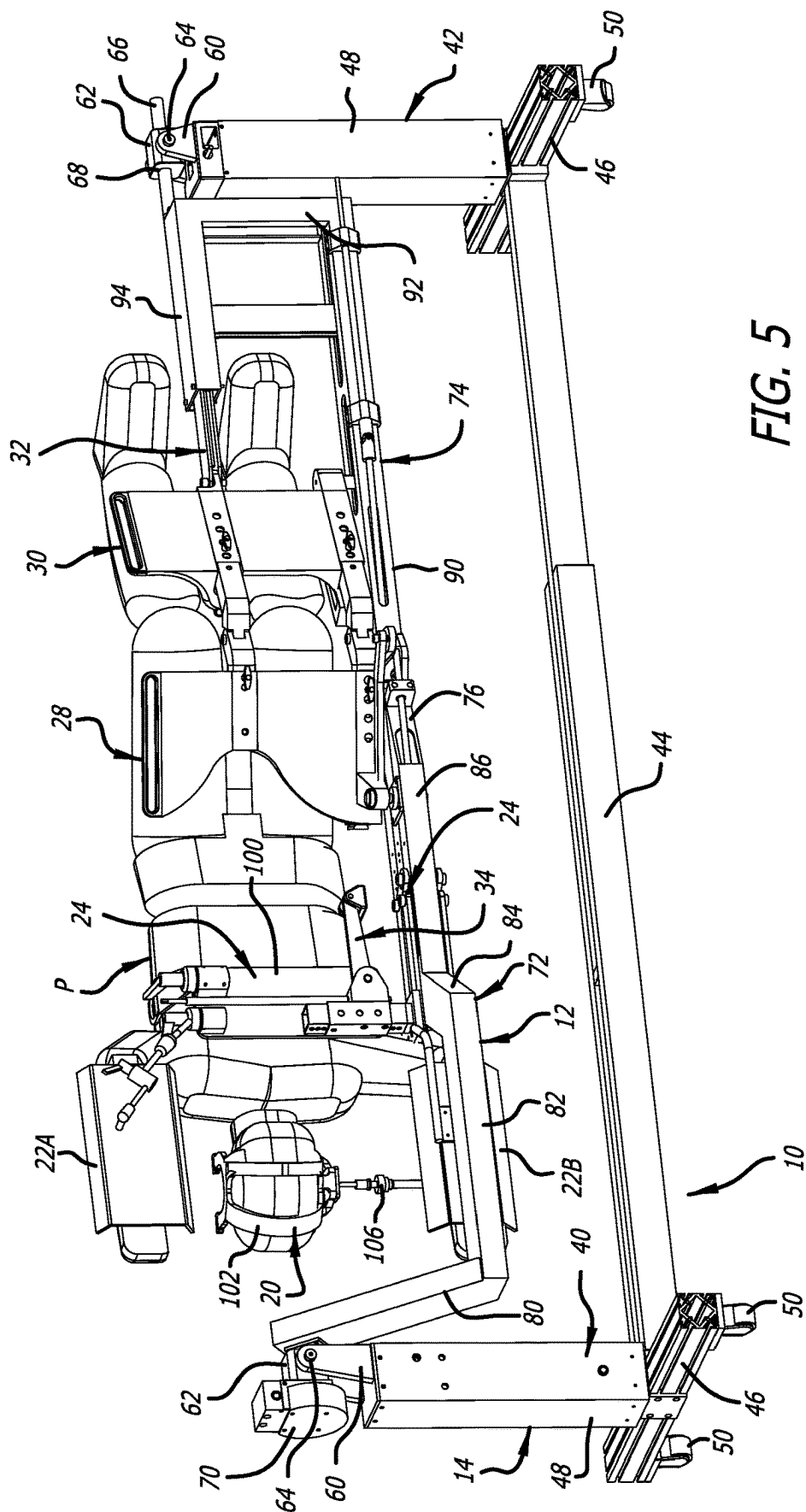


FIG. 5

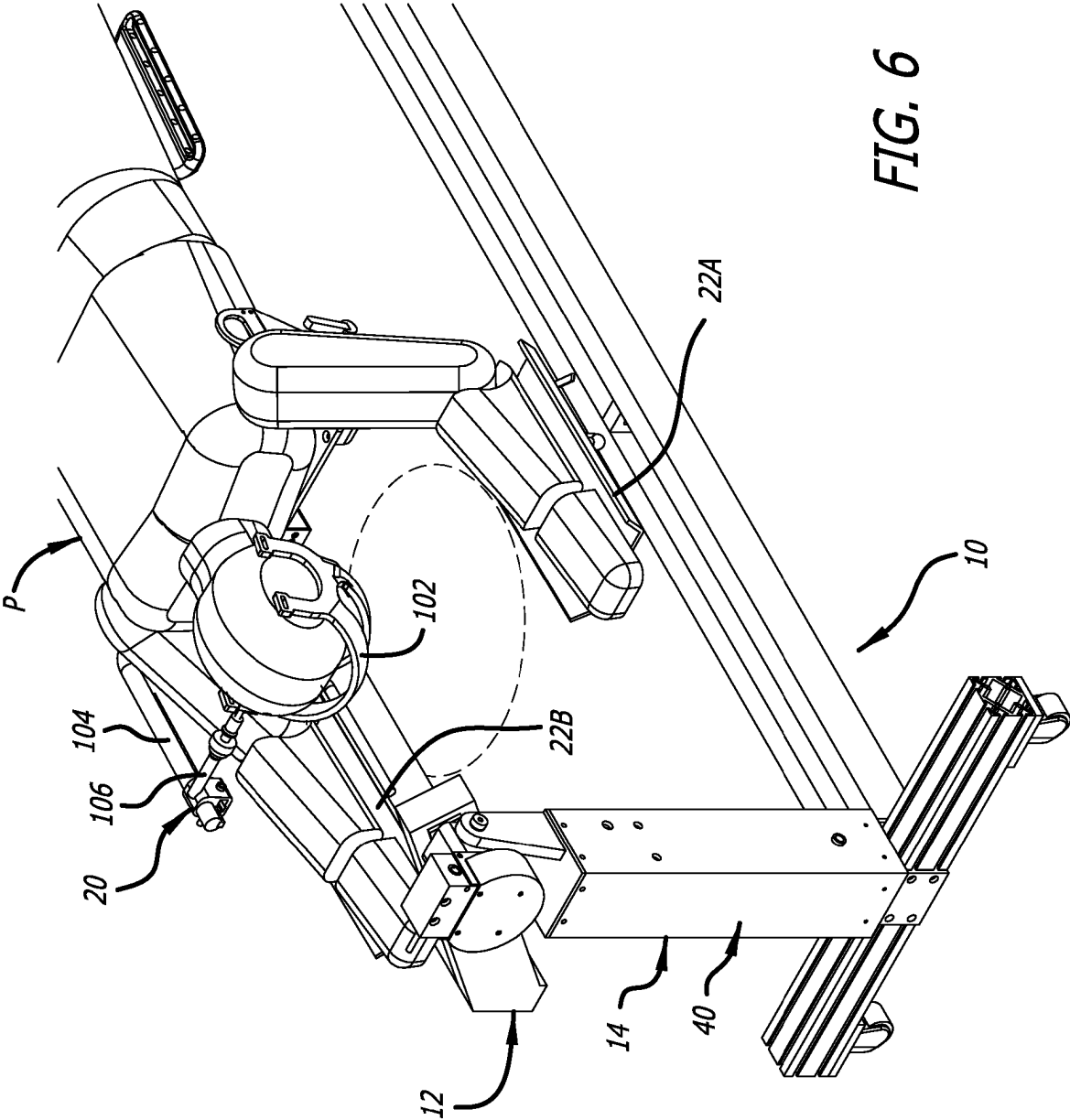
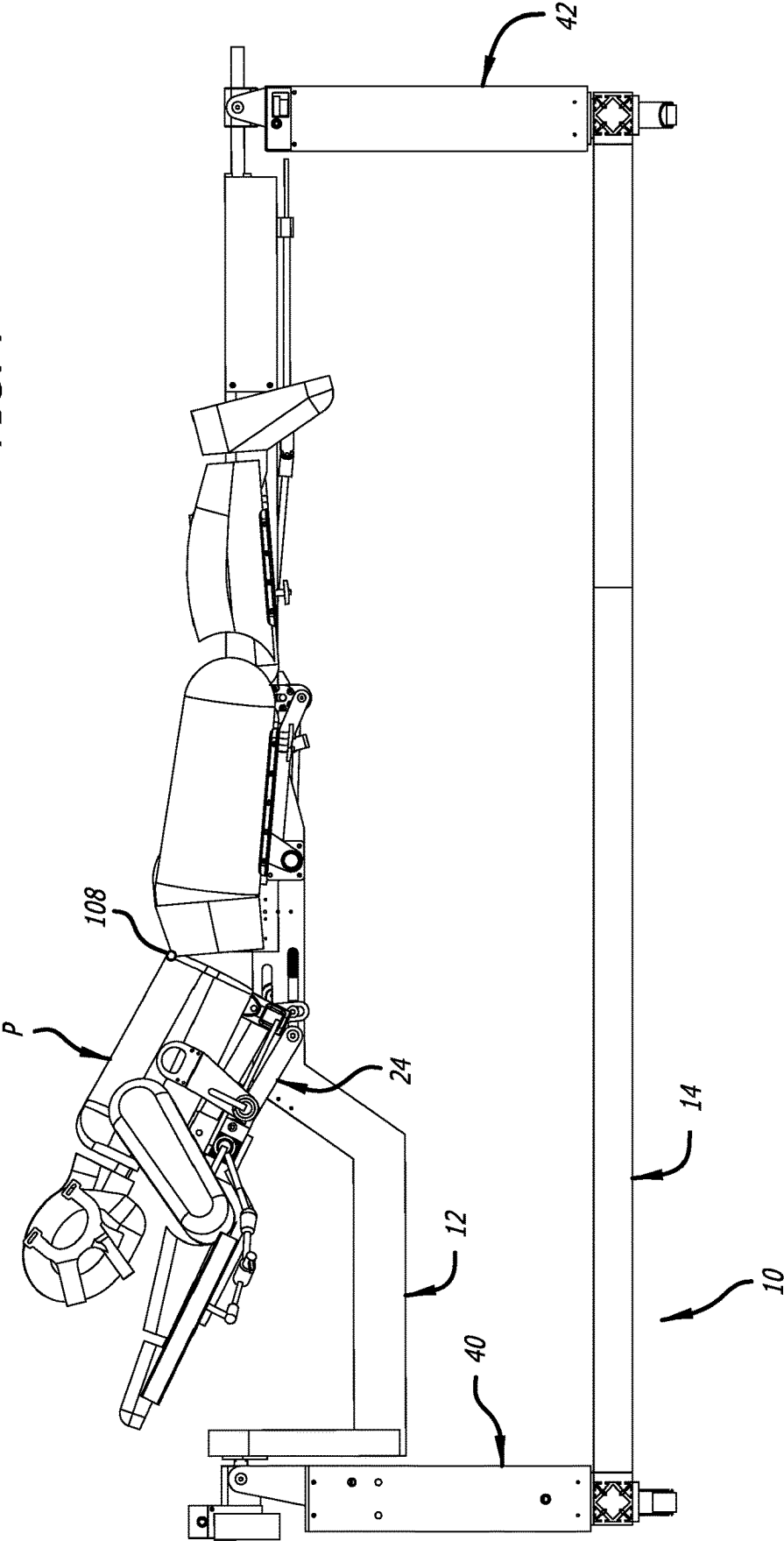


FIG. 7



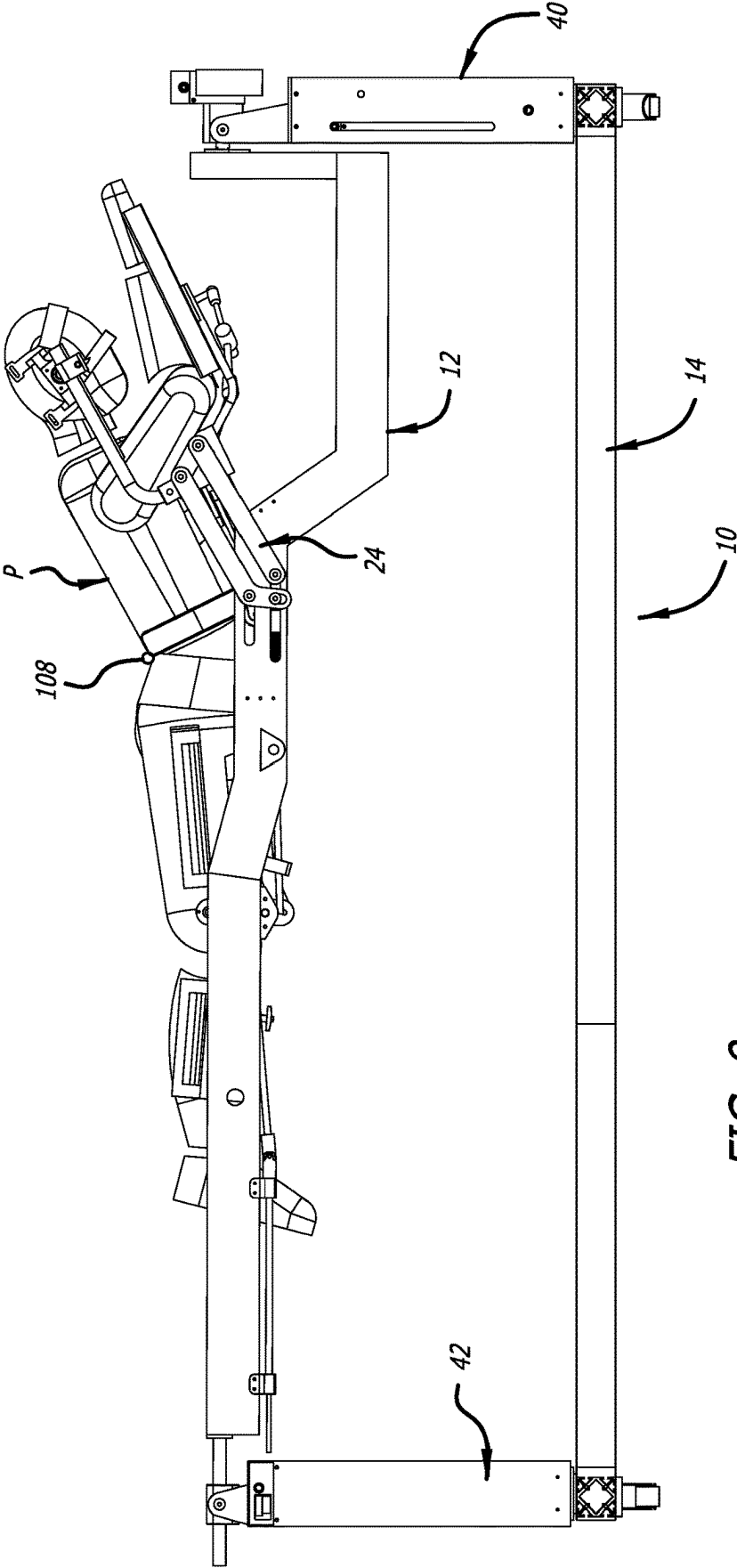
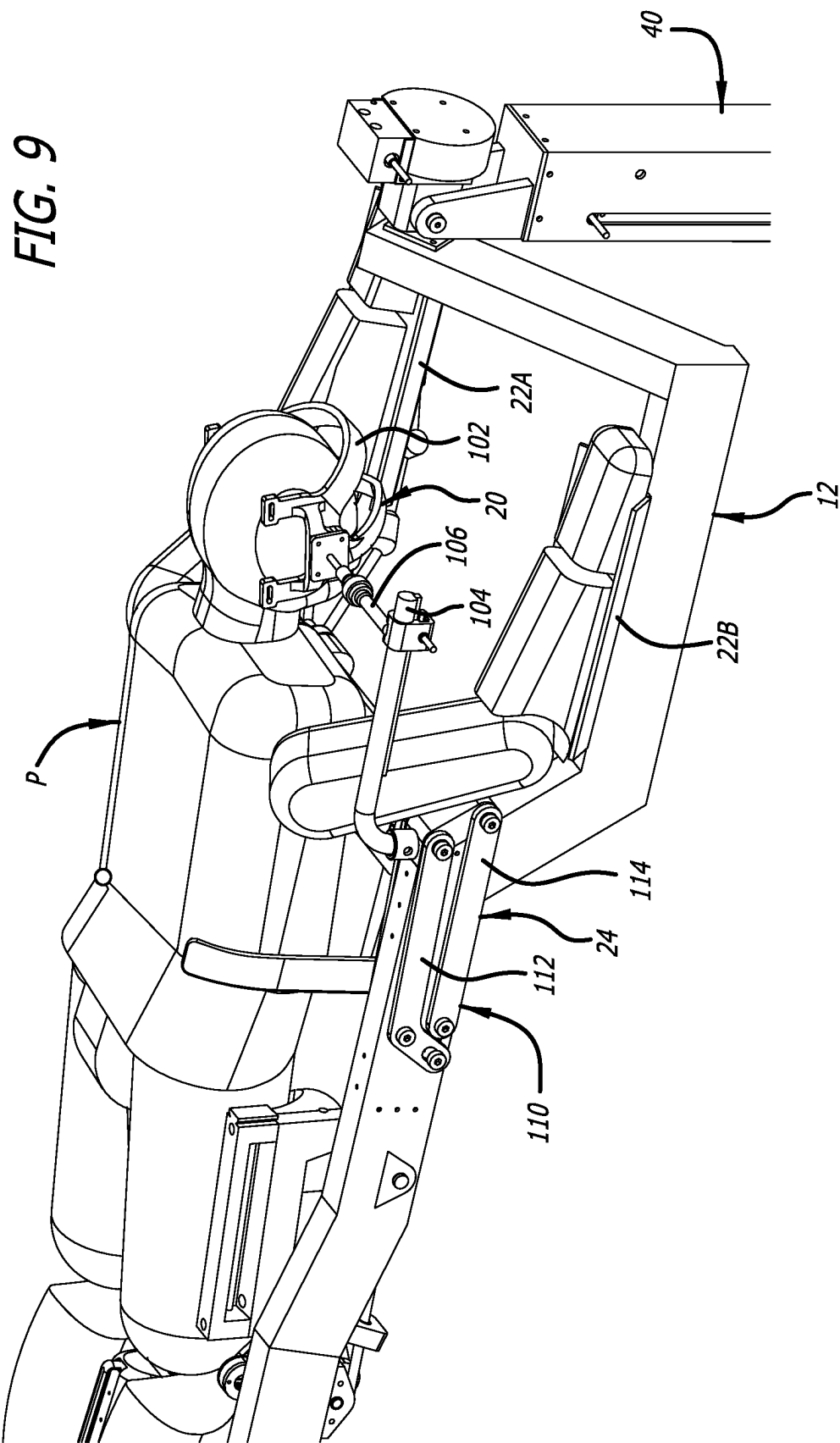


FIG. 8



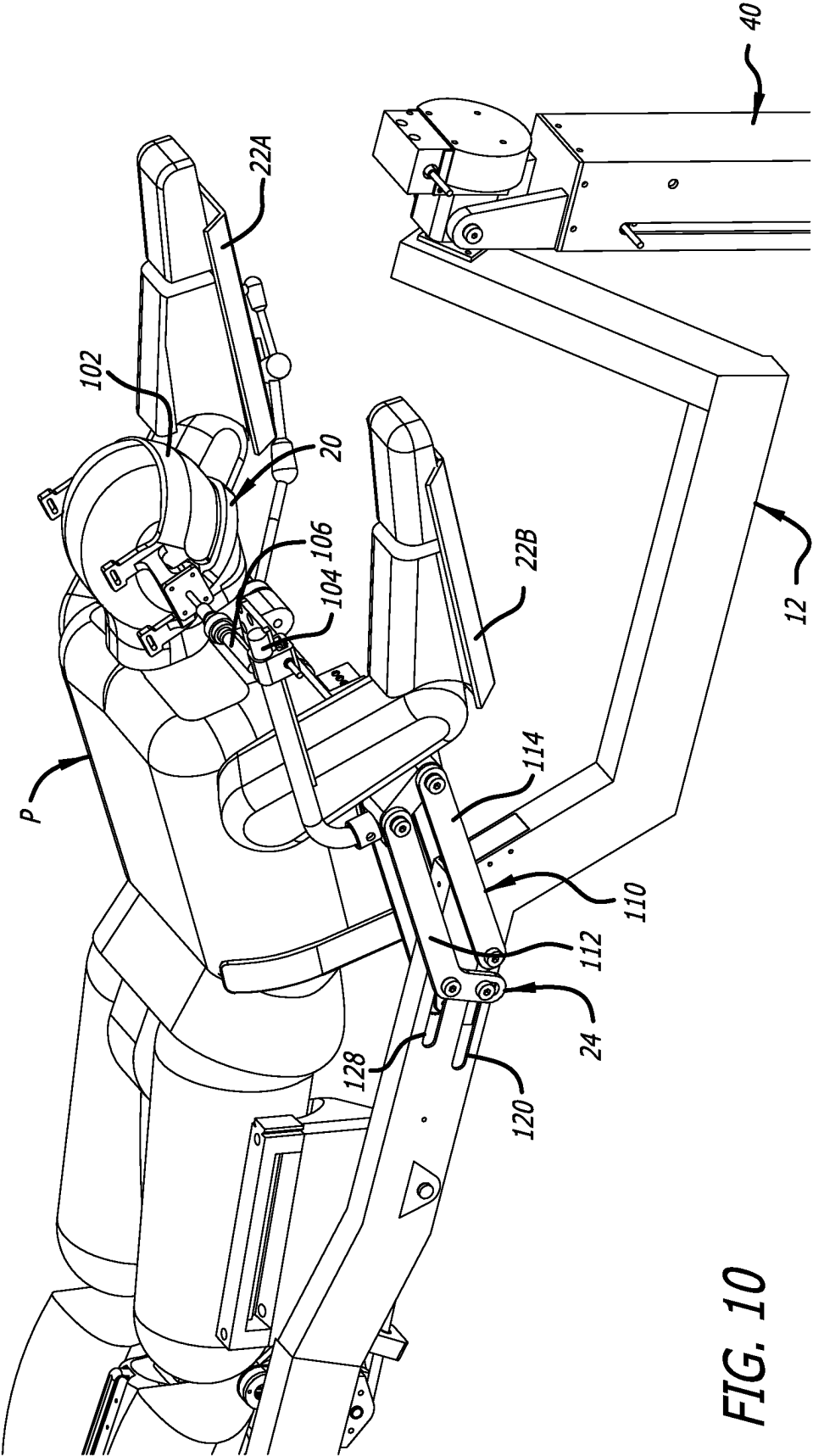


FIG. 10

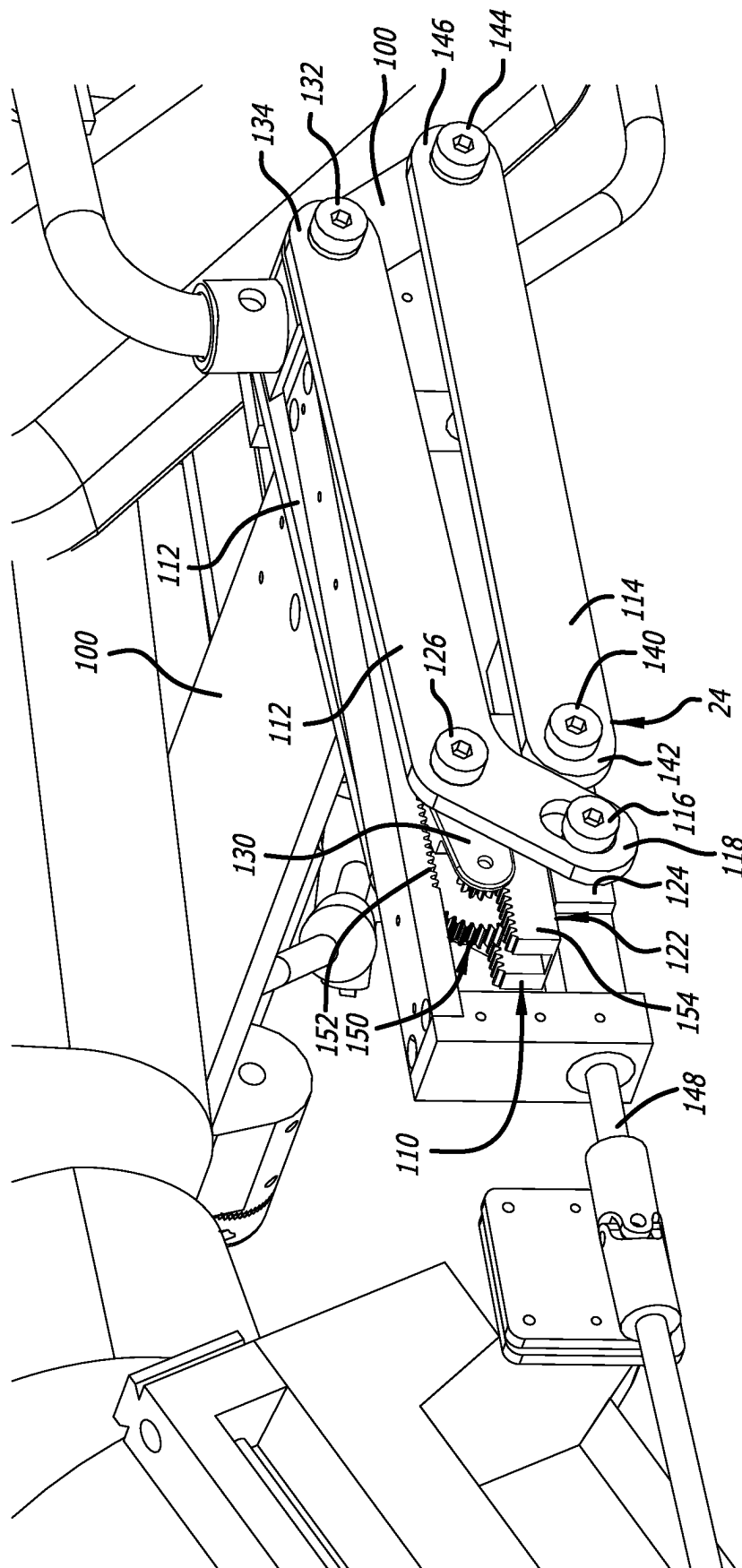


FIG. 11

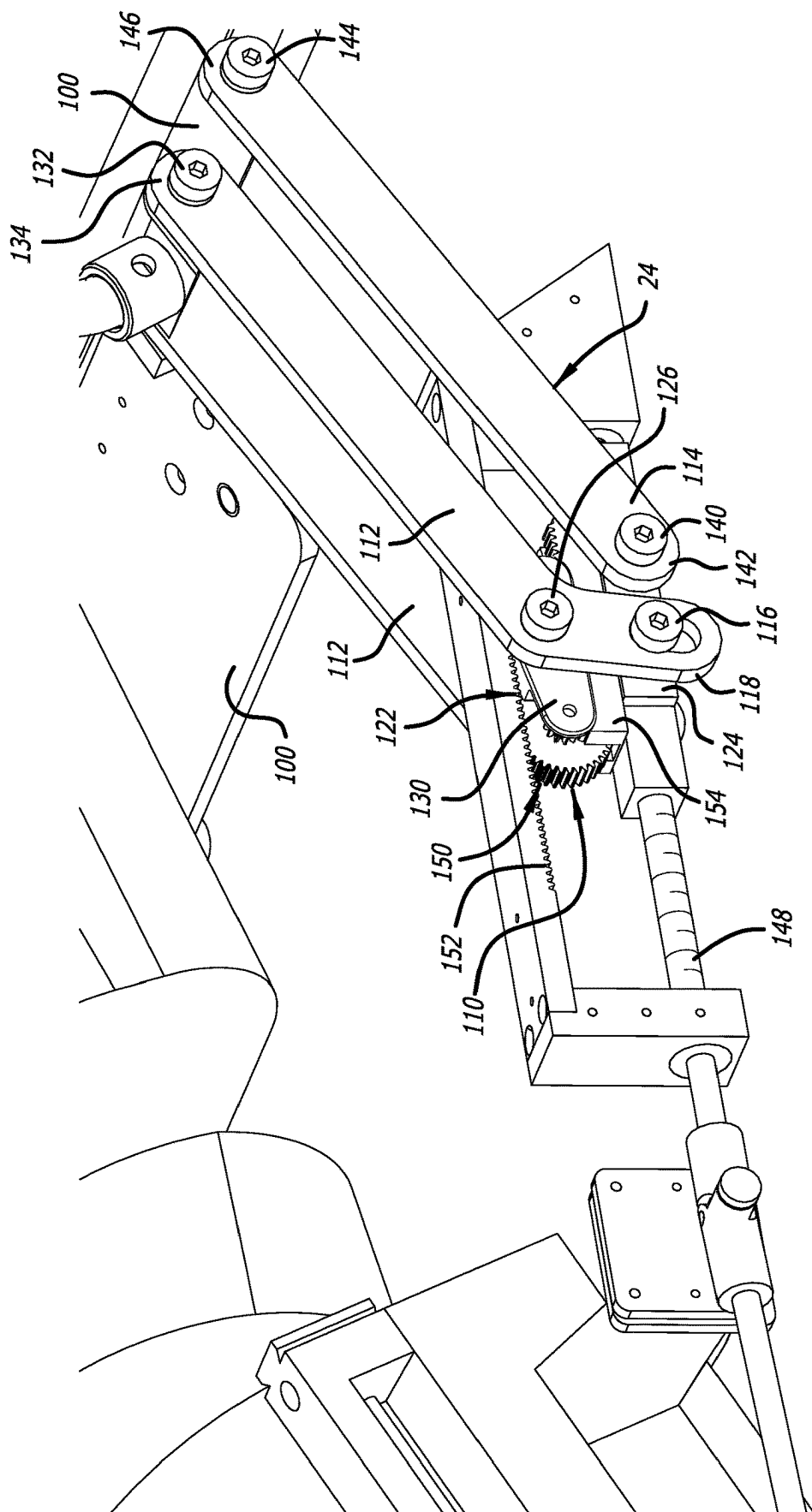
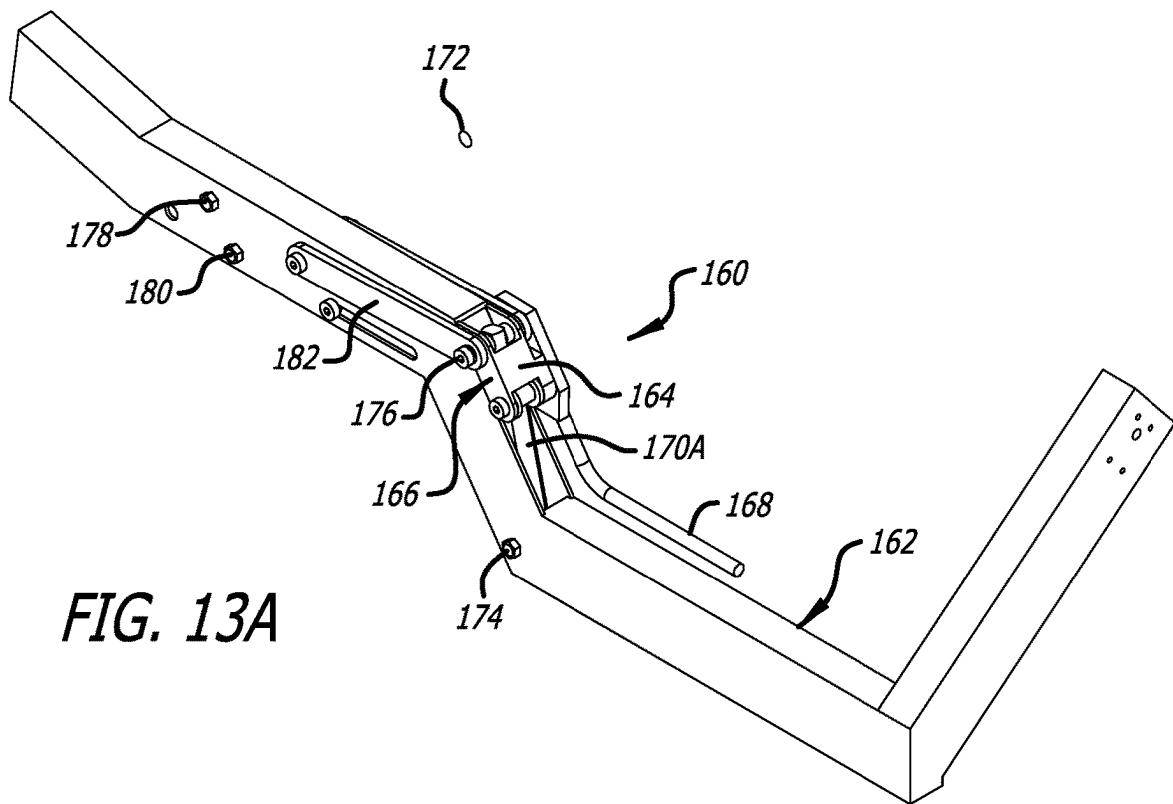


FIG. 12



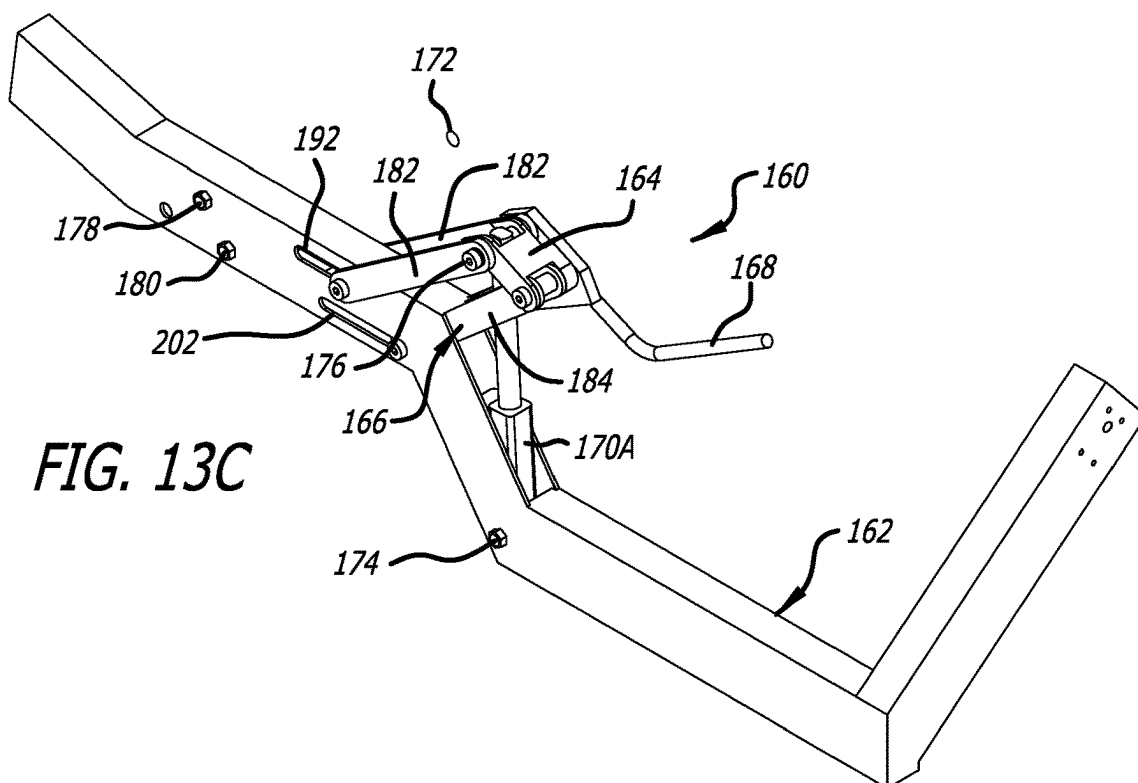
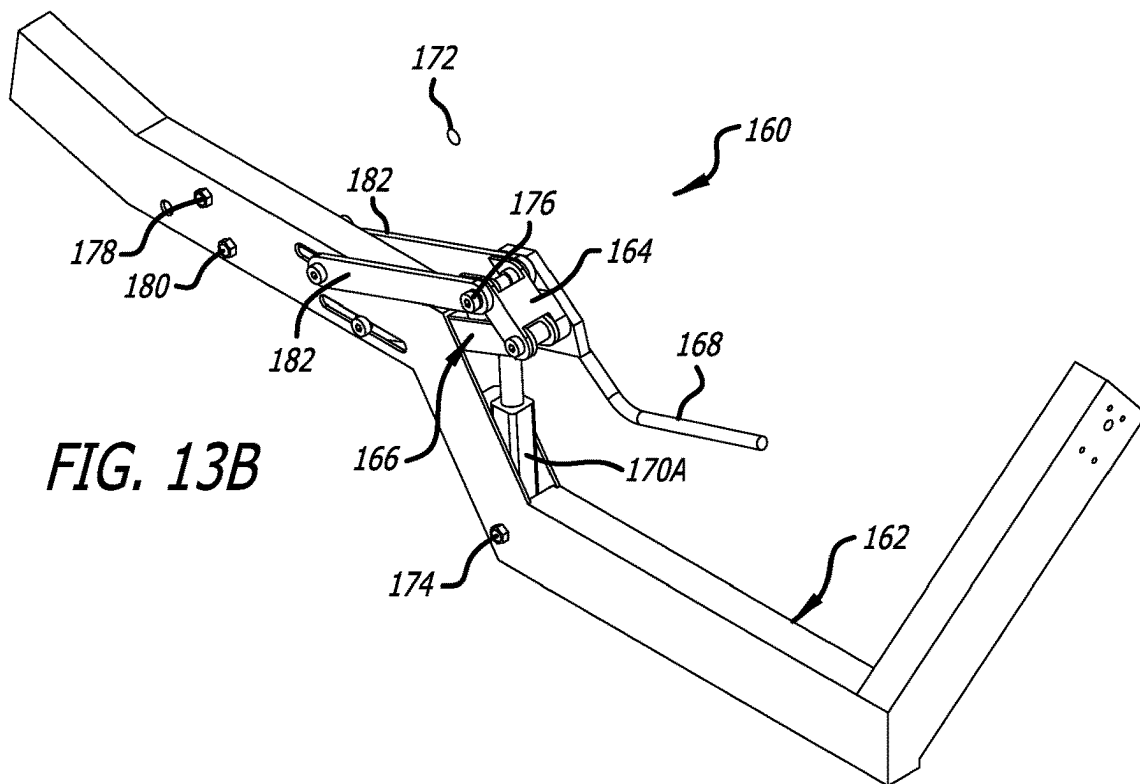


FIG. 14

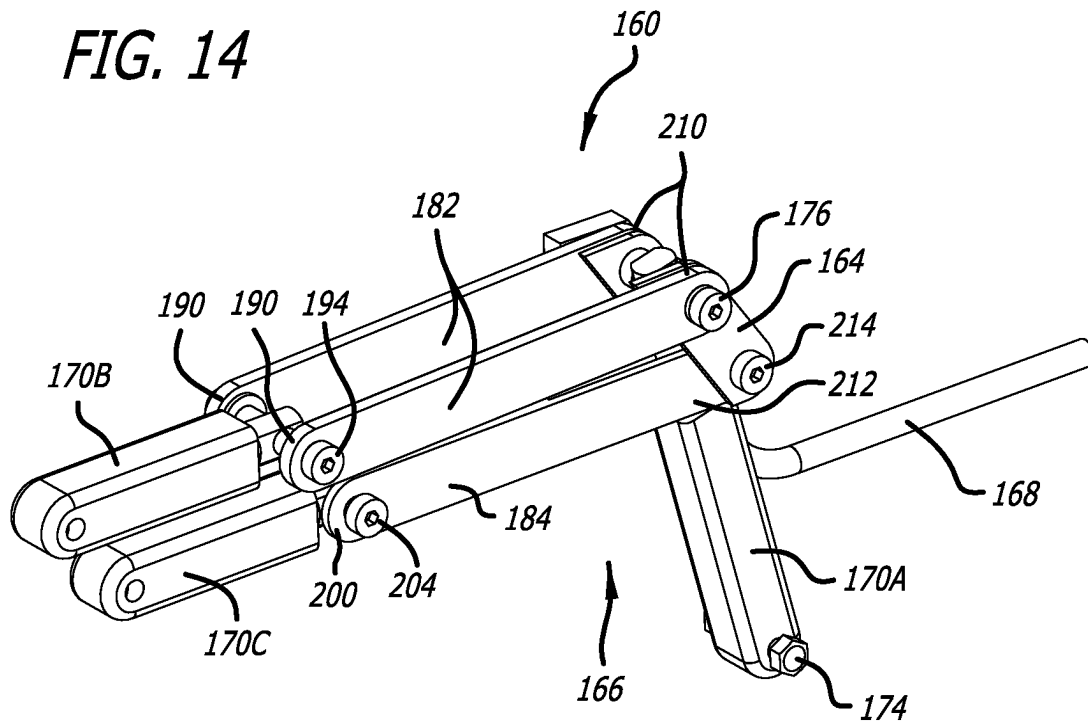


FIG. 15

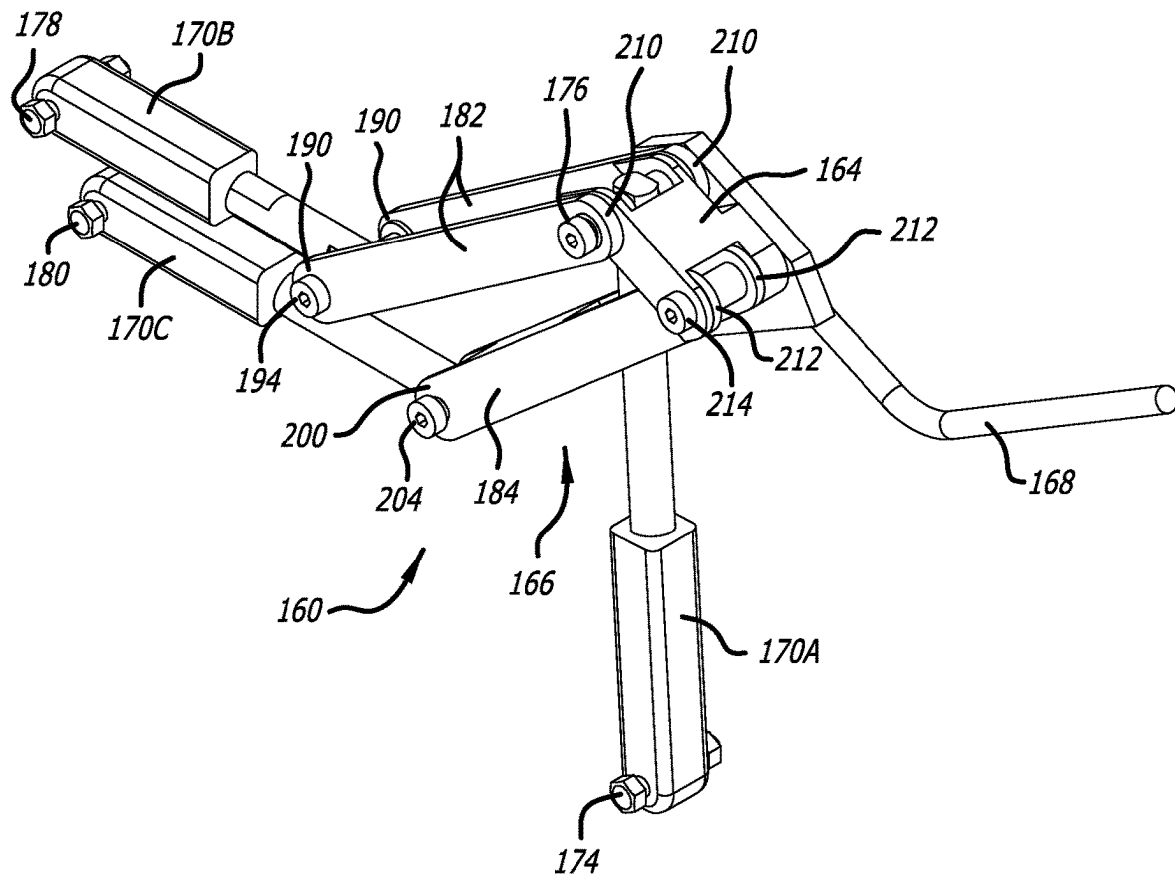
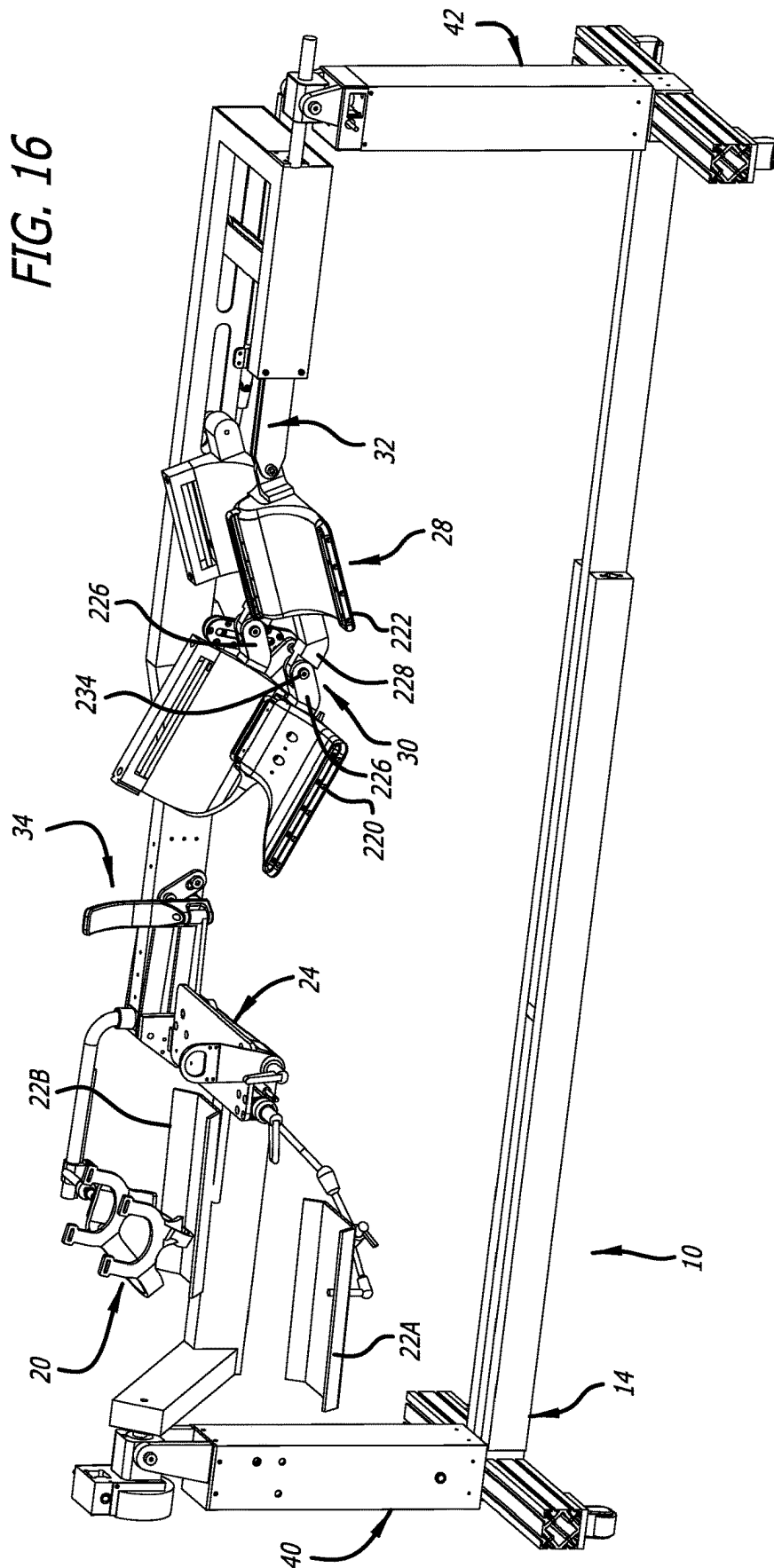


FIG. 16



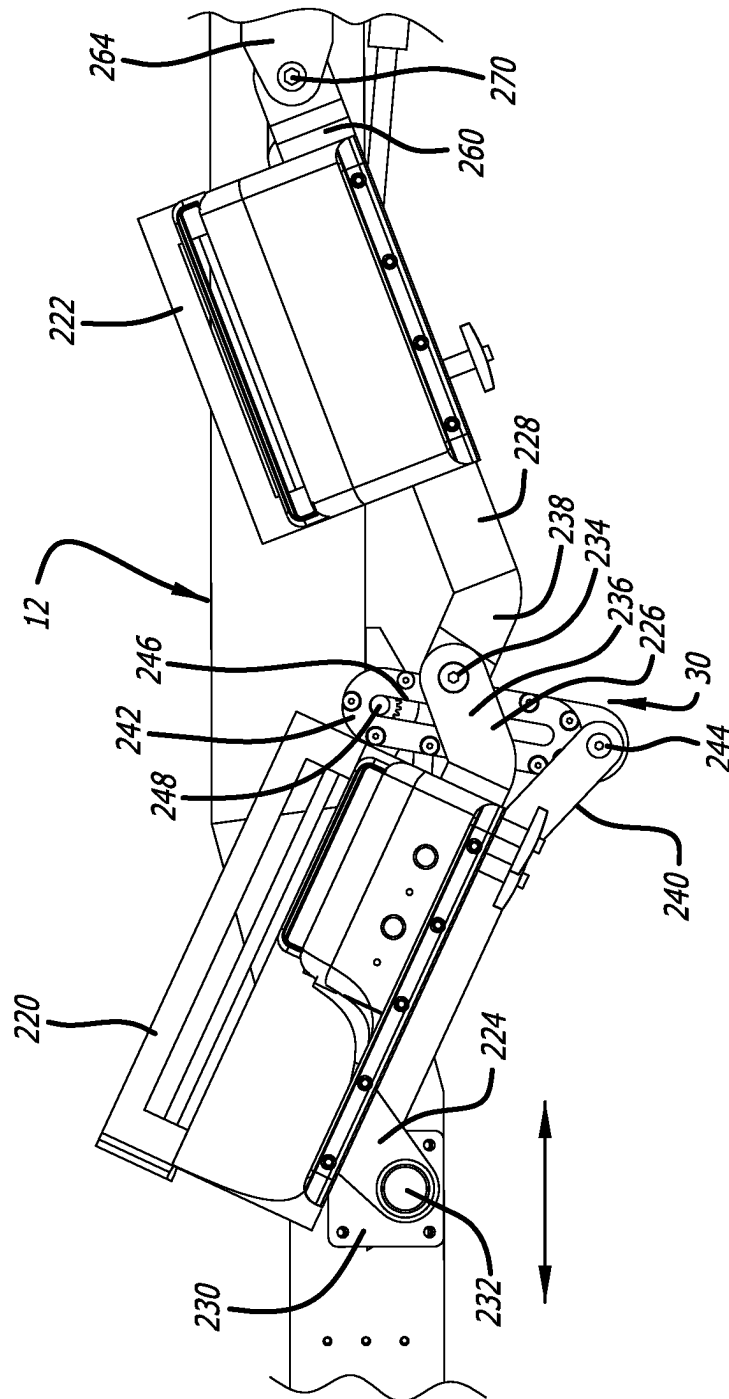
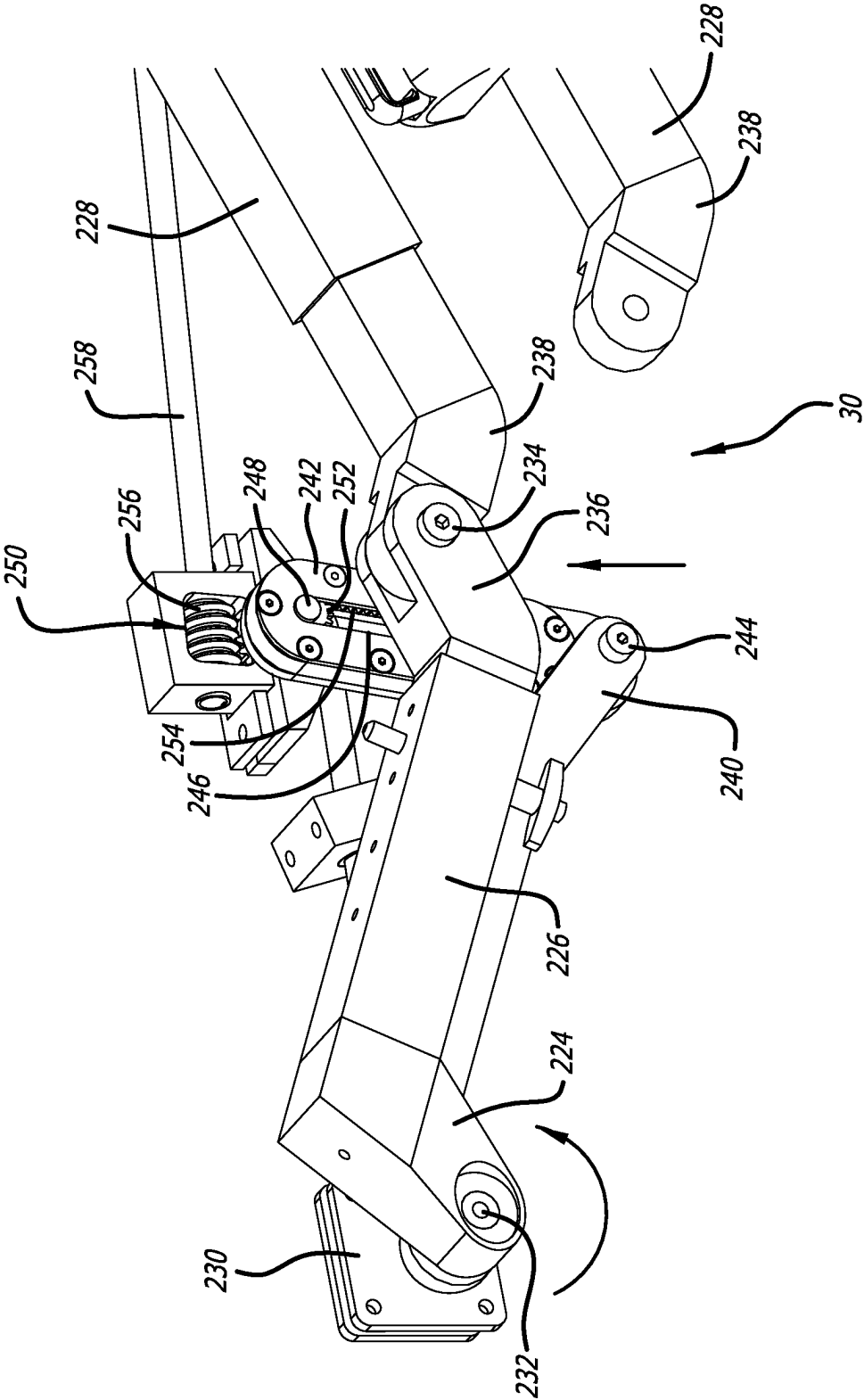
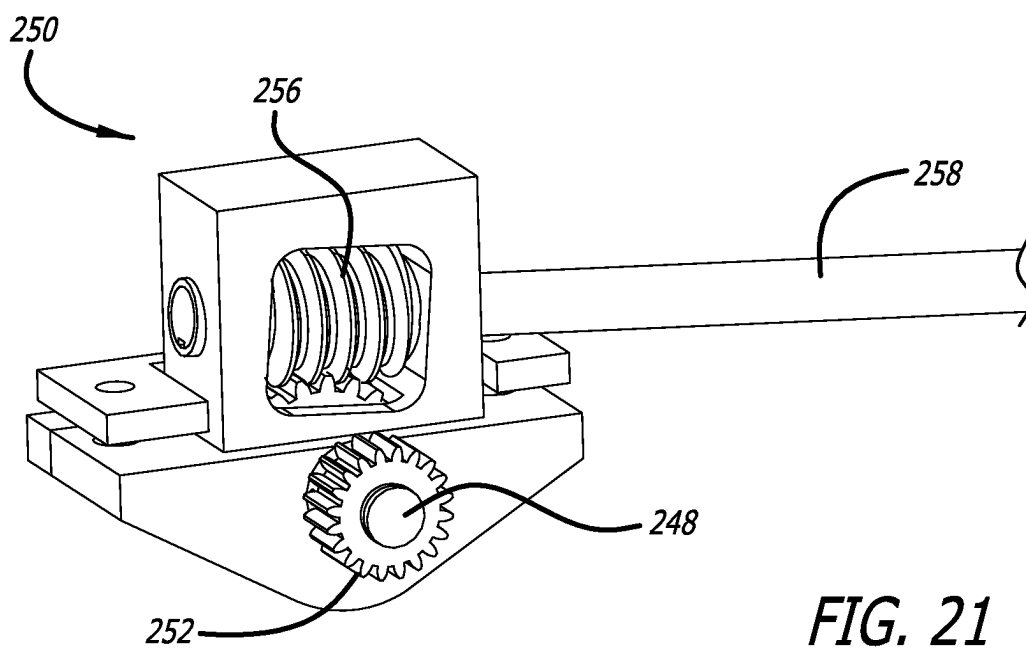
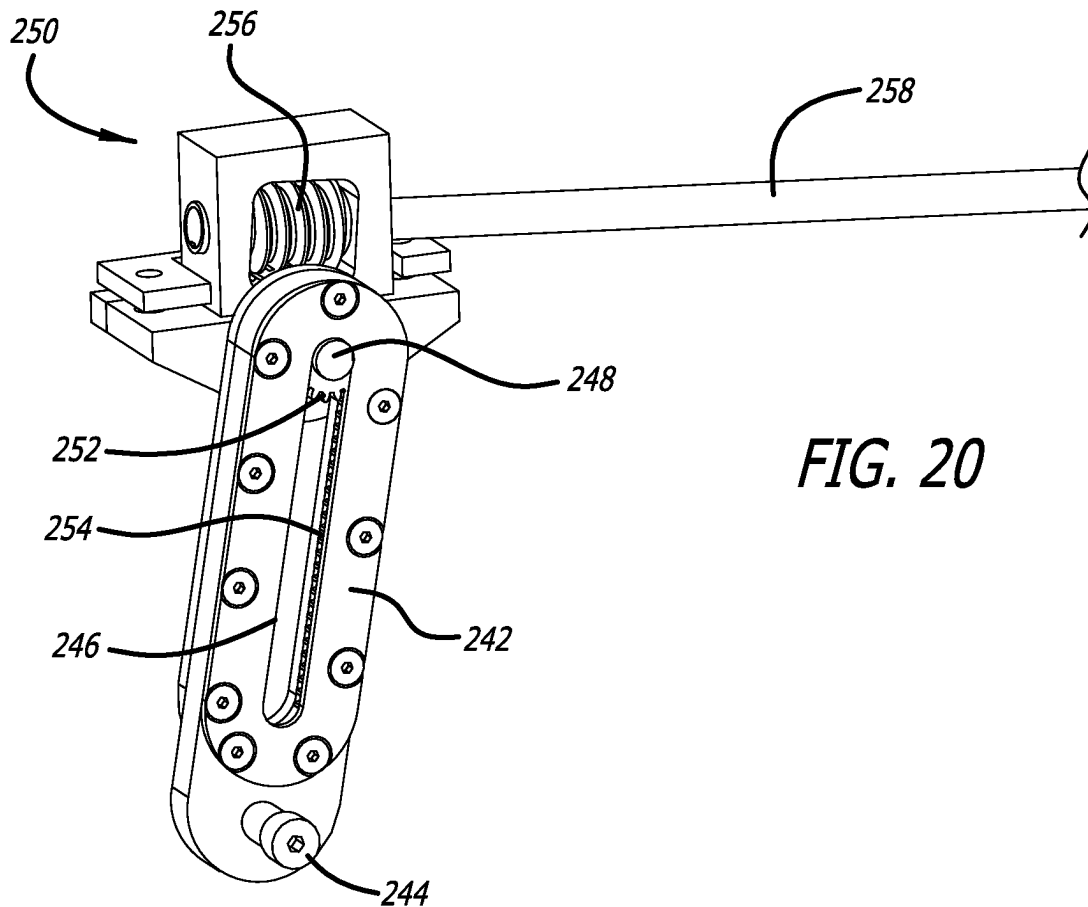


FIG. 18

FIG. 19





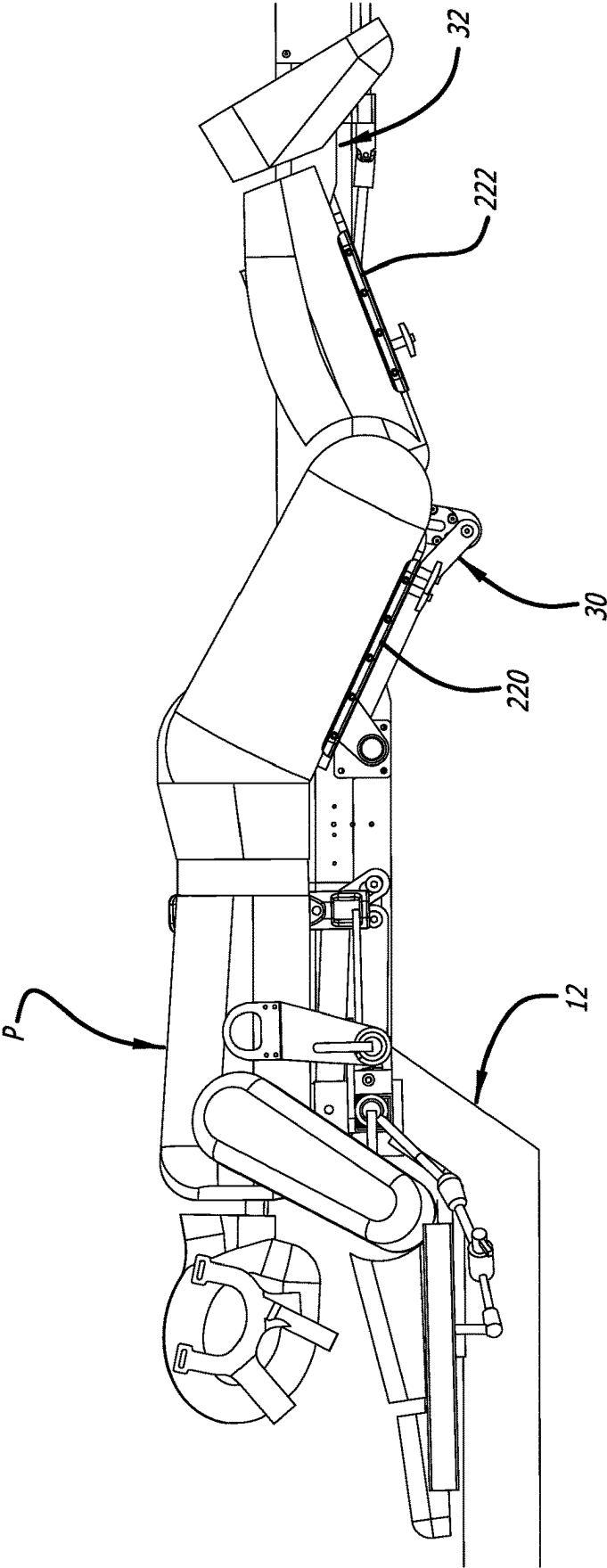


FIG. 22

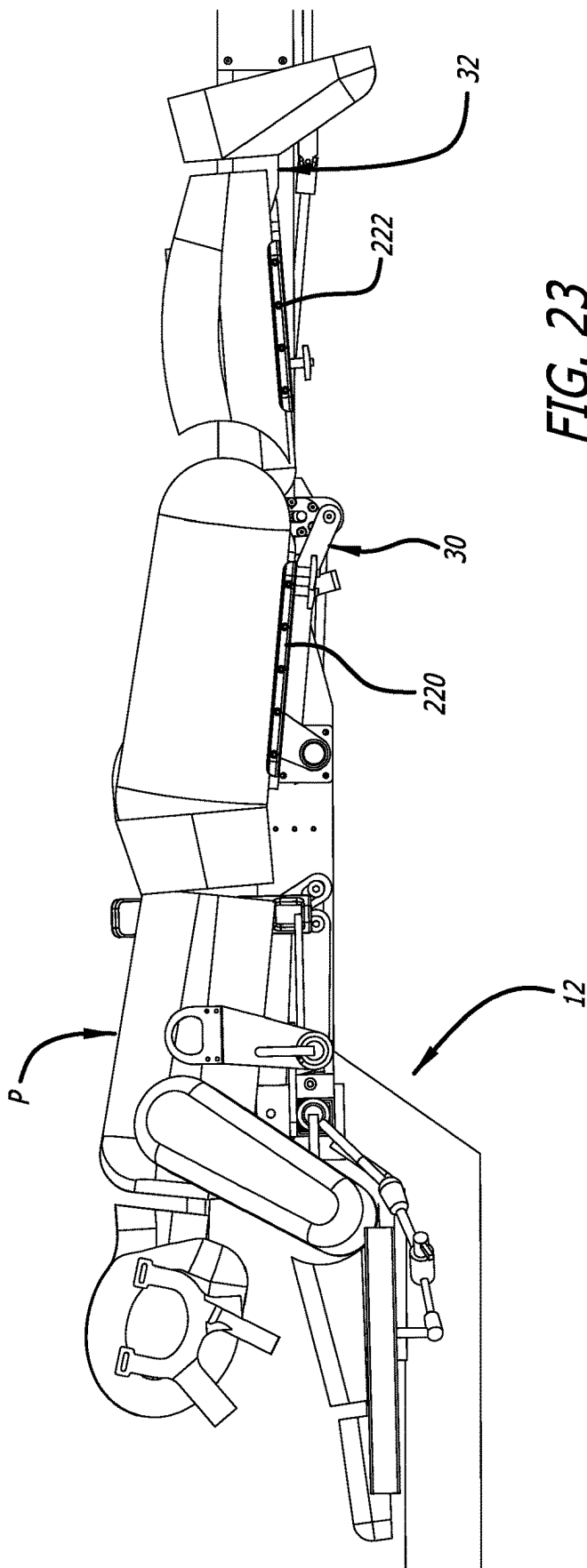
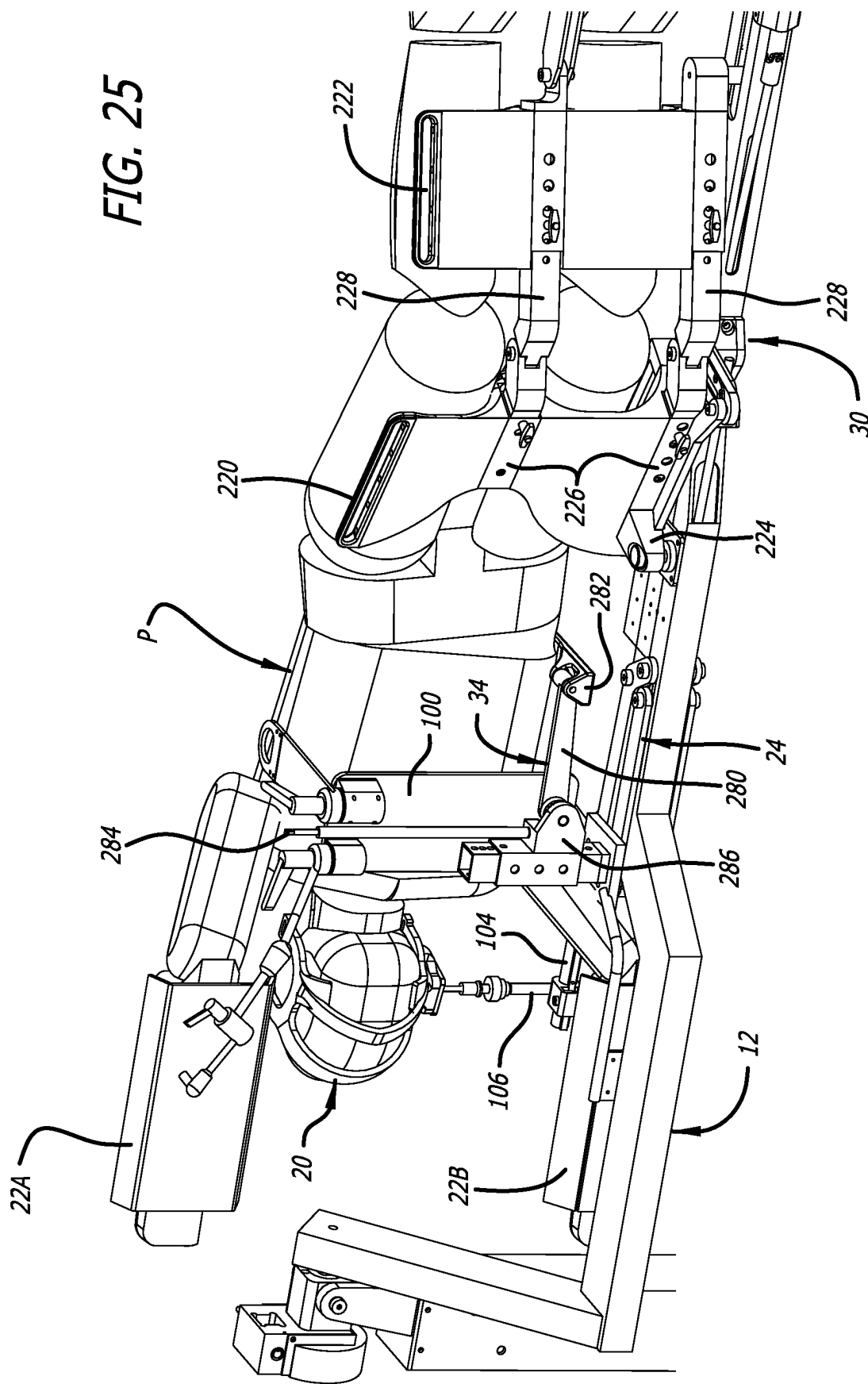
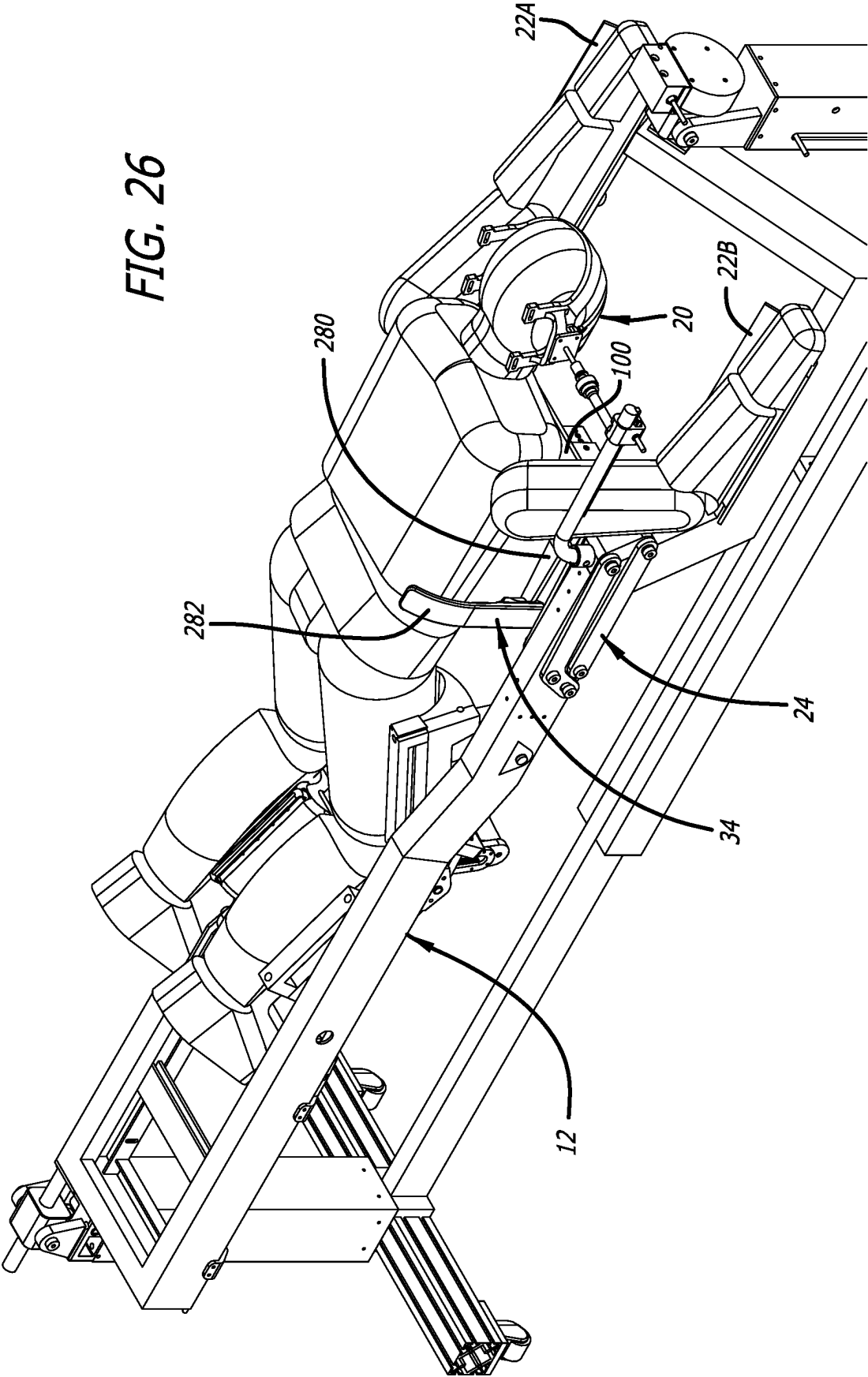


FIG. 25





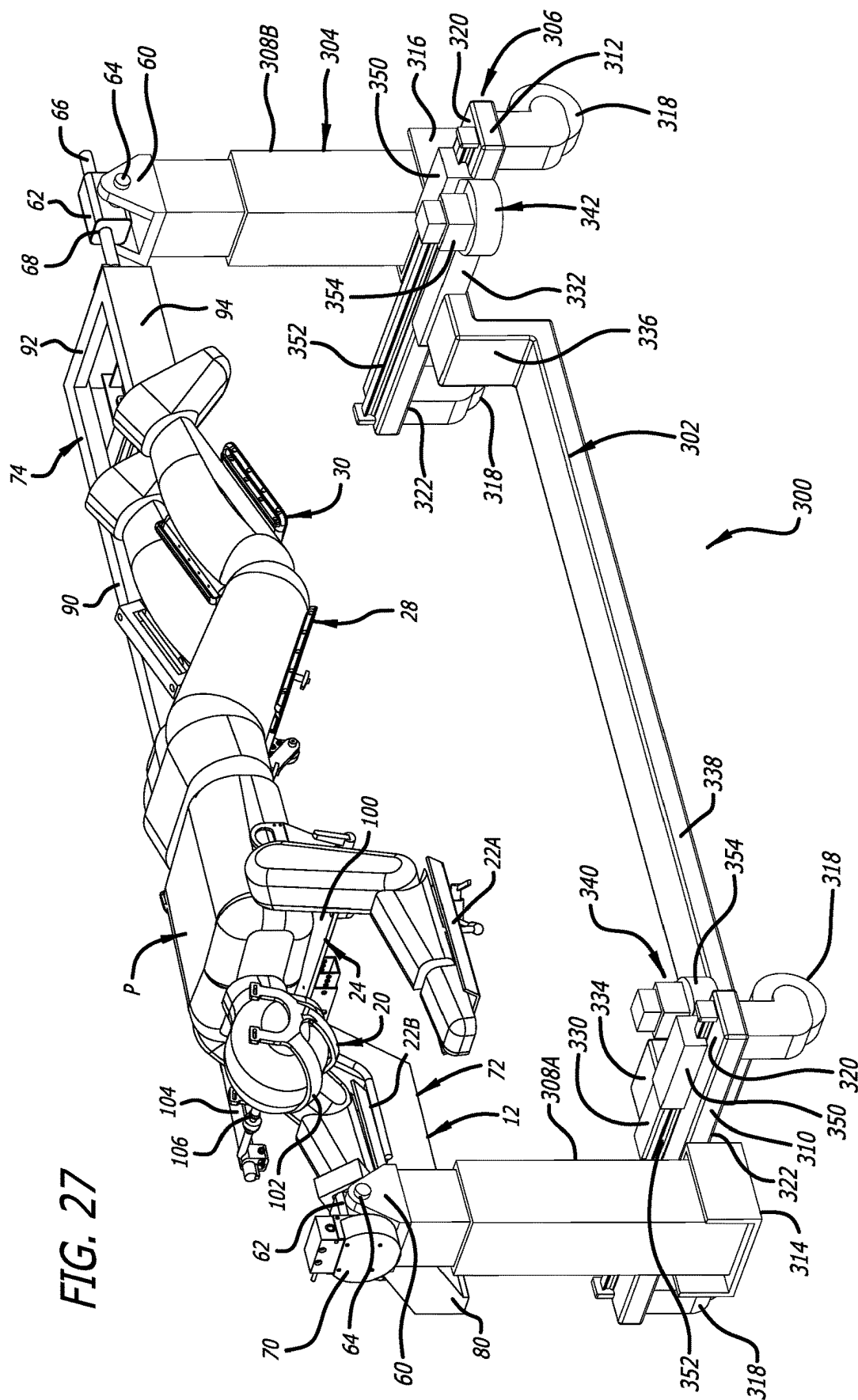


FIG. 27

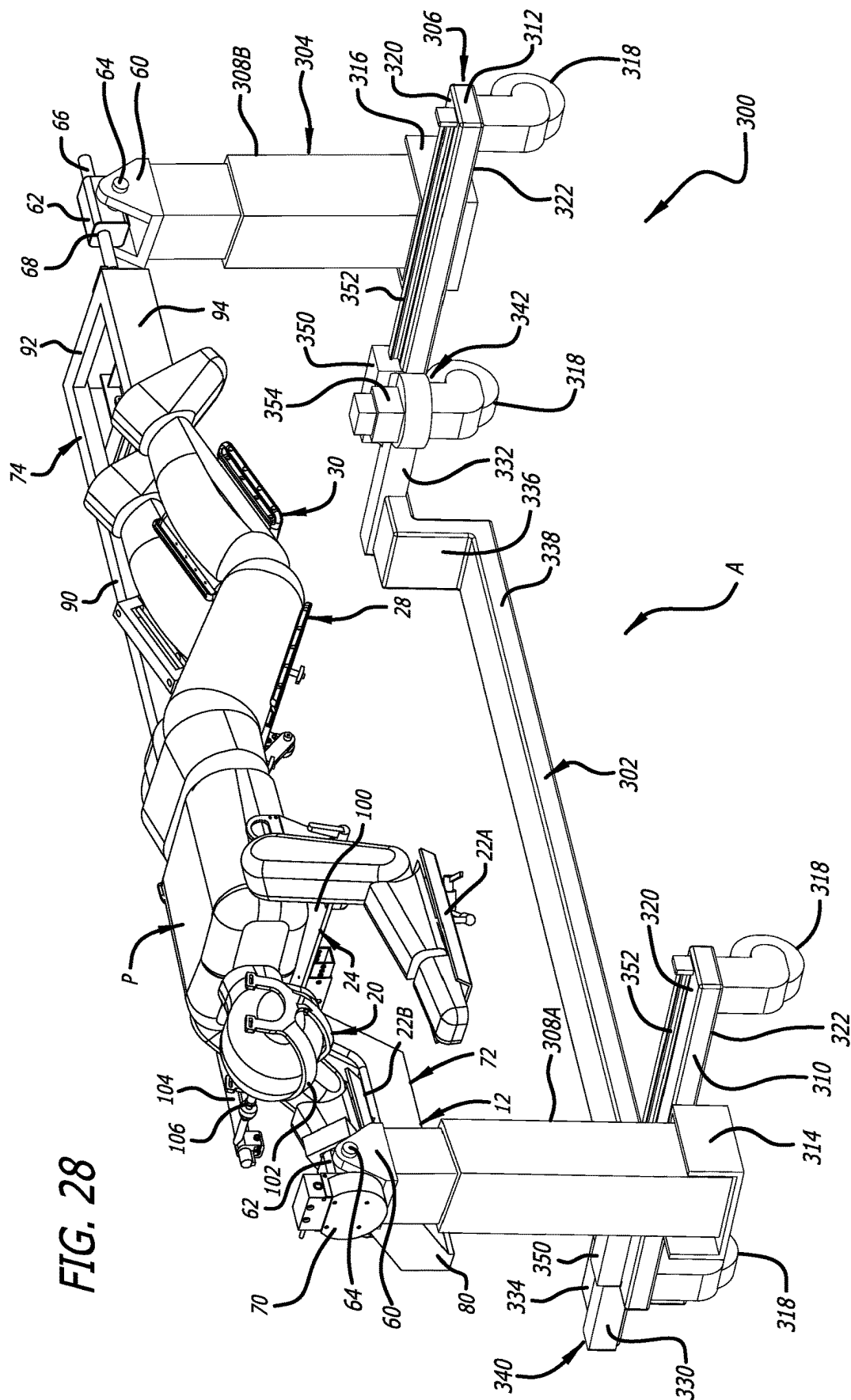
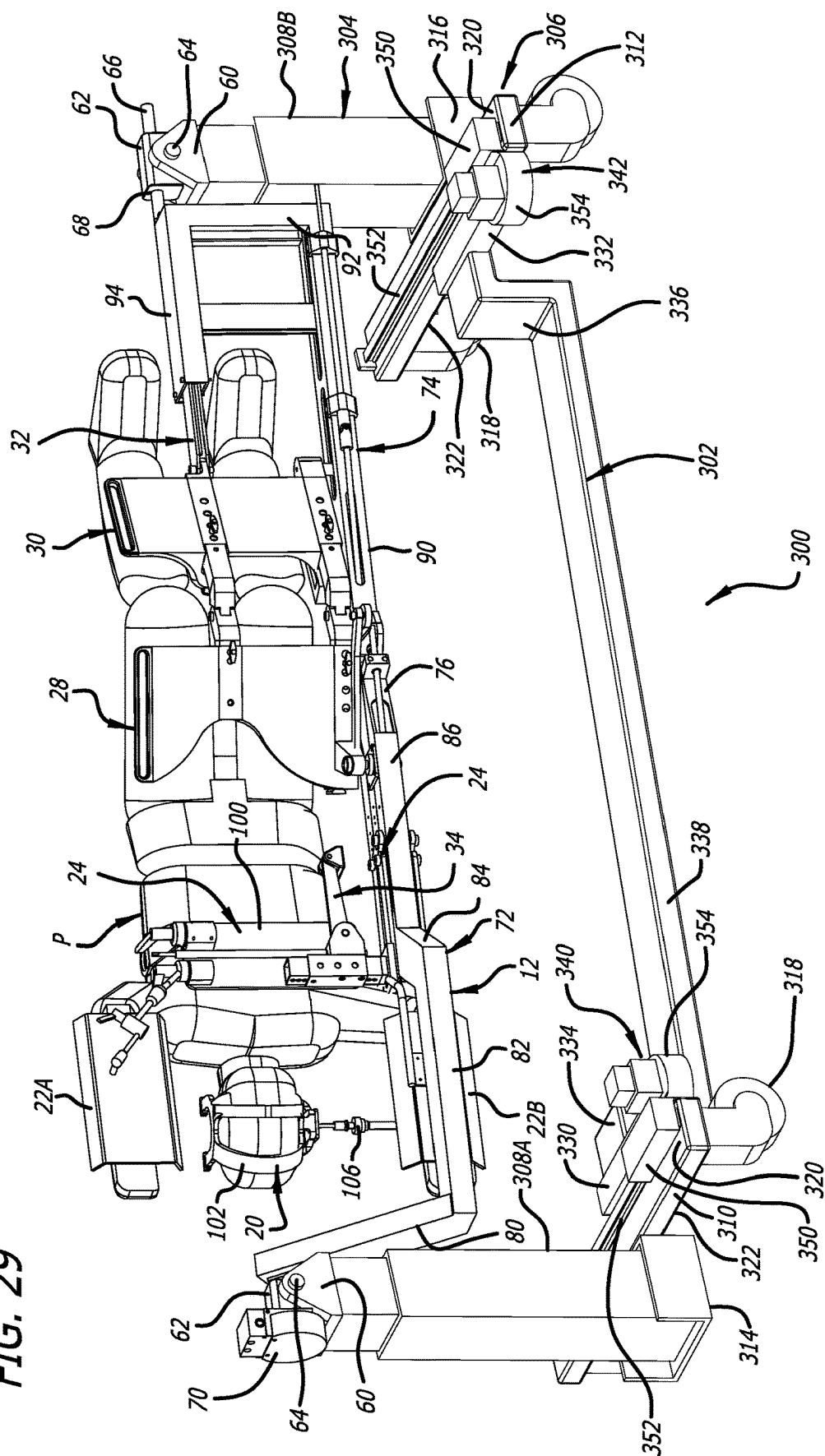
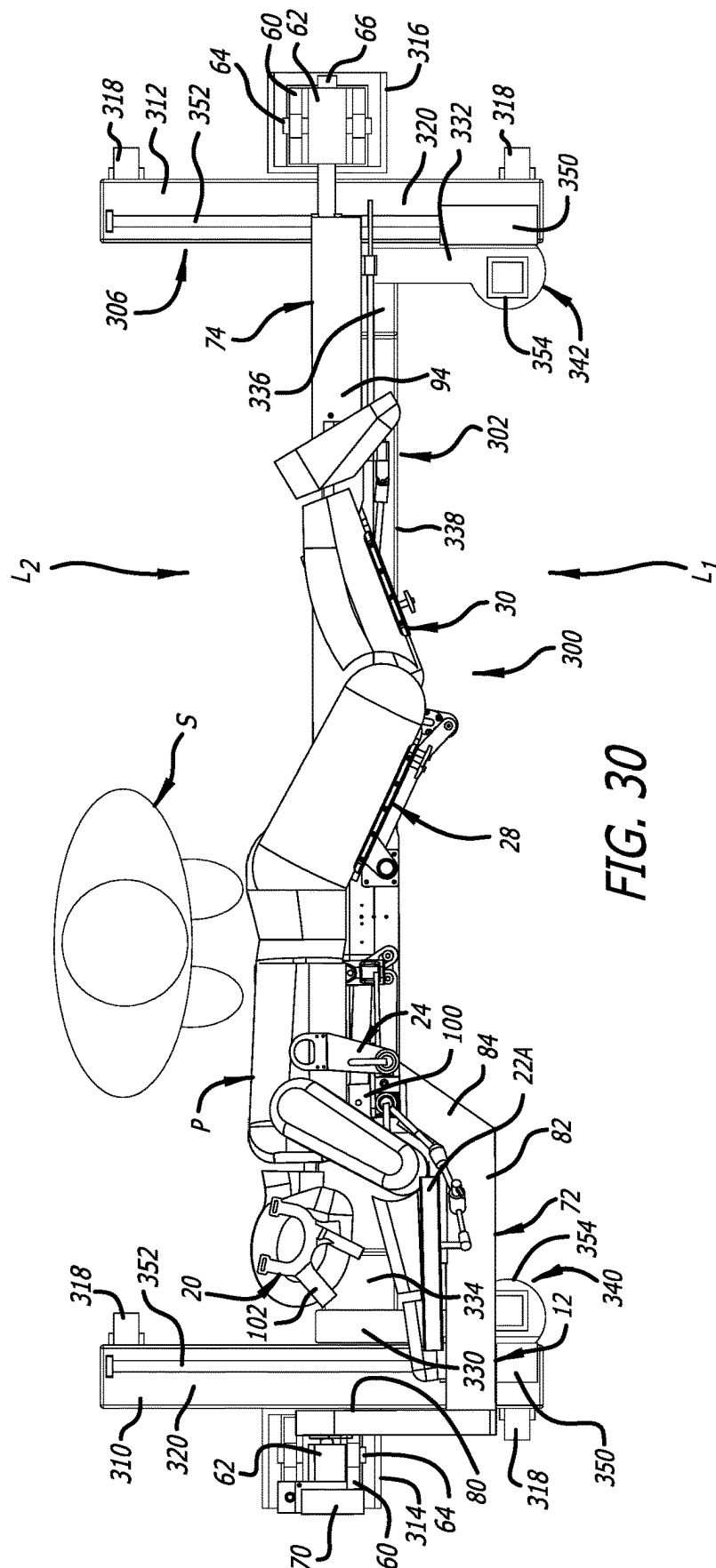
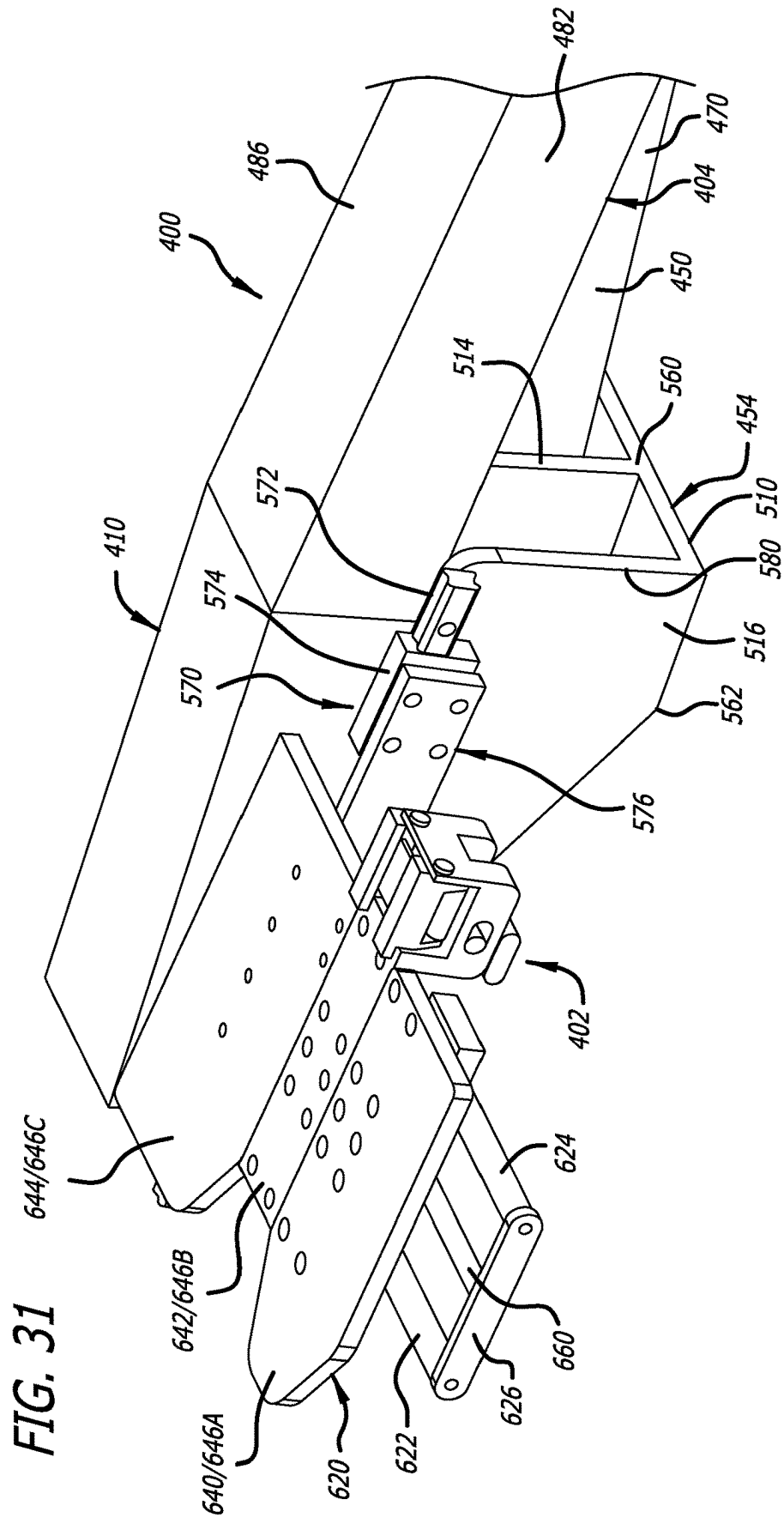
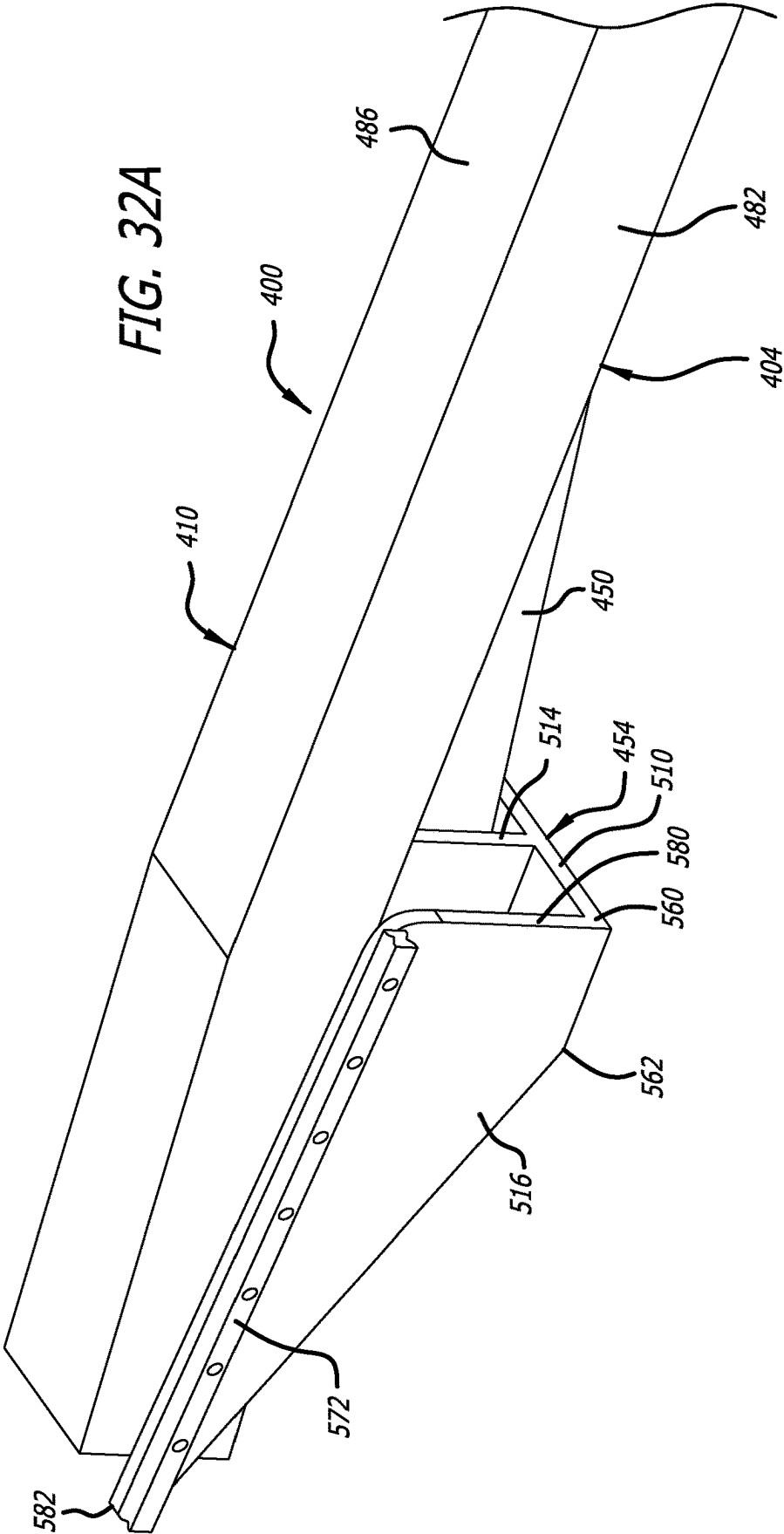


FIG. 29









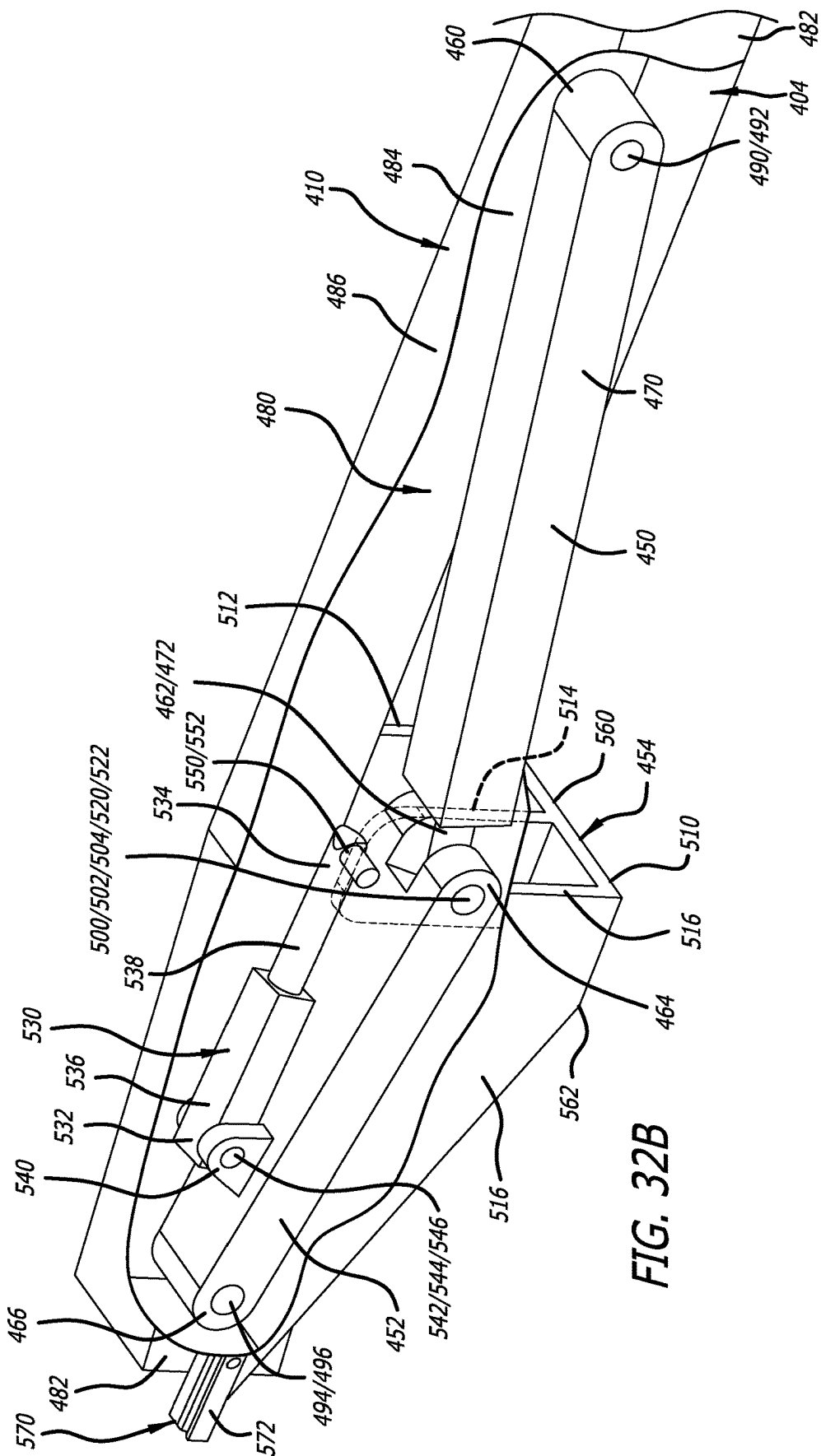
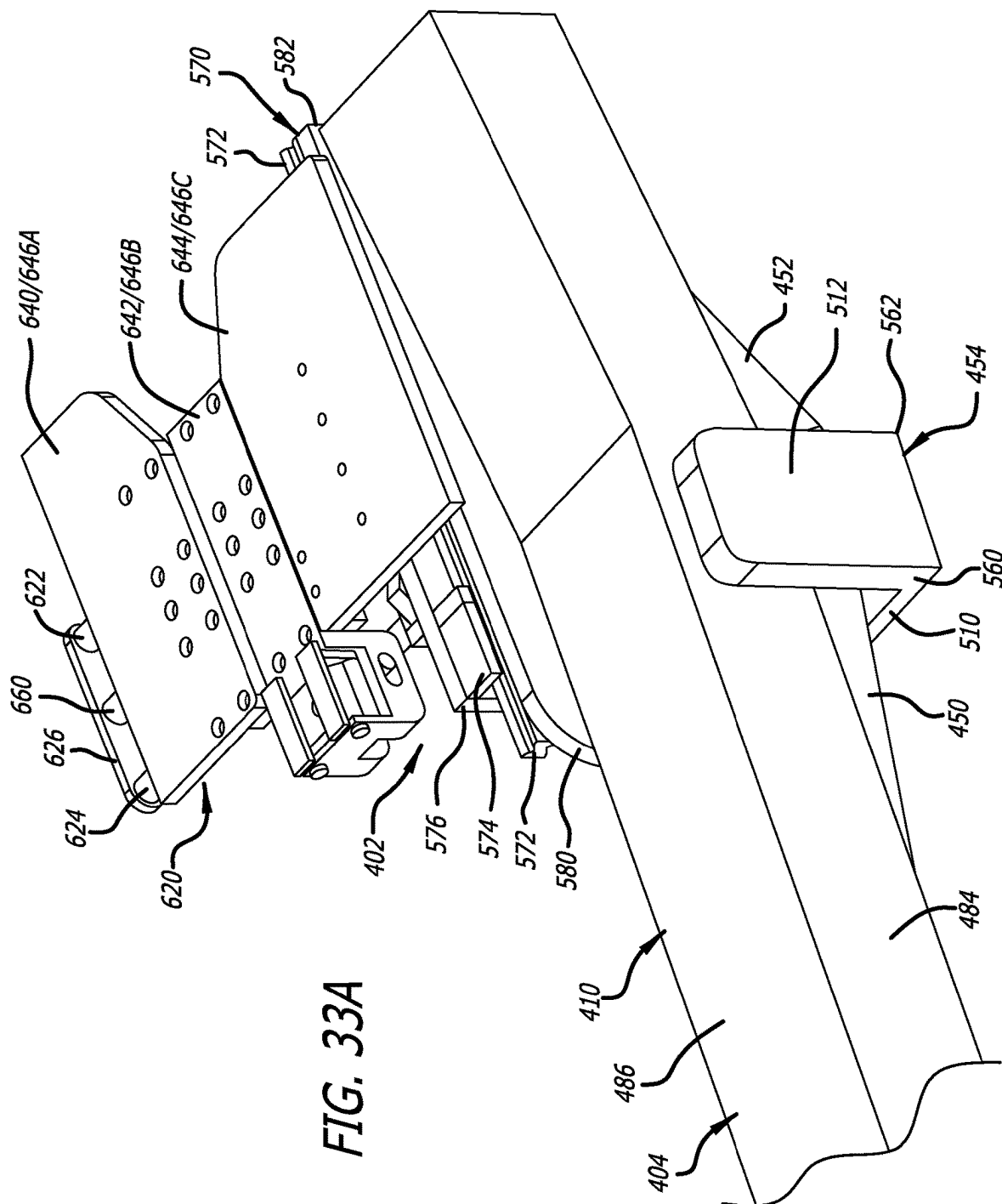
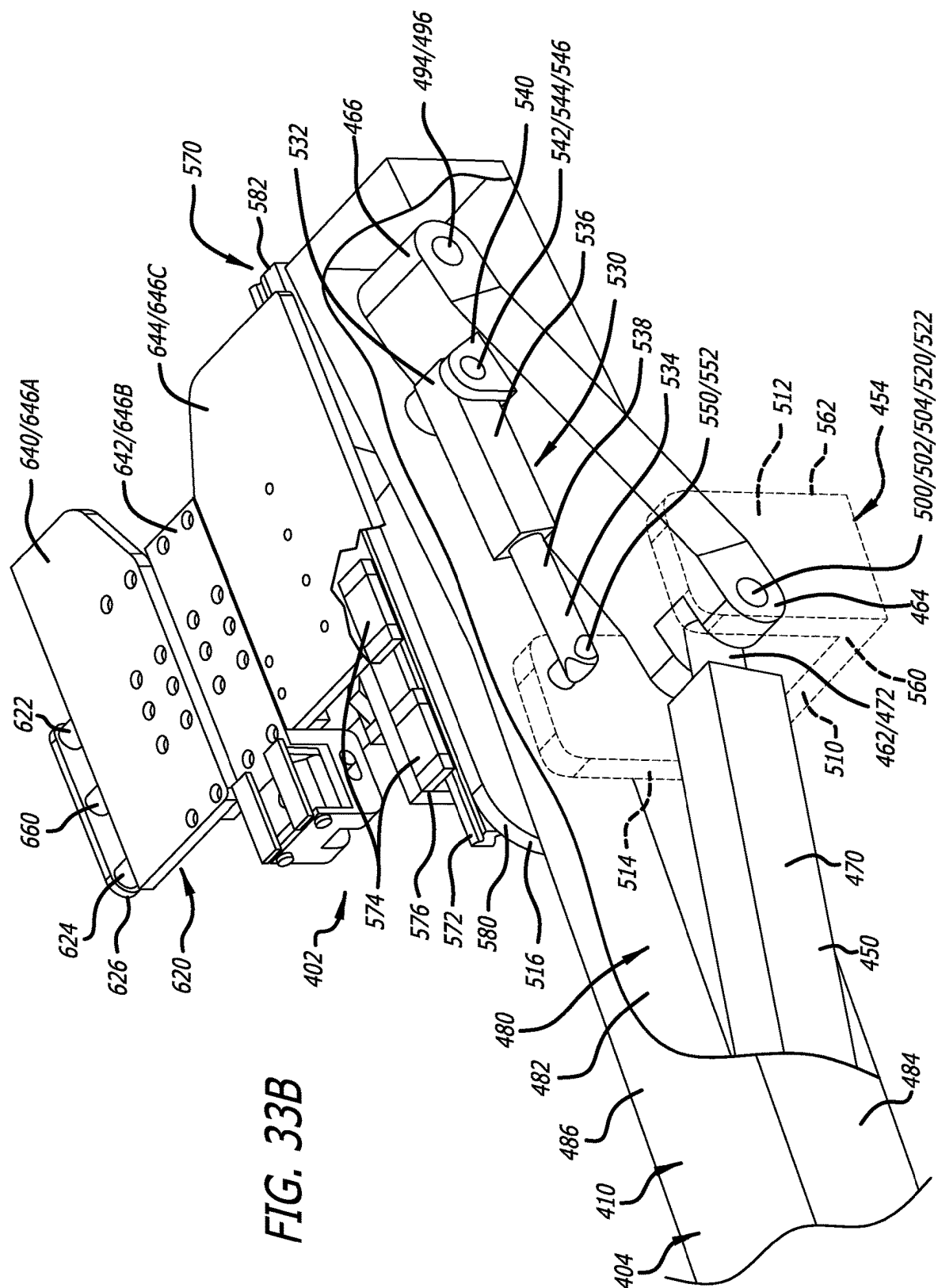


FIG. 32B





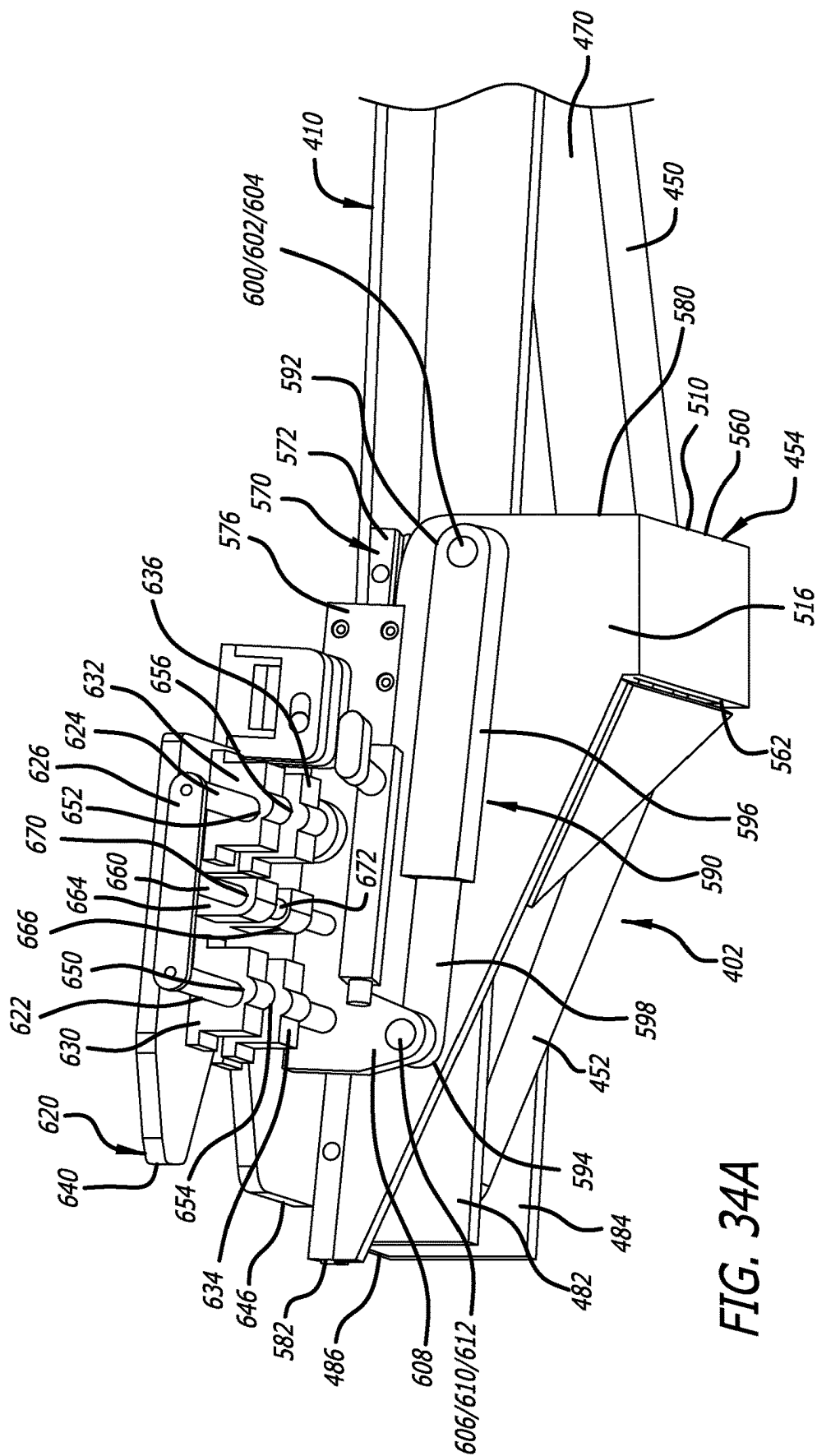


FIG. 34A

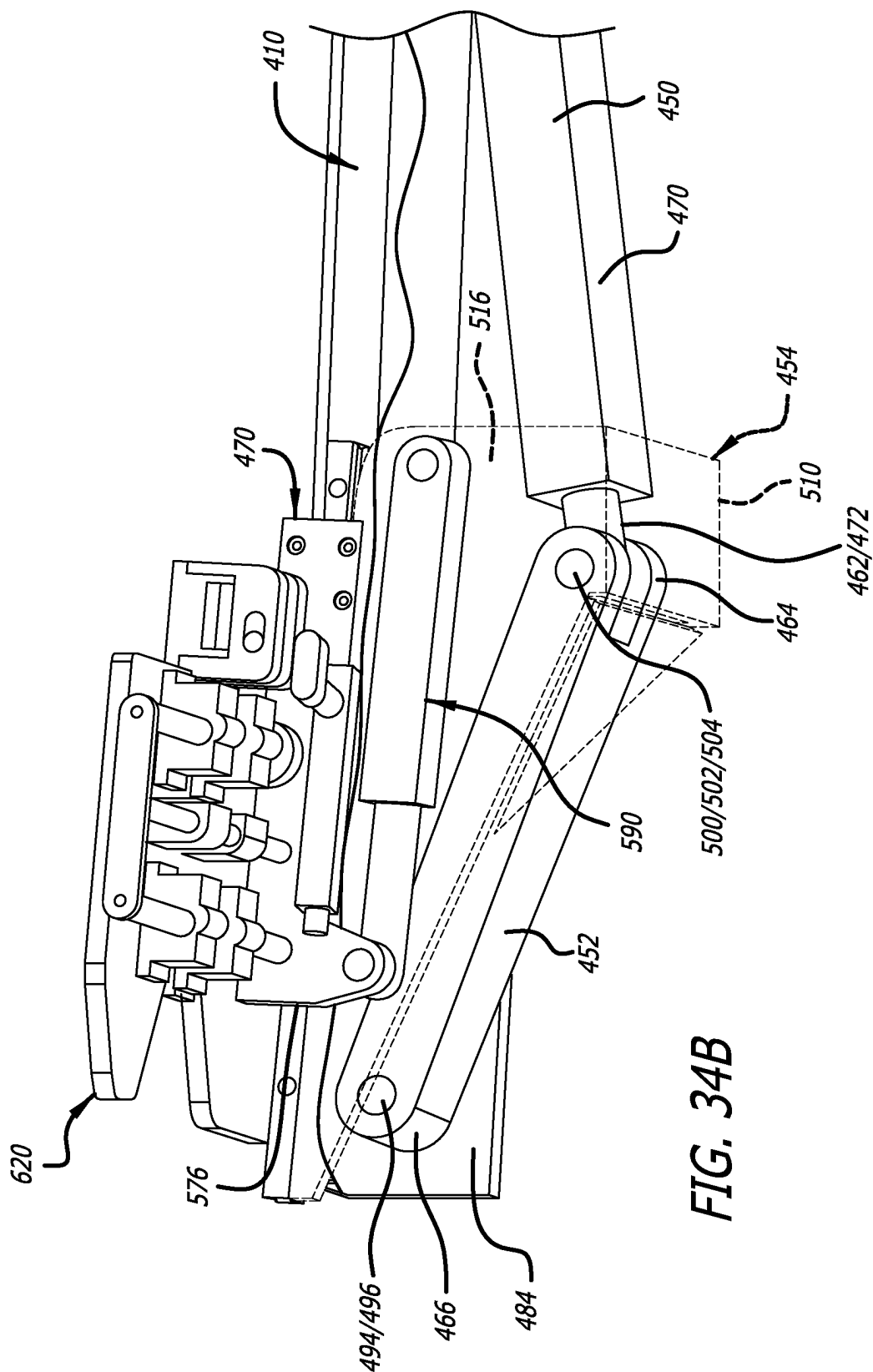


FIG. 34B

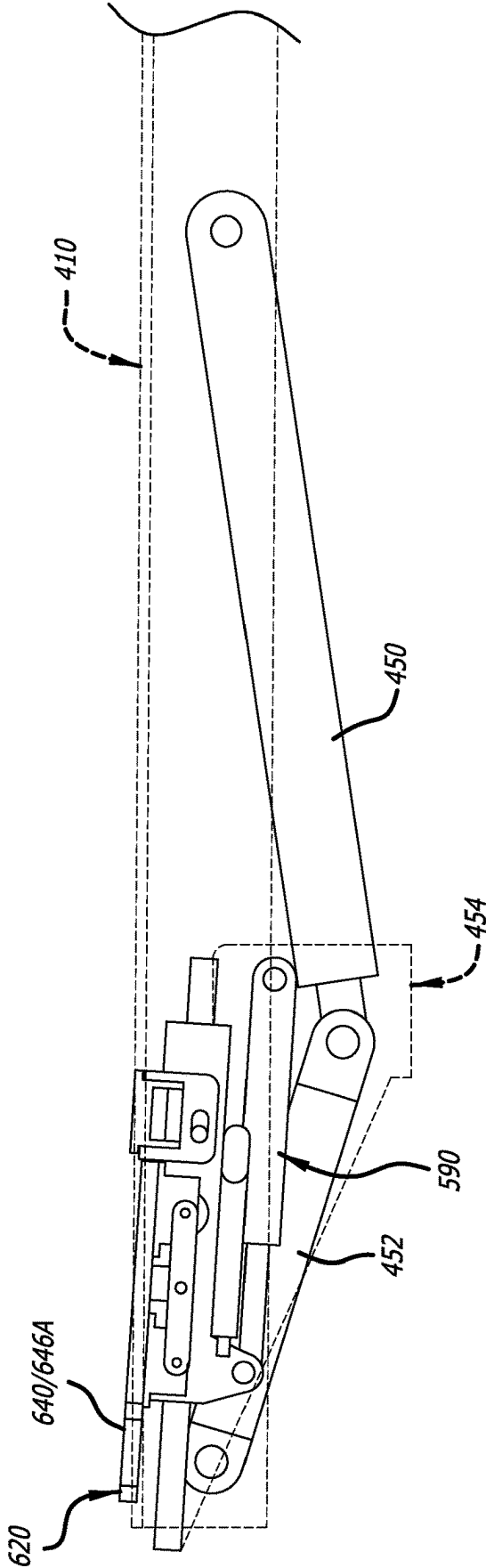


FIG. 35

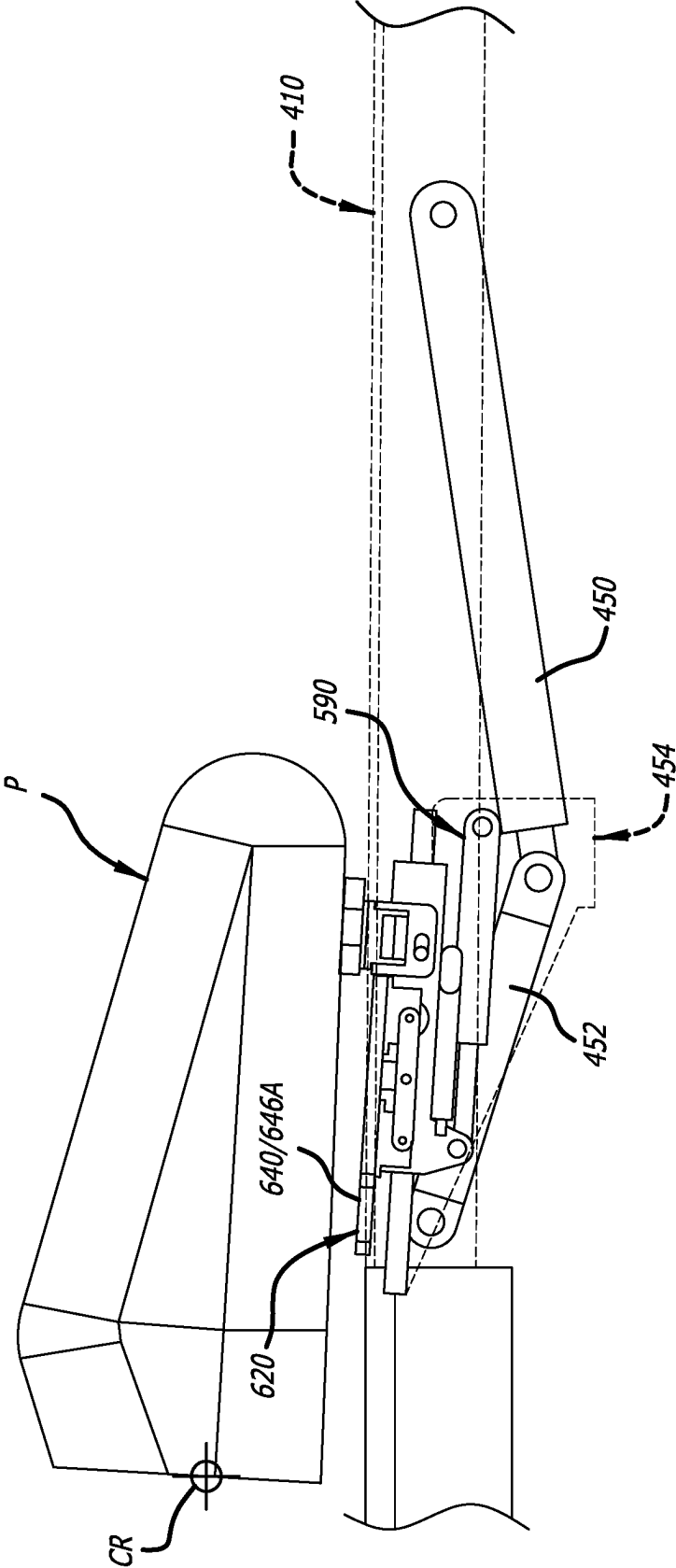
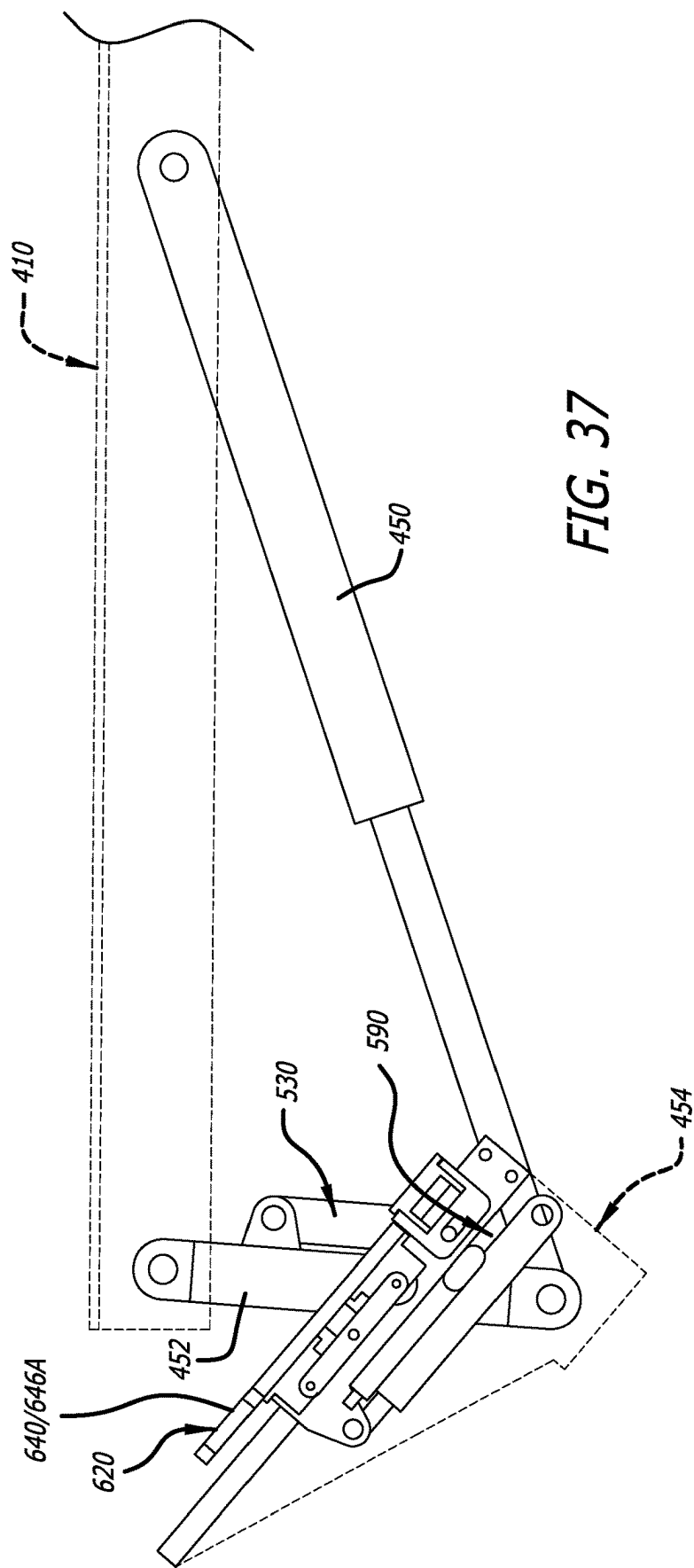
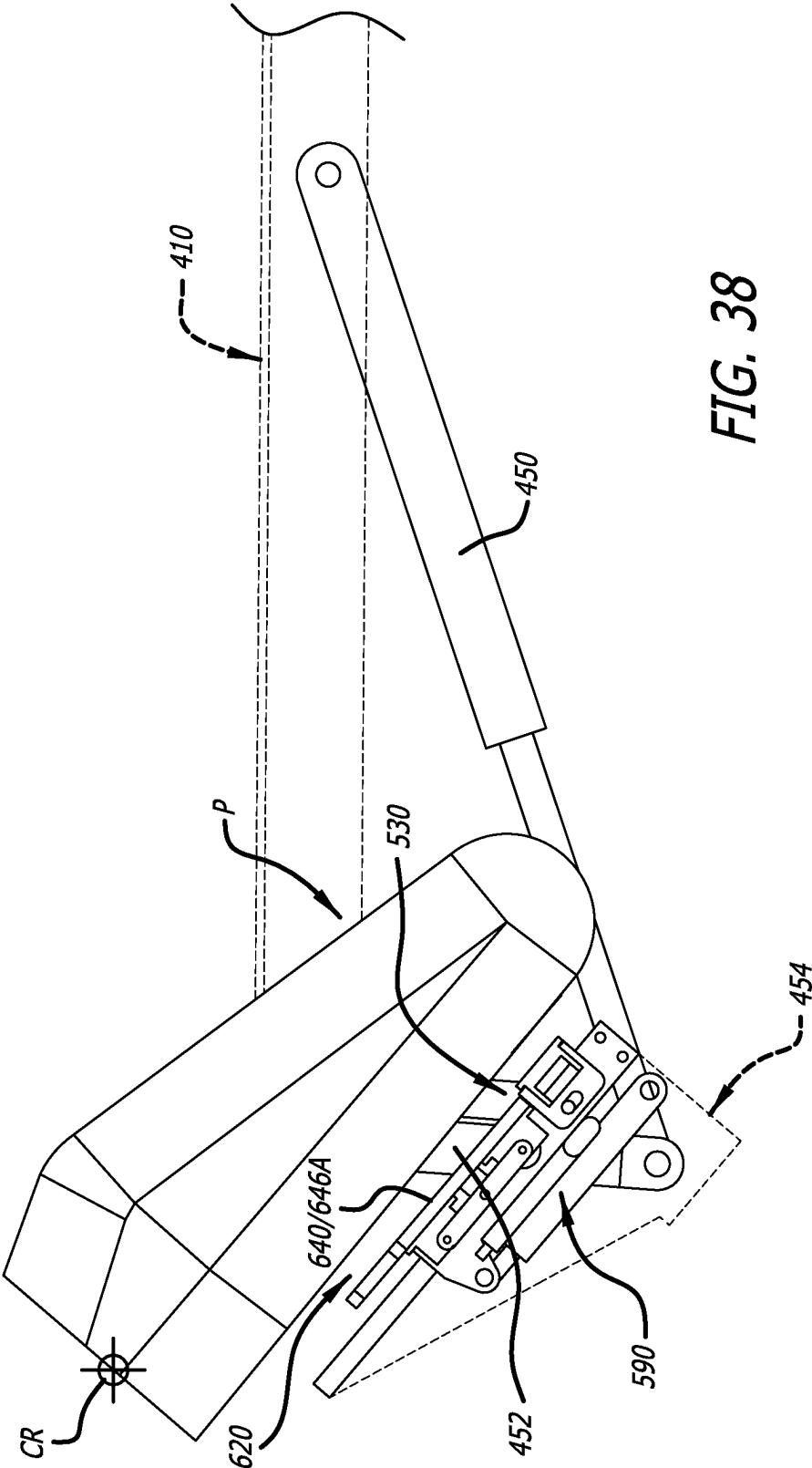


FIG. 36





Height 6-3"	Neutral 3 deg	30 deg flex 40 deg	length change
Cylinder 1	24.862	40.3749	15.5129
Cylinder 2	8.63941	6.65196	-1.98745
Cylinder 3	11.9626	9.54042	-2.42218

FIG. 39

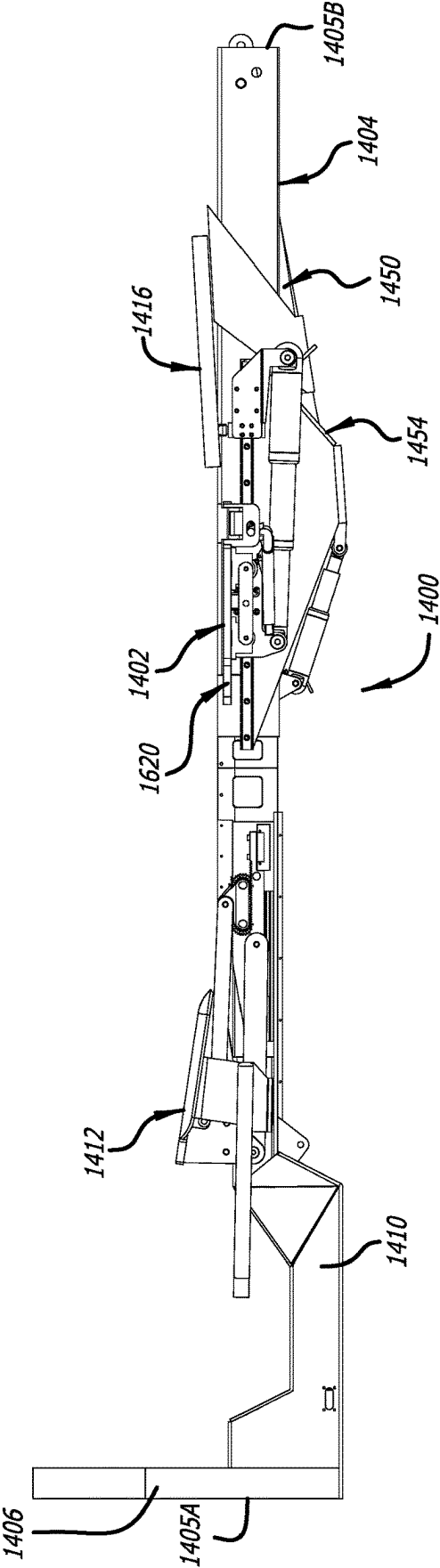


FIG. 40

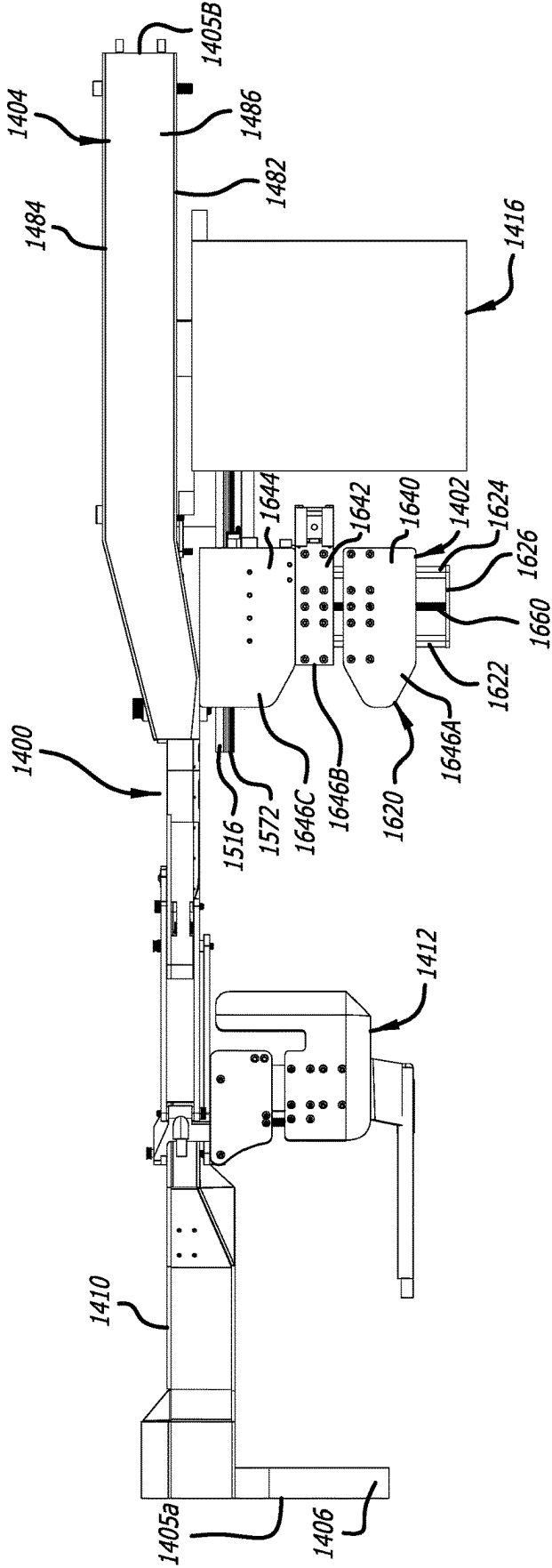


FIG. 41

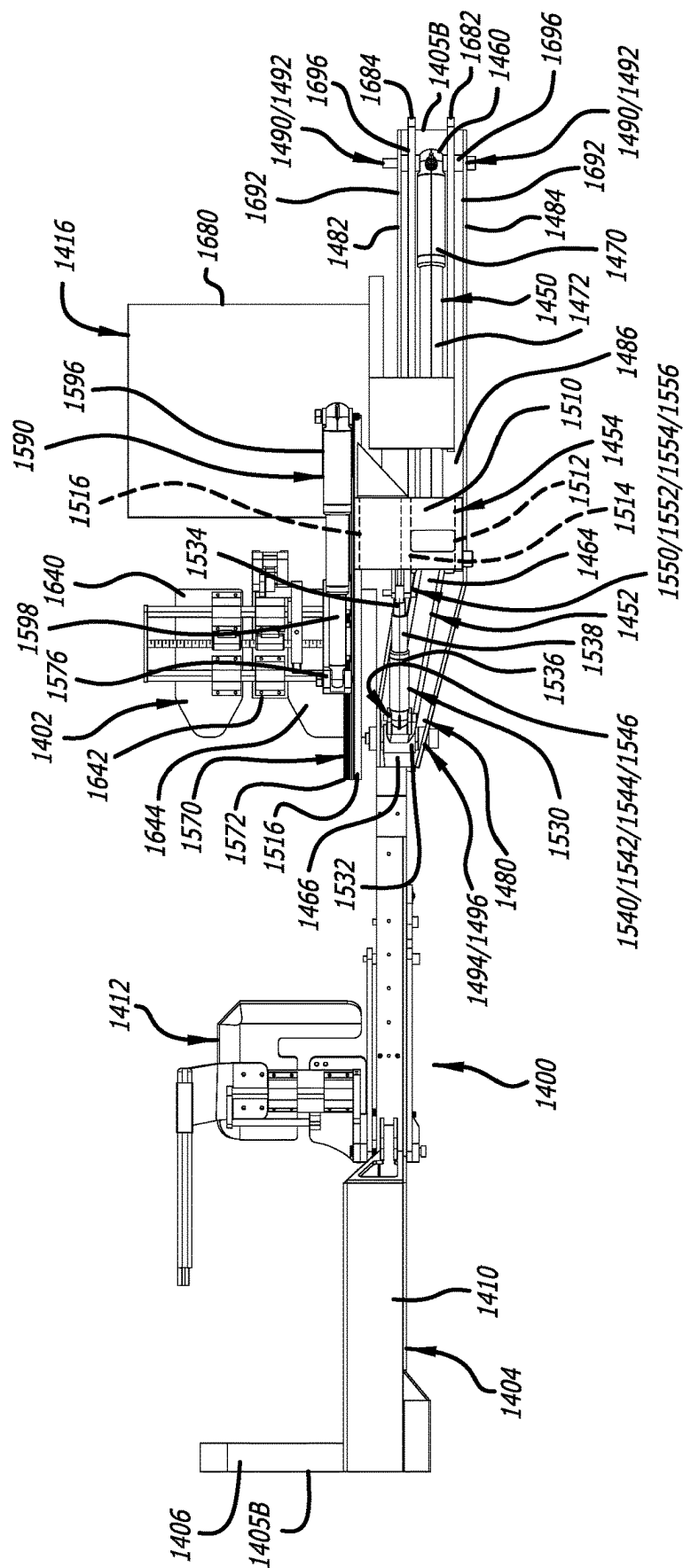
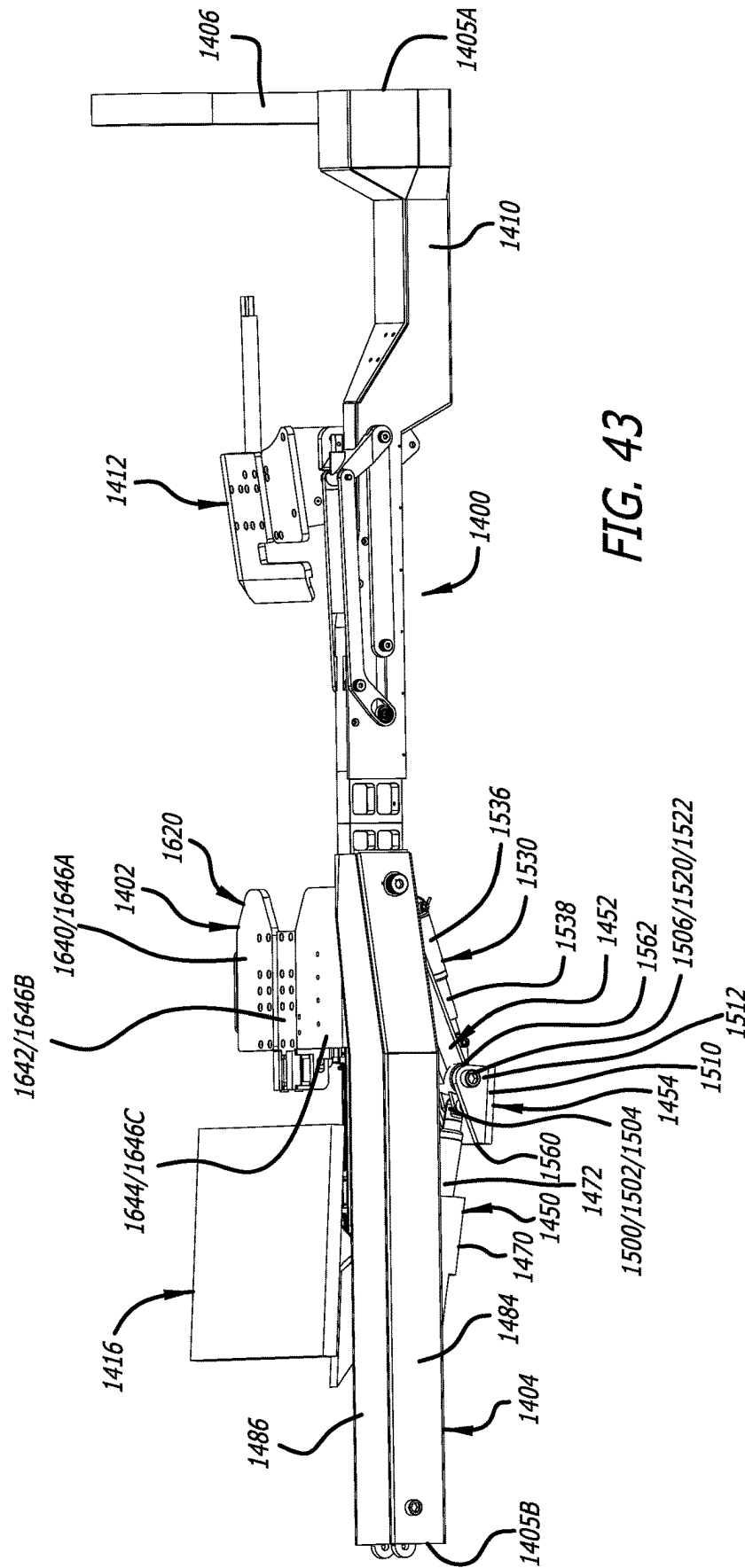


FIG. 42



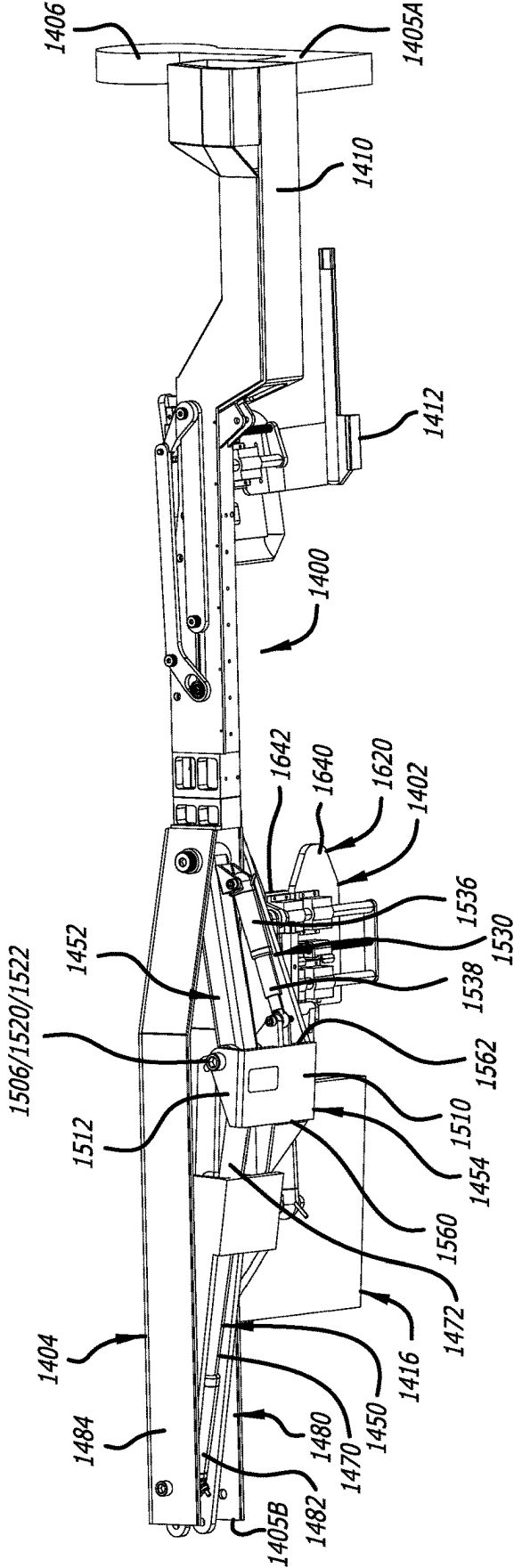
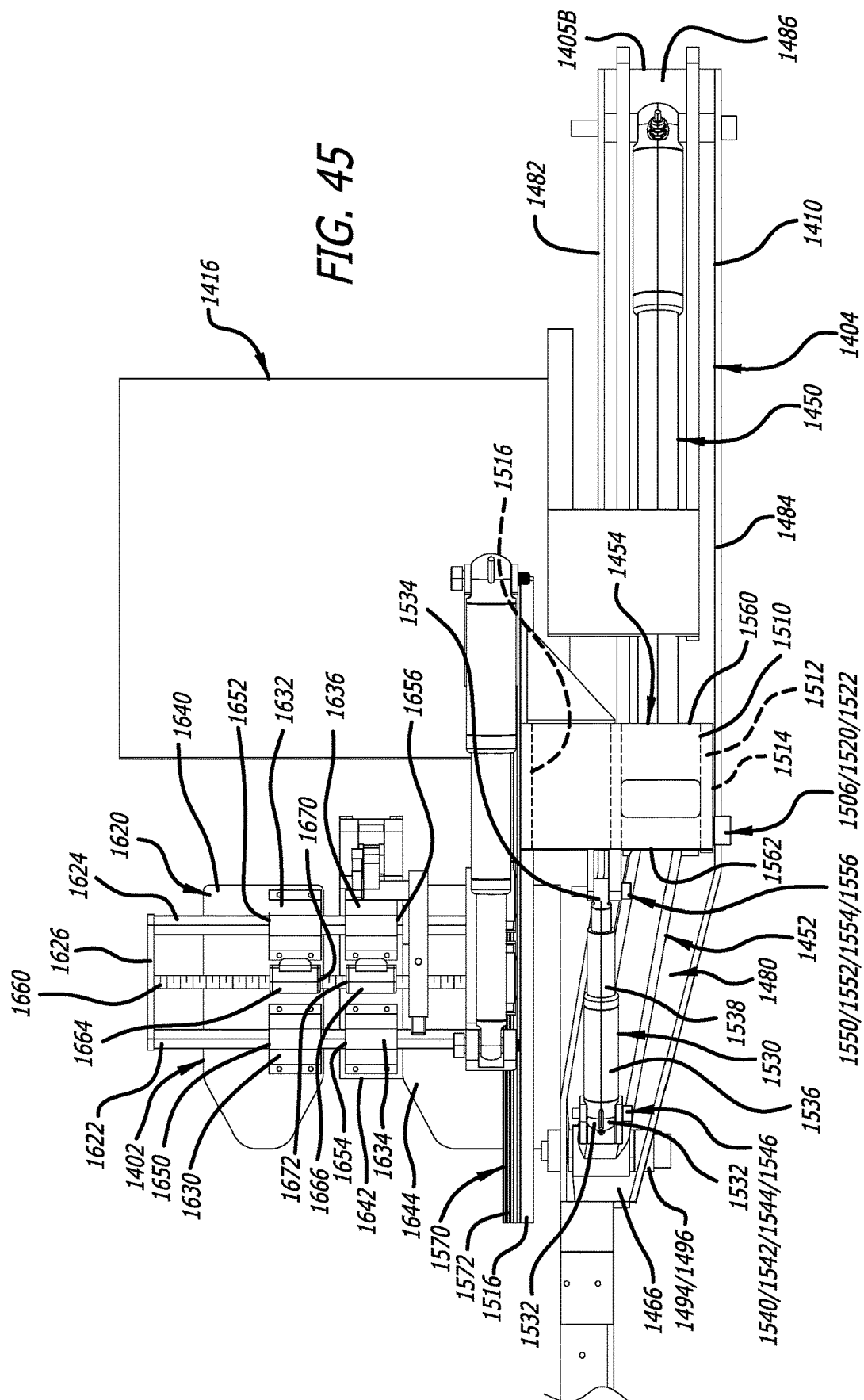


FIG. 44



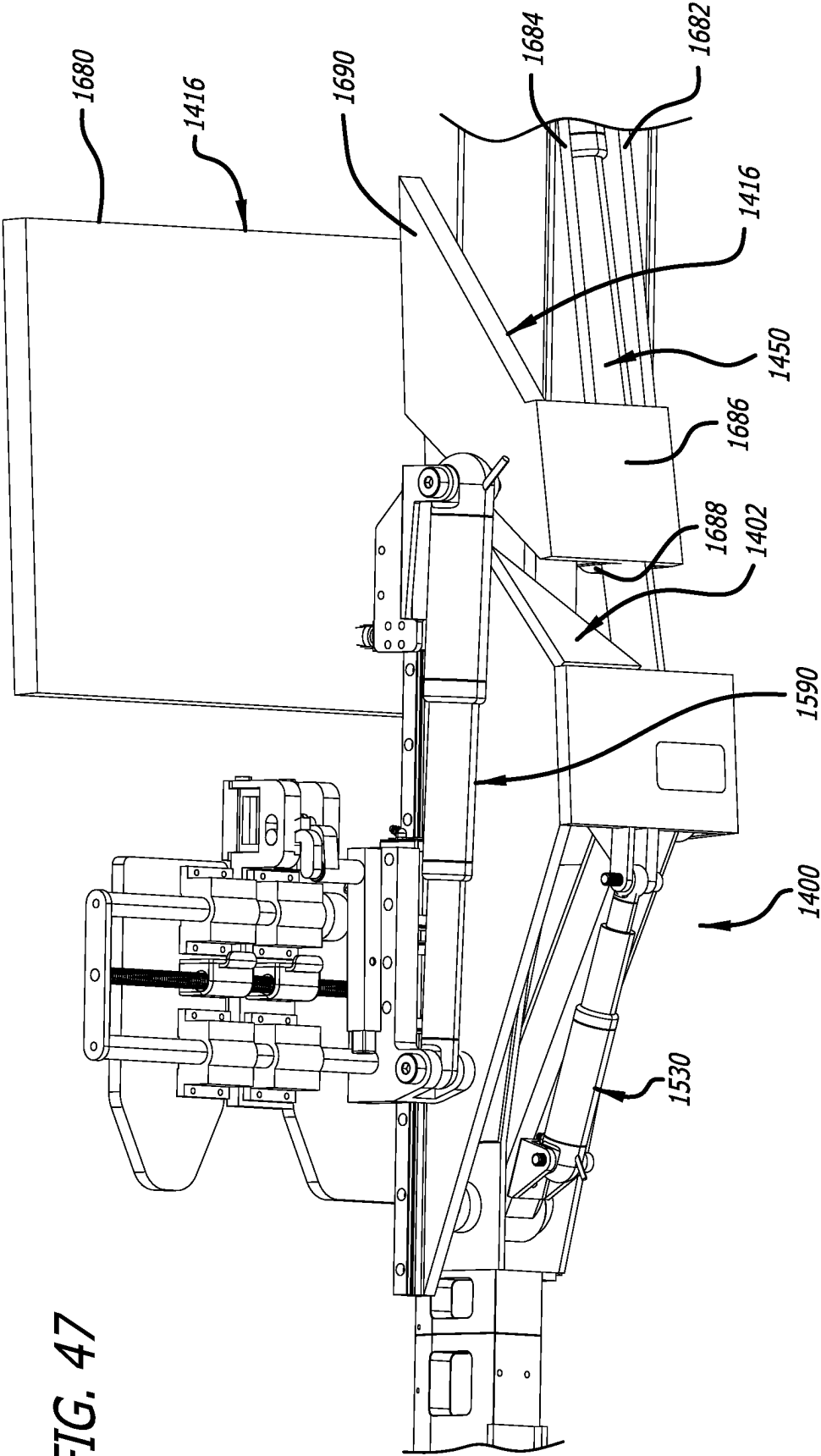


FIG. 47

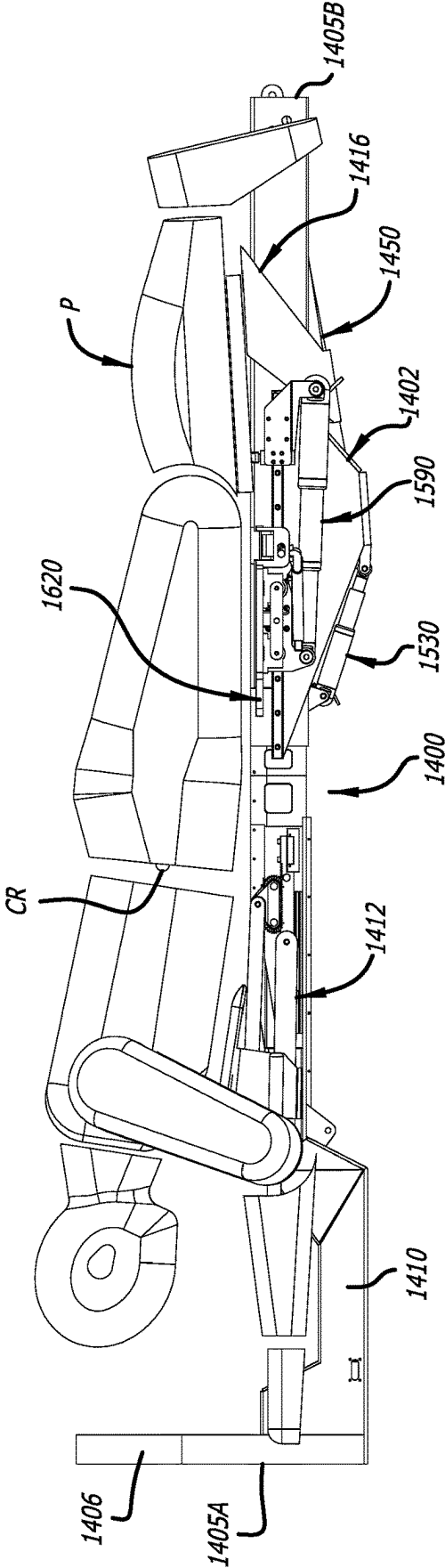
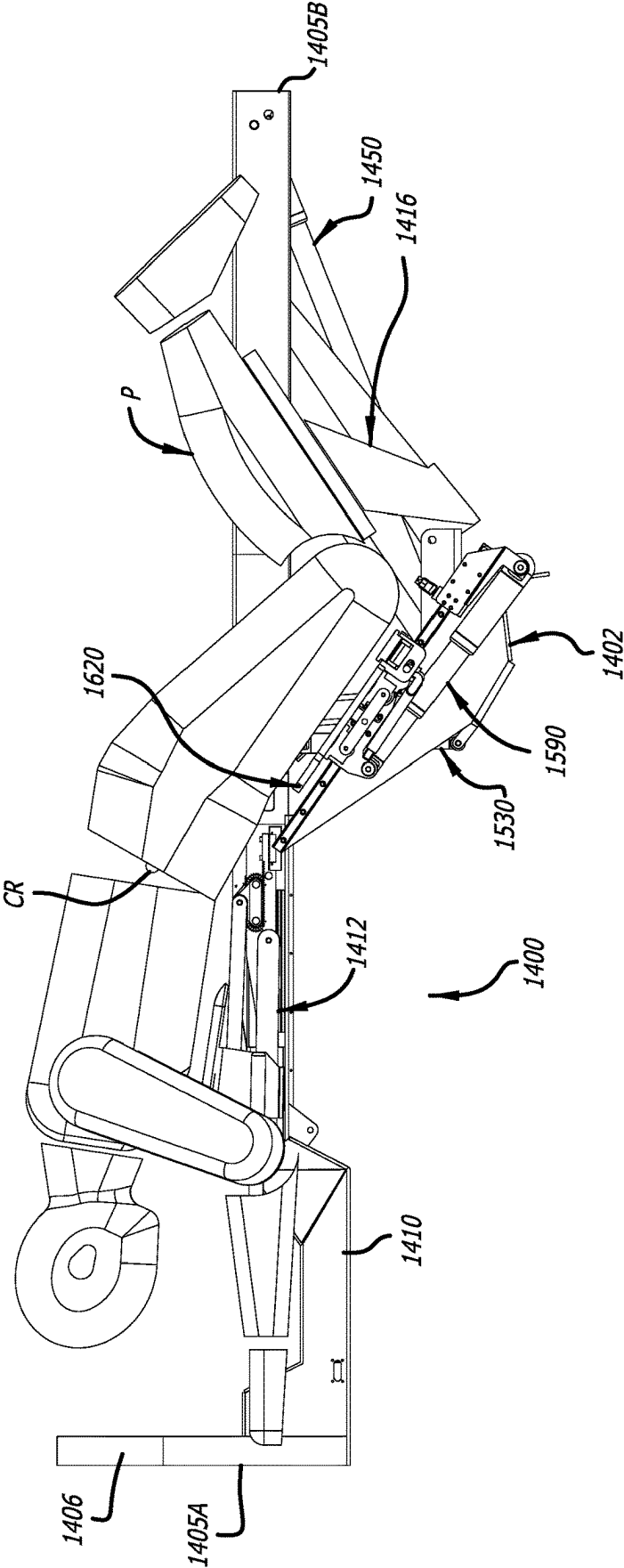


FIG. 48

FIG. 49



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RECONFIGURABLE UPPER LEG SUPPORT FOR A SURGICAL FRAME

The present application is a continuation of U.S. Ser. No. 16/927,219, filed Jul. 13, 2020; which claims benefit of U.S. Provisional Application No. 62/905,770, filed Sep. 25, 2019; all of which are incorporated by reference herein.

FIELD

The present technology generally relates to a reconfigurable upper leg support for use with a surgical frame incorporating a main beam capable of rotation.

BACKGROUND

Access to a patient is of paramount concern during surgery. Surgical frames have been used to position and reposition patients during surgery. For example, surgical frames have been configured to manipulate the rotational position of the patient before, during, and even after surgery. Such surgical frames include support structures to facilitate the rotational movement of the patient. Typical support structures can include main beams supported at either end thereof for rotational movement about axes of rotation extending along the lengths of the surgical frames. The main beams can be positioned and repositioned to afford various positions of the patients positioned thereon. To illustrate, the main beams can be rotated for positioning a patient in prone positions, lateral positions, and positions 45° between the prone and lateral positions. In addition to the rotational positioning afforded by the main beams, the patients can be further manipulated by support structures attached relative to the main beam. To illustrate, an upper leg support can be provided to support portions of upper legs, hips, and the lower back of the patient. Such an upper leg support can be moveable with respect to the main beam to facilitate positioning and repositioning of the upper legs, the hips, and the lower back of the patient to facilitate access to the patient during surgery. However, patients have different sizes and it is desirous to inhibit torsion of the patient's spine during use of surgical frame. Therefore, there is a need for a reconfigurable upper leg support that via reconfiguration thereof can accommodate patients of different sizes, can provide flexure of the patient's lumbar spine to facilitate surgical access thereto, and can prevent unwanted torsion of a patient's spine during such reconfiguration.

SUMMARY

The techniques of this disclosure generally relate to a reconfigurable upper leg support attached relative to a rotatable main beam that is articulable to adjust the position of the upper legs of a patient to correspondingly affect the flexure of the lumbar spine of a patient, while simultaneously inhibiting unwanted torsion of the patient's spine caused by reconfiguration of the upper leg support.

In one aspect, the present disclosure provides a method of adjusting a position of a patient supported on a surgical frame, the method including positioning the patient on the surgical frame by supporting upper legs of the patient on a support plate; extending a first arm portion relative to a second arm portion to adjust a position of a platform portion relative to a portion of the surgical frame, the first arm portion including a first end attached relative to the portion of the surgical frame and a second end attached relative to the platform portion, and the second arm portion including

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a first end attached relative to the platform portion and a second end attached relative to the portion of the surgical frame; extending a first telescoping shaft to adjust a position of the platform portion relative to at least one of the first arm portion and the second arm portion, the first telescoping shaft including a first end attached relative to the second arm portion and a second end attached relative to the platform portion; extending a second telescoping shaft to adjust a position of the support plate relative to the platform portion, the second telescoping shaft including a first end attached relative to the platform portion and a second end attached to a support bracket moveably attached relative to the support platform, and the support plate being supported relative to the support bracket; adjusting a center of rotation of a lumbar portion of a spine of the patient by coordinating the extension of the first arm portion, the first telescoping shaft, and the second telescoping shaft.

In one aspect, the present disclosure provides a method of adjusting a position of a patient supported on a surgical frame, the method including supporting upper legs of the patient on a support plate; extending a first arm portion to adjust a position of a platform portion relative to a portion of the surgical frame, the first arm portion including a first end attached relative to the portion of the surgical frame and a second end attached relative to the platform portion; extending a telescoping shaft to adjust a position of the support plate relative to the platform portion, the telescoping shaft including a first end attached relative to the platform portion and a second end attached to a support bracket moveably attached relative to the support platform, and the support plate being supported relative to the support bracket; and adjusting a center of rotation of a lumbar portion of a spine of the patient by coordinating the extension of the first arm portion and the telescoping shaft.

In one aspect, the present disclosure provides a method of adjusting a position of a patient supported on a surgical frame, the method including positioning the patient on the surgical frame by supporting upper legs of the patient on a support plate; pivoting an arm portion relative to a portion of the surgical frame to adjust a position of a platform portion relative to a portion of the surgical frame, the arm portion including a first end pivotally attached relative to the portion of the surgical frame and a second end attached relative to the platform portion; extending a first telescoping shaft to adjust a position of the platform portion relative to the arm portion, the first telescoping shaft including a first end attached relative to the arm portion and a second end attached relative to the platform portion; and extending a second telescoping shaft to adjust a position of the support plate relative to the platform portion, the second telescoping shaft including a first end attached relative to the platform portion and a second end attached to a support bracket moveably attached relative to the support platform, and the support plate being supported relative to the support bracket.

The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques described in this disclosure will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a top perspective view that illustrates a prior art surgical frame with a patient positioned thereon in a prone position;

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FIG. 2 is a side elevational view that illustrates the surgical frame of FIG. 1 with the patient positioned thereon in a prone position;

FIG. 3 is another side elevational view that illustrates the surgical frame of FIG. 1 with the patient positioned thereon in a prone position;

FIG. 4 is a top plan view that illustrates the surgical frame of FIG. 1 with the patient positioned thereon in a prone position;

FIG. 5 is a top perspective view that illustrates the surgical frame of FIG. 1 with the patient positioned thereon in a lateral position;

FIG. 6 is a top perspective view that illustrates portions of the surgical frame of FIG. 1 showing an area of access to the head of the patient positioned thereon in a prone position;

FIG. 7 is a side elevational view that illustrates the surgical frame of FIG. 1 showing a torso-lift support supporting the patient in a lifted position;

FIG. 8 is another side elevational view that illustrates the surgical frame of FIG. 1 showing the torso-lift support supporting the patient in the lifted position;

FIG. 9 is an enlarged top perspective view that illustrates portions of the surgical frame of FIG. 1 showing the torso-lift support supporting the patient in an unlifted position;

FIG. 10 is an enlarged top perspective view that illustrates portions of the surgical frame of FIG. 1 showing the torso-lift support supporting the patient in the lifted position;

FIG. 11 is an enlarged top perspective view that illustrates componentry of the torso-lift support in the unlifted position;

FIG. 12 is an enlarged top perspective view that illustrates the componentry of the torso-lift support in the lifted position;

FIG. 13A is a perspective view of an embodiment that illustrates a structural offset main beam for use with another embodiment of a torso-lift support showing the torso-lift support in a retracted position;

FIG. 13B is a perspective view similar to FIG. 13A showing the torso-lift support at half travel;

FIG. 13C is a perspective view similar to FIGS. 13A and 13B showing the torso-lift support at full travel;

FIG. 14 is a perspective view that illustrates a chest support lift mechanism of the torso-lift support of FIGS. 13A-13C with actuators thereof retracted;

FIG. 15 is another perspective view that illustrates a chest support lift mechanism of the torso-lift support of FIGS. 13A-13C with the actuators thereof extended;

FIG. 16 is a top perspective view that illustrates the surgical frame of FIG. 1;

FIG. 17 is an enlarged top perspective view that illustrates portions of the surgical frame of FIG. 1 showing a sagittal adjustment assembly including a pelvic-tilt mechanism and leg adjustment mechanism;

FIG. 18 is an enlarged side elevational view that illustrates portions of the surgical frame of FIG. 1 showing the pelvic-tilt mechanism;

FIG. 19 is an enlarged perspective view that illustrates componentry of the pelvic-tilt mechanism;

FIG. 20 is an enlarged perspective view that illustrates a captured rack and a worm gear assembly of the componentry of the pelvic-tilt mechanism;

FIG. 21 is an enlarged perspective view that illustrates the worm gear assembly of FIG. 20;

FIG. 22 is a side elevational view that illustrates portions of the surgical frame of FIG. 1 showing the patient posi-

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tioned thereon and the pelvic-tilt mechanism of the sagittal adjustment assembly in the flexed position;

FIG. 23 is another side elevational view that illustrates portions of the surgical frame of FIG. 1 showing the patient positioned thereon and the pelvic-tilt mechanism of the sagittal adjustment assembly in the fully extended position;

FIG. 24 is an enlarged top perspective view that illustrates portions of the surgical frame of FIG. 1 showing a coronal adjustment assembly;

FIG. 25 is a top perspective view that illustrates portions of the surgical frame of FIG. 1 showing operation of the coronal adjustment assembly;

FIG. 26 is a top perspective view that illustrates a portion of the surgical frame of FIG. 1 showing operation of the coronal adjustment assembly;

FIG. 27 is a top perspective view that illustrates a prior art surgical frame in accordance with an embodiment of the present invention with the patient positioned thereon in a prone position showing a translating beam thereof in a first position;

FIG. 28 is another top perspective view that illustrates the surgical frame of FIG. 27 with the patient in a prone position showing the translating beam thereof in a second position;

FIG. 29 is yet another top perspective view that illustrates the surgical frame of FIG. 27 with the patient in a lateral position showing the translating beam thereof in a third position;

FIG. 30 is a top plan view that illustrates the surgical frame of FIG. 27 with the patient in a lateral position showing the translating beam thereof in the third position;

FIG. 31 is a top, side perspective view that illustrates a portion of a main beam of a surgical frame, and a portion of a reconfigurable upper leg support of a first embodiment of the present disclosure;

FIG. 32A is a top, side perspective view similar to FIG. 31 that illustrates a portion of the reconfigurable upper leg support of FIG. 31 relative to the main beam;

FIG. 32B is a fragmentary, top, side perspective view similar to FIG. 32A that illustrates a portion of the reconfigurable upper leg support of FIG. 31 relative to the main beam;

FIG. 33A is a top, side perspective view that illustrates a portion of the main beam, and a portion of the reconfigurable upper leg support of FIG. 31;

FIG. 33B is a fragmentary, top, side perspective view similar to FIG. 33A that illustrates a portion of the reconfigurable upper leg support of FIG. 31 relative to the main beam;

FIG. 34A is a bottom, side perspective view that illustrates a portion of the main beam, and a portion of the reconfigurable upper leg support of FIG. 31;

FIG. 34B is a fragmentary bottom, side perspective view similar to FIG. 34A that illustrates a portion of the reconfigurable upper leg support of FIG. 31 relative to the main beam;

FIG. 35 is a side elevational view that illustrates the reconfigurable upper leg support of FIG. 31 with a first arm portion, a first telescoping shaft portion, and a second telescoping shaft portion adjusted to position the upper leg support in a first position;

FIG. 36 is a side elevational view that illustrates the reconfigurable upper leg support of FIG. 31 showing a position of upper legs, hips, and lower back of a patient supported thereby with the upper leg support in the first position;

FIG. 37 is a side elevational view that illustrates the reconfigurable upper leg support of FIG. 31 with the first

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arm portion, the first telescoping shaft portion, and the second telescoping shaft portion adjusted to position the upper leg support in a second position;

FIG. 38 is a side elevational view that illustrates the reconfigurable upper leg support of FIG. 31 showing a position of the upper legs, hips, and lower back of the patient supported thereby with the upper leg support in the second position;

FIG. 39 is a table illustrating extension amounts for the first arm portion, the first telescoping shaft portion, and the second telescoping shaft portion;

FIG. 40 is a side elevational view that illustrates a portion of a main beam of a surgical frame, and a reconfigurable upper leg support of second embodiment of the present disclosure;

FIG. 41 is a top plan view that illustrates the reconfigurable upper leg support of FIG. 40 relative to the main beam;

FIG. 42 is a bottom plan view that illustrates the reconfigurable upper leg support of FIG. 40 relative to the main beam;

FIG. 43 is a side, top perspective view that illustrates the reconfigurable upper leg support of FIG. 40 relative to the main beam;

FIG. 44 is a side, bottom perspective view that illustrates the reconfigurable upper leg support of FIG. 40 relative to the main beam;

FIG. 45 is an enlarged, bottom plan view that illustrates the reconfigurable upper leg support of FIG. 40 relative to the main beam;

FIG. 46 is an enlarged, bottom perspective view that illustrates the reconfigurable upper leg support of FIG. 40 relative to the main beam;

FIG. 47 is another enlarged, bottom perspective view that illustrates the reconfigurable upper leg support of FIG. 40 relative to the main beam;

FIG. 48 is a side elevational view that illustrates the reconfigurable upper leg support of FIG. 40 and a patient partially supported thereby with a first arm portion, a first telescoping shaft portion, and a second telescoping shaft portion adjusted to position the upper leg support and the patient in a first position; and

FIG. 49 is a side elevational view that illustrates the reconfigurable upper leg support of FIG. 40 and the patient partially supported thereby with the first arm portion, the first telescoping shaft portion, and the second telescoping shaft portion adjusted to position the upper leg support and the patient in a second position.

The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques described in this disclosure will be apparent from the description and drawings, and from the claims.

DETAILED DESCRIPTION

FIGS. 1-26 depict a prior art embodiment and components of a surgical support frame generally indicated by the numeral 10. FIGS. 1-26 were previously described in U.S. Ser. No. 15/239,256, which is hereby incorporated by reference herein in its entirety. Furthermore, FIGS. 27-30 were previously described in U.S. Ser. No. 15/639,080, which is hereby incorporated by reference herein in its entirety.

As discussed below, the surgical frame 10 serves as an exoskeleton to support the body of the patient P as the patient's body is manipulated thereby, and, in doing so, serves to support the patient P such that the patient's spine does not experience unnecessary torsion.

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The surgical frame 10 is configured to provide a relatively minimal amount of structure adjacent the patient's spine to facilitate access thereto and to improve the quality of imaging available before and during surgery. Thus, the surgeon's workspace and imaging access are thereby increased. Furthermore, radiolucent or low magnetic susceptibility materials can be used in constructing the structural components adjacent the patient's spine in order to further enhance imaging quality.

The surgical frame 10 has a longitudinal axis and a length therealong. As depicted in FIGS. 1-5, for example, the surgical frame 10 includes an offset structural main beam 12 and a support structure 14. The offset main beam 12 is spaced from the ground by the support structure 14. As discussed below, the offset main beam 12 is used in supporting the patient P on the surgical frame 10 and various support components of the surgical frame 10 that directly contact the patient P (such as a head support 20, arm supports 22A and 22B, torso-lift supports 24 and 160, a sagittal adjustment assembly 28 including a pelvic-tilt mechanism 30 and a leg adjustment mechanism 32, and a coronal adjustment assembly 34). As discussed below, an operator such as a surgeon can control actuation of the various support components to manipulate the position of the patient's body. Soft straps (not shown) are used with these various support components to secure the patient P to the frame and to enable either manipulation or fixation of the patient P. Reusable soft pads can be used on the load-bearing areas of the various support components.

The offset main beam 12 is used to facilitate rotation of the patient P. The offset main beam 12 can be rotated a full 360° before and during surgery to facilitate various positions of the patient P to afford various surgical pathways to the patient's spine depending on the surgery to be performed. For example, the offset main beam 12 can be positioned to place the patient P in a prone position (e.g., FIGS. 1-4), a lateral position (e.g., FIG. 5), and in a position 45° between the prone and lateral positions. Furthermore, the offset main beam 12 can be rotated to afford anterior, posterior, lateral, anterolateral, and posterolateral pathways to the spine. As such, the patient's body can be flipped numerous times before and during surgery without compromising sterility or safety. The various support components of the surgical frame 10 are strategically placed to further manipulate the patient's body into position before and during surgery. Such intraoperative manipulation and positioning of the patient P affords a surgeon significant access to the patient's body. To illustrate, when the offset main beam 12 is rotated to position the patient P in a lateral position, as depicted in FIG. 5, the head support 20, the arm supports 22A and 22B, the torso-lift support 24, the sagittal adjustment assembly 28, and/or the coronal adjustment assembly 34 can be articulated such that the surgical frame 10 is OLIF-capable or DLIF-capable.

As depicted in FIG. 1, for example, the support structure 14 includes a first support portion 40 and a second support portion 42 interconnected by a cross member 44. Each of the first and second support portions 40 and 42 include a horizontal portion 46 and a vertical support post 48. The horizontal portions 46 are connected to the cross member 44, and casters 50 can be attached to the horizontal portions 46 to facilitate movement of the surgical frame 10.

The vertical support posts 48 can be adjustable to facilitate expansion and contraction of the heights thereof. Expansion and contraction of the vertical support posts 48 facilitates raising and lowering, respectively, of the offset main beam 12. As such, the vertical support posts 48 can be adjusted to have equal or different heights. For example, the

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vertical support posts **48** can be adjusted such that the vertical support post **48** of the second support portion **42** is raised 12 inches higher than the vertical support post **48** of the first support portion **40** to place the patient P in a reverse Trendelenburg position.

Furthermore, cross member **44** can be adjustable to facilitate expansion and contraction of the length thereof. Expansion and contraction of the cross member **44** facilitates lengthening and shortening, respectively, of the distance between the first and second support portions **40** and **42**.

The vertical support post **48** of the first and second support portions **40** and **42** have heights at least affording rotation of the offset main beam **12** and the patient P positioned thereon. Each of the vertical support posts **48** include a clevis **60**, a support block **62** positioned in the clevis **60**, and a pin **64** pinning the clevis **60** to the support block **62**. The support blocks **62** are capable of pivotal movement relative to the clevises **60** to accommodate different heights of the vertical support posts **48**. Furthermore, axles **66** extending outwardly from the offset main beam **12** are received in apertures **68** formed on the support blocks **62**. The axles **66** define an axis of rotation of the offset main beam **12**, and the interaction of the axles **66** with the support blocks **62** facilitate rotation of the offset main beam **12**.

Furthermore, a servomotor **70** can be interconnected with the axle **66** received in the support block **62** of the first support portion **40**. The servomotor **70** can be computer controlled and/or operated by the operator of the surgical frame **10** to facilitate controlled rotation of the offset main beam **12**. Thus, by controlling actuation of the servomotor **70**, the offset main beam **12** and the patient P supported thereon can be rotated to afford the various surgical pathways to the patient's spine.

As depicted in FIGS. 1-5, for example, the offset main beam **12** includes a forward portion **72** and a rear portion **74**. The forward portion **72** supports the head support **20**, the arm supports **22A** and **22B**, the torso-lift support **24**, and the coronal adjustment assembly **34**, and the rear portion **74** supports the sagittal adjustment assembly **28**. The forward and rear portions **72** and **74** are connected to one another by connection member **76** shared therebetween. The forward portion **72** includes a first portion **80**, a second portion **82**, a third portion **84**, and a fourth portion **86**. The first portion **80** extends transversely to the axis of rotation of the offset main beam **12**, and the second and fourth portions **82** and **86** are aligned with the axis of rotation of the offset main beam **12**. The rear portion **74** includes a first portion **90**, a second portion **92**, and a third portion **94**. The first and third portions **90** and **94** are aligned with the axis of rotation of the offset main beam **12**, and the second portion **92** extends transversely to the axis of rotation of the offset main beam **12**.

The axles **66** are attached to the first portion **80** of the forward portion **72** and to the third portion **94** of the rear portion **74**. The lengths of the first portion **80** of the forward portion **72** and the second portion **92** of the rear portion **74** serve in offsetting portions of the forward and rear portions **72** and **74** from the axis of rotation of the offset main beam **12**. This offset affords positioning of the cranial-caudal axis of patient P approximately aligned with the axis of rotation of the offset main beam **12**.

Programmable settings controlled by a computer controller (not shown) can be used to maintain an ideal patient height for a working position of the surgical frame **10** at a near-constant position through rotation cycles, for example, between the patient positions depicted in FIGS. 1 and 5. This allows for a variable axis of rotation between the first portion **40** and the second portion **42**.

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As depicted in FIG. 5, for example, the head support **20** is attached to a chest support plate **100** of the torso-lift support **24** to support the head of the patient P. If the torso-lift support **24** is not used, the head support **20** can be directly attached to the forward portion **72** of the offset main beam **12**. As depicted in FIGS. 4 and 6, for example, the head support **20** further includes a facial support cradle **102**, an axially adjustable head support beam **104**, and a temple support portion **106**. Soft straps (not shown) can be used to secure the patient P to the head support **20**. The facial support cradle **102** includes padding across the forehead and cheeks, and provides open access to the mouth of the patient P. The head support **20** also allows for imaging access to the cervical spine. Adjustment of the head support **20** is possible via adjusting the angle and the length of the head support beam **104** and the temple support portion **106**.

As depicted in FIG. 5, for example, the arm supports **22A** and **22B** contact the forearms and support the remainder of the arms of the patient P, with the first arm support **22A** and the second arm support **22B** attached to the chest support plate **100** of the torso-lift support **24**. If the torso-lift support **24** is not used, the arm supports **22A** and **22B** can both be directly attached to the offset main beam **12**. The arm supports **22A** and **22B** are positioned such that the arms of the patient P are spaced away from the remainder of the patient's body to provide access (FIG. 6) to at least portions of the face and neck of the patient P, thereby providing greater access to the patient.

As depicted in FIGS. 7-12, for example, the surgical frame **10** includes a torso-lift capability for lifting and lowering the torso of the patient P between an uplifted position and a lifted position, which is described in detail below with respect to the torso-lift support **24**. As depicted in FIGS. 7 and 8, for example, the torso-lift capability has an approximate center of rotation ("COR") **108** that is located at a position anterior to the patient's spine about the L2 of the lumbar spine, and is capable of elevating the upper body of the patient at least an additional six inches when measured at the chest support plate **100**.

As depicted in FIGS. 9-12, for example, the torso-lift support **24** includes a "crawling" four-bar mechanism **110** attached to the chest support plate **100**. Soft straps (not shown) can be used to secure the patient P to the chest support plate **100**. The head support **20** and the arm supports **22A** and **22B** are attached to the chest support plate **100**, thereby moving with the chest support plate **100** as the chest support plate **100** is articulated using the torso-lift support **24**. The fixed COR **108** is defined at the position depicted in FIGS. 7 and 8. Appropriate placement of the COR **108** is important so that spinal cord integrity is not compromised (i.e., overly compressed or stretched) during the lift maneuver performed by the torso-lift support **24**.

As depicted in FIGS. 10-12, for example, the four-bar mechanism **110** includes first links **112** pivotally connected between offset main beam **12** and the chest support plate **100**, and second links **114** pivotally connected between the offset main beam **12** and the chest support plate **100**. As depicted in FIGS. 11 and 12, for example, in order to maintain the COR **108** at the desired fixed position, the first and second links **112** and **114** of the four-bar mechanism **110** crawl toward the first support portion **40** of the support structure **14**, when the patient's upper body is being lifted. The first and second links **112** and **114** are arranged such that neither the surgeon's workspace nor imaging access are compromised while the patient's torso is being lifted.

As depicted in FIGS. 11 and 12, for example, each of the first links **112** define an L-shape, and includes a first pin **116**

at a first end 118 thereof. The first pin 116 extends through first elongated slots 120 defined in the offset main beam 12, and the first pin 116 connects the first links 112 to a dual rack and pinion mechanism 122 via a drive nut 124 provided within the offset main beam 12, thus defining a lower pivot point thereof. Each of the first links 112 also includes a second pin 126 positioned proximate the corner of the L-shape. The second pin 126 extends through second elongated slots 128 defined in the offset main beam 12, and is linked to a carriage 130 of rack and pinion mechanism 122. Each of the first links 112 also includes a third pin 132 at a second end 134 that is pivotally attached to chest support plate 100, thus defining an upper pivot point thereof.

As depicted in FIGS. 11 and 12, for example, each of the second links 114 includes a first pin 140 at a first end 142 thereof. The first pin 140 extends through the first elongated slot 120 defined in the offset main beam 12, and the first pin 140 connects the second links 114 to the drive nut 124 of the rack and pinion mechanism 122, thus defining a lower pivot point thereof. Each of the second links 114 also includes a second pin 144 at a second end 146 that is pivotally connected to the chest support plate 100, thus defining an upper pivot point thereof.

As depicted in FIGS. 11 and 12, the rack and pinion mechanism 122 includes a drive screw 148 engaging the drive nut 124. Coupled gears 150 are attached to the carriage 130. The larger of the gears 150 engage an upper rack 152 (fixed within the offset main beam 12), and the smaller of the gears 150 engage a lower rack 154. The carriage 130 is defined as a gear assembly that floats between the two racks 152 and 154.

As depicted in FIGS. 11 and 12, the rack and pinion mechanism 122 converts rotation of the drive screw 148 into linear translation of the first and second links 112 and 114 in the first and second elongated slots 120 and 128 toward the first portion 40 of the support structure 14. As the drive nut 124 translates along drive screw 148 (via rotation of the drive screw 148), the carriage 130 translates towards the first portion 40 with less travel due to the different gear sizes of the coupled gears 150. The difference in travel, influenced by different gear ratios, causes the first links 112 pivotally attached thereto to lift the chest support plate 100. Lowering of the chest support plate 100 is accomplished by performing this operation in reverse. The second links 114 are "idler" links (attached to the drive nut 124 and the chest support plate 100) that controls the tilt of the chest support plate 100 as it is being lifted and lowered. All components associated with lifting while tilting the chest plate predetermine where COR 108 resides. Furthermore, a servomotor (not shown) interconnected with the drive screw 148 can be computer controlled and/or operated by the operator of the surgical frame 10 to facilitate controlled lifting and lowering of the chest support plate 100. A safety feature can be provided, enabling the operator to read and limit a lifting and lowering force applied by the torso-lift support 24 in order to prevent injury to the patient P. Moreover, the torso-lift support 24 can also include safety stops (not shown) to prevent over-extension or compression of the patient P, and sensors (not shown) programmed to send patient position feedback to the safety stops.

An alternative preferred embodiment of a torso-lift support is generally indicated by the numeral 160 in FIGS. 13A-15. As depicted in FIGS. 13A-13C, an alternate offset main beam 162 is utilized with the torso-lift support 160. Furthermore, the torso-lift support 160 has a support plate 164 pivotally linked to the offset main beam 162 by a chest support lift mechanism 166. An arm support rod/plate 168 is

connected to the support plate 164, and the second arm support 22B. The support plate 164 is attached to the chest support plate 100, and the chest support lift mechanism 166 includes various actuators 170A, 170B, and 170C used to facilitate positioning and repositioning of the support plate 164 (and hence, the chest support plate 100).

As discussed below, the torso-lift support 160 depicted in FIGS. 13A-15 enables a COR 172 thereof to be programmatically altered such that the COR 172 can be a fixed COR or a variable COR. As their names suggest, the fixed COR stays in the same position as the torso-lift support 160 is actuated, and the variable COR moves between a first position and a second position as the torso-lift support 160 is actuated between its initial position and final position at full travel thereof. Appropriate placement of the COR 172 is important so that spinal cord integrity is not compromised (i.e., overly compressed or stretched). Thus, the support plate 164 (and hence, the chest support plate 100) follows a path coinciding with a predetermined COR 172 (either fixed or variable). FIG. 13A depicts the torso-lift support 160 retracted, FIG. 13B depicts the torso-lift support 160 at half travel, and FIG. 13C depicts the torso-lift support 160 at full travel.

As discussed above, the chest support lift mechanism 166 includes the actuators 170A, 170B, and 170C to position and reposition the support plate 164 (and hence, the chest support plate 100). As depicted in FIGS. 14 and 15, for example, the first actuator 170A, the second actuator 170B, and the third actuator 170C are provided. Each of the actuators 170A, 170B, and 170C are interconnected with the offset main beam 12 and the support plate 164, and each of the actuators 170A, 170B, and 170C are moveable between a retracted and extended position. As depicted in FIGS. 13A-13C, the first actuator 170A is pinned to the offset main beam 162 using a pin 174 and pinned to the support plate 164 using a pin 176. Furthermore, the second and third actuators 170B and 170C are received within the offset main beam 162. The second actuator 170B is interconnected with the offset main beam 162 using a pin 178, and the third actuator 170C is interconnected with the offset main beam 162 using a pin 180.

The second actuator 170B is interconnected with the support plate 164 via first links 182, and the third actuator 170C is interconnected with the support plate 164 via second links 184. First ends 190 of the first links 182 are pinned to the second actuator 170B and elongated slots 192 formed in the offset main beam 162 using a pin 194, and first ends 200 of the second links 184 are pinned to the third actuator 170C and elongated slots 202 formed in the offset main beam 162 using a pin 204. The pins 194 and 204 are moveable within the elongated slots 192 and 202. Furthermore, second ends 210 of the first links 182 are pinned to the support plate 164 using the pin 176, and second ends 212 of the second links 184 are pinned to the support plate 164 using a pin 214. To limit interference therebetween, as depicted in FIGS. 13A-13C, the first links 182 are provided on the exterior of the offset main beam 162, and, depending on the position thereof, the second links 184 are positioned on the interior of the offset main beam 162.

Actuation of the actuators 170A, 170B, and 170C facilitates movement of the support plate 164. Furthermore, the amount of actuation of the actuators 170A, 170B, and 170C can be varied to affect different positions of the support plate 164. As such, by varying the amount of actuation of the actuators 170A, 170B, and 170C, the COR 172 thereof can be controlled. As discussed above, the COR 172 can be predetermined, and can be either fixed or varied. Furthermore, the actuation of the actuators 170A, 170B, and 170C

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can be computer controlled and/or operated by the operator of the surgical frame 10, such that the COR 172 can be programmed by the operator. As such, an algorithm can be used to determine the rates of extension of the actuators 170A, 170B, and 170C to control the COR 172, and the computer controls can handle implementation of the algorithm to provide the predetermined COR. A safety feature can be provided, enabling the operator to read and limit a lifting force applied by the actuators 170A, 170B, and 170C in order to prevent injury to the patient P. Moreover, the torso-lift support 160 can also include safety stops (not shown) to prevent over-extension or compression of the patient P, and sensors (not shown) programmed to send patient position feedback to the safety stops.

FIGS. 16-23 depict portions of the sagittal adjustment assembly 28. The sagittal adjustment assembly 28 can be used to distract or compress the patient's lumbar spine during or after lifting or lowering of the patient's torso by the torso-lift supports. The sagittal adjustment assembly 28 supports and manipulates the lower portion of the patient's body. In doing so, the sagittal adjustment assembly 28 is configured to make adjustments in the sagittal plane of the patient's body, including tilting the pelvis, controlling the position of the upper and lower legs, and lordosing the lumbar spine.

As depicted in FIGS. 16 and 17, for example, the sagittal adjustment assembly 28 includes the pelvic-tilt mechanism 30 for supporting the thighs and lower legs of the patient P. The pelvic-tilt mechanism 30 includes a thigh cradle 220 configured to support the patient's thighs, and a lower leg cradle 222 configured to support the patient's shins. Different sizes of thigh and lower leg cradles can be used to accommodate different sizes of patients, i.e., smaller thigh and lower leg cradles can be used with smaller patients, and larger thigh and lower leg cradles can be used with larger patients. Soft straps (not shown) can be used to secure the patient P to the thigh cradle 220 and the lower leg cradle 222. The thigh cradle 220 and the lower leg cradle 222 are moveable and pivotal with respect to one another and to the offset main beam 12. To facilitate rotation of the patient's hips, the thigh cradle 220 and the lower leg cradle 222 can be positioned anterior and inferior to the patient's hips.

As depicted in FIGS. 18 and 25, for example, a first support strut 224 and second support struts 226 are attached to the thigh cradle 220. Furthermore, third support struts 228 are attached to the lower leg cradle 222. The first support strut 224 is pivotally attached to the offset main beam 12 via a support plate 230 and a pin 232, and the second support struts 226 are pivotally attached to the third support struts 228 via pins 234. The pins 234 extend through angled end portions 236 and 238 of the second and third support struts 226 and 228, respectively. Furthermore, the lengths of second and third support struts 226 and 228 are adjustable to facilitate expansion and contraction of the lengths thereof.

To accommodate patients with different torso lengths, the position of the thigh cradle 220 can be adjustable by moving the support plate 230 along the offset main beam 12. Furthermore, to accommodate patients with different thigh and lower leg lengths, the lengths of the second and third support struts 226 and 228 can be adjusted.

To control the pivotal angle between the second and third support struts 226 and 228 (and hence, the pivotal angle between the thigh cradle 220 and lower leg cradle 222), a link 240 is pivotally connected to a captured rack 242 via a pin 244. The captured rack 242 includes an elongated slot 246, through which is inserted a worm gear shaft 248 of a worm gear assembly 250. The worm gear shaft 248 is

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attached to a gear 252 provided on the interior of the captured rack 242. The gear 252 contacts teeth 254 provided inside the captured rack 242, and rotation of the gear 252 (via contact with the teeth 254) causes motion of the captured rack 242 upwardly and downwardly. The worm gear assembly 250, as depicted in FIGS. 19-21, for example, includes worm gears 256 which engage a drive shaft 258, and which are connected to the worm gear shaft 248.

The worm gear assembly 250 also is configured to function as a brake, which prevents unintentional movement of the sagittal adjustment assembly 28. Rotation of the drive shaft 258 causes rotation of the worm gears 256, thereby causing reciprocal vertical motion of the captured rack 242. The vertical reciprocal motion of the captured rack 242 causes corresponding motion of the link 240, which in turn pivots the second and third support struts 226 and 228 to correspondingly pivot the thigh cradle 220 and lower leg cradle 222. A servomotor (not shown) interconnected with the drive shaft 258 can be computer controlled and/or operated by the operator of the surgical frame 10 to facilitate controlled reciprocal motion of the captured rack 242.

The sagittal adjustment assembly 28 also includes the leg adjustment mechanism 32 facilitating articulation of the thigh cradle 220 and the lower leg cradle 222 with respect to one another. In doing so, the leg adjustment mechanism 32 accommodates the lengthening and shortening of the patient's legs during bending thereof. As depicted in FIG. 17, for example, the leg adjustment mechanism 32 includes a first bracket 260 and a second bracket 262 attached to the lower leg cradle 222. The first bracket 260 is attached to a first carriage portion 264, and the second bracket 262 is attached to a second carriage portion 266 via pins 270 and 272, respectively. The first carriage portion 264 is slidable within third portion 94 of the rear portion 74 of the offset main beam 12, and the second carriage portion 266 is slidable within the first portion 90 of the rear portion 74 of the offset main beam 12. An elongated slot 274 is provided in the first portion 90 to facilitate engagement of the second bracket 262 and the second carriage portion 266 via the pin 272. As the thigh cradle 220 and the lower leg cradle 222 articulate with respect to one another (and the patient's legs bend accordingly), the first carriage 264 and the second carriage 266 can move accordingly to accommodate such movement.

The pelvic-tilt mechanism 30 is movable between a flexed position and a fully extended position. As depicted in FIG. 22, in the flexed position, the lumbar spine is hypo-lordosed. This opens the posterior boundaries of the lumbar vertebral bodies and allows for easier placement of any interbody devices. The lumbar spine stretches slightly in this position. As depicted in FIG. 23, in the extended position, the lumbar spine is lordosed. This compresses the lumbar spine. When posterior fixation devices, such as rods and screws, are placed, optimal sagittal alignment can be achieved. During sagittal alignment, little to negligible angle change occurs between the thighs and the pelvis. The pelvic-tilt mechanism 30 also can hyper-extend the hips as a means of lordosing the spine, in addition to tilting the pelvis. One of ordinary skill will recognize, however, that straightening the patient's legs does not lordose the spine. Leg straightening is a consequence of rotating the pelvis while maintaining a fixed angle between the pelvis and the thighs.

The sagittal adjustment assembly 28, having the configuration described above, further includes an ability to compress and distract the spine dynamically while in the lordosed or flexed positions. The sagittal adjustment assembly 28 also includes safety stops (not shown) to prevent over-

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extension or compression of the patient, and sensors (not shown) programmed to send patient position feedback to the safety stops.

As depicted in FIGS. 24-26, for example, the coronal adjustment assembly 34 is configured to support and manipulate the patient's torso, and further to correct a spinal deformity, including but not limited to a scoliotic spine. As depicted in FIGS. 24-26, for example, the coronal adjustment assembly 34 includes a lever 280 linked to an arcuate radiolucent paddle 282. As depicted in FIGS. 24 and 25, for example, a rotatable shaft 284 is linked to the lever 280 via a transmission 286, and the rotatable shaft 284 projects from an end of the chest support plate 100. Rotation of the rotatable shaft 284 is translated by the transmission 286 into rotation of the lever 280, causing the paddle 282, which is linked to the lever 280, to swing in an arc. Furthermore, a servomotor (not shown) interconnected with the rotatable shaft 284 can be computer controlled and/or operated by the operator of the surgical frame 10 to facilitate controlled rotation of the lever 280.

As depicted in FIG. 24, for example, adjustments can be made to the position of the paddle 282 to manipulate the torso and straighten the spine. As depicted in FIG. 25, when the offset main beam 12 is positioned such that the patient P is positioned in a lateral position, the coronal adjustment assembly 34 supports the patient's torso. As further depicted in FIG. 26, when the offset main beam 12 is positioned such that the patient P is positioned in a prone position, the coronal adjustment assembly 34 can move the torso laterally, to correct a deformity, including but not limited to a scoliotic spine. When the patient is strapped in via straps (not shown) at the chest and legs, the torso is relatively free to move and can be manipulated. Initially, the paddle 282 is moved by the lever 280 away from the offset main beam 12. After the paddle 282 has been moved away from the offset main beam 12, the torso can be pulled with a strap towards the offset main beam 12. The coronal adjustment assembly 34 also includes safety stops (not shown) to prevent over-extension or compression of the patient, and sensors (not shown) programmed to send patient position feedback to the safety stops.

A preferred embodiment of a surgical frame incorporating a translating beam is generally indicated by the numeral 300 in FIGS. 27-30. Like the surgical frame 10, the surgical frame 300 serves as an exoskeleton to support the body of the patient P as the patient's body is manipulated thereby. In doing so, the surgical frame 300 serves to support the patient P such that the patient's spine does not experience unnecessary stress/torsion.

The surgical frame 300 includes translating beam 302 that is generally indicated by the numeral 302 in FIGS. 27-30. The translating beam 302 is capable of translating motion affording it to be positioned and repositioned with respect to portions of the remainder of the surgical frame 300. As discussed below, the positioning and repositioning of the translating beam 302, for example, affords greater access to a patient receiving area A defined by the surgical frame 300, and affords greater access to the patient P by a surgeon and/or a surgical assistant (generally indicated by the letter S in FIG. 30) via access to either of the lateral sides L_1 and L_2 (FIG. 30) of the surgical frame 300.

As discussed below, by affording greater access to the patient receiving area A, the surgical frame 300 affords transfer of the patient P from and to a surgical table/gurney. Using the surgical frame 300, the surgical table/gurney can be conventional, and there is no need to lift the surgical

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table/gurney over portions of the surgical frame 300 to afford transfer of the patient P thereto.

The surgical frame 300 is configured to provide a relatively minimal amount of structure adjacent the patient's spine to facilitate access thereto and to improve the quality of imaging available before, during, and even after surgery. Thus, the workspace of a surgeon and/or a surgical assistant and imaging access are thereby increased. The workspace, as discussed below, can be further increased by positioning and repositioning the translating beam 302. Furthermore, radiolucent or low magnetic susceptibility materials can be used in constructing the structural components adjacent the patient's spine in order to further enhance imaging quality.

The surgical frame 300, as depicted in FIGS. 27-30, is similar to the surgical frame 10 except that surgical frame 300 includes a support structure 304 having a support platform 306 incorporating the translating beam 302. The surgical frame 300 incorporates the offset main beam 12 and the features associated therewith from the surgical table 300. As such, the element numbering used to describe the surgical frame 10 is also applicable to portions of the surgical frame 300.

Rather than including the cross member 44, and the horizontal portions 46 and the vertical portions 48 of the first and second support portions 40 and 42, the support structure 304 includes the support platform 306, a first vertical support post 308A, and a second vertical support post 308B. As depicted in FIGS. 27-30, the support platform 306 extends from adjacent one longitudinal end to adjacent the other longitudinal end of the surgical frame 300, and the support platform 306 supports the first vertical support post 308A at the one longitudinal end and supports the second vertical support post 308B at the other longitudinal end.

As depicted in FIGS. 27-30, the support platform 306 (in addition to the translating beam 302) includes a first end member 310, a second end member 312, a first support bracket 314, and a second support bracket 316. Casters 318 are attached to the first and second end members 310 and 312. The first end member 310 and the second end member 312 each include an upper surface 320 and a lower surface 322. The casters 318 can be attached to the lower surface of each of the first and second end members 310 and 312 at each end thereof, and the casters 318 can be spaced apart from one another to afford stable movement of the surgical frame 300. Furthermore, the first support bracket 314 supports the first vertical support post 308A, and the second support bracket 316 supports the vertical second support post 308B.

The translating beam 302 is interconnected with the first and second end members 310 and 312 of the support platform 306, and as depicted in FIGS. 27-30, the translating beam 302 is capable of movement with respect to the first and second end members 310 and 312. The translating beam 302 includes a first end member 330, a second end member 332, a first L-shaped member 334, a second L-shaped member 336, and a cross member 338. The first L-shaped member 334 is attached to the first end member 330 and the cross member 338, and the second L-shaped member 336 is attached to the second end member 332 and the cross member 338. Portions of the first and second L-shaped members 334 and 336 extend downwardly relative to the first and second end members 330 and 332 such that the cross member 338 is positioned vertically below the first and second end member 330 and 332. The vertical position of the cross member 338 relative to the remainder of the surgical

frame **300** lowers the center of gravity of the surgical frame **300**, and in doing so, serves in adding to the stability of the surgical frame **300**.

The translating beam **302**, as discussed above, is capable of being positioned and repositioned with respect to portions of the remainder of the surgical frame **300**. To that end, the support platform **306** includes a first translation mechanism **340** and a second translation mechanism **342**. The first translation mechanism **340** facilitates attachment between the first end members **310** and **330**, and the second translation mechanism **342** facilitates attachment between the second end members **312** and **332**. The first and second translation mechanism **340** and **342** also facilitate movement of the translating beam **302** relative to the first end member **310** and the second end member **312**.

The first and second translation mechanisms **340** and **342** can each include a transmission **350** and a track **352** for facilitating movement of the translating beam **302**. The tracks **352** are provided on the upper surface **320** of the first and second end members **310** and **312**, and the transmissions **350** are interoperable with the tracks **352**. The first and second translation mechanisms **340** and **342** can each include an electrical motor **354** or a hand crank (not shown) for driving the transmissions **350**. Furthermore, the transmissions **350** can include, for example, gears or wheels driven thereby for contacting the tracks **352**. The interoperability of the transmissions **350**, the tracks **352**, and the motors **354** or hand cranks form a drive train for moving the translating beam **302**. The movement afforded by the first and second translation mechanism **340** and **342** allows the translating beam **302** to be positioned and repositioned relative to the remainder of the surgical frame **300**.

The surgical frame **300** can be configured such that operation of the first and second translation mechanism **340** and **342** can be controlled by an operator such as a surgeon and/or a surgical assistant. As such, movement of the translating beam **302** can be effectuated by controlled automation. Furthermore, the surgical frame **300** can be configured such that movement of the translating beam **302** automatically coincides with the rotation of the offset main beam **12**. By tying the position of the translating beam **302** to the rotational position of the offset main beam **12**, the center of gravity of the surgical frame **300** can be maintained in positions advantageous to the stability thereof.

During use of the surgical frame **300**, access to the patient receiving area **A** and the patient **P** can be increased or decreased by moving the translating beam **302** between the lateral sides L_1 and L_2 of the surgical frame **300**. Affording greater access to the patient receiving area **A** facilitates transfer of the patient **P** between the surgical table/gurney and the surgical frame **300**. Furthermore, affording greater access to the patient **P** facilitates ease of access by a surgeon and/or a surgical assistant to the surgical site on the patient **P**.

The translating beam **302** is moveable using the first and second translation mechanisms **340** and **342** between a first terminal position (FIG. **28**) and a second terminal position (FIGS. **29** and **30**). The translating beam **302** is positionable at various positions (FIG. **27**) between the first and second terminal positions. When the translating beam **302** is in the first terminal position, as depicted in FIG. **28**, the translating beam **302** and its cross member **338** are positioned on the lateral side L_1 of the surgical frame **300**. Furthermore, when the translating beam **302** is in the second terminal position, as depicted in FIGS. **29** and **30**, the translating beam **302** and its cross member **338** are positioned in the middle of the surgical frame **300**.

With the translating beam **302** and its cross member **338** moved to be positioned at the lateral side L_1 , the surgical table/gurney and the patient **P** positioned thereon can be positioned under the offset main beam **12** in the patient receiving area **A** to facilitate transfer of the patient **P** to or from the offset main beam **12**. As such, the position of the translating beam **302** at the lateral side L_1 enlarges the patient receiving area **A** so that the surgical table/gurney can be received therein to allow such transfer to or from the offset main beam **12**.

Furthermore, with the translating beam **302** and its cross member **338** moved to be in the middle of the surgical frame **300** (FIGS. **29** and **30**), a surgeon and/or a surgical assistant can have access to the patient **P** from either of the lateral sides L_1 or L_2 . As such, the position of the translating beam **302** in the middle of the surgical frame **300** allows a surgeon and/or a surgical assistant to get close to the patient **P** supported by the surgical frame **300**. As depicted in FIG. **30**, for example, a surgeon and/or a surgical assistant can get close to the patient **P** from the lateral side L_2 without interference from the translating beam **302** and its cross member **338**. The position of the translating beam **302** can be selected to accommodate access by both a surgeon and/or a surgical assistant by avoiding contact thereof with the feet and legs of a surgeon and/or a surgical assistant.

The position of the translating beam **302** and its cross member **338** can also be changed according to the rotational position of the offset main beam **12**. To illustrate, the offset main beam **12** can be rotated a full 360° before, during, and even after surgery to facilitate various positions of the patient to afford various surgical pathways to the patient's spine depending on the surgery to be performed. For example, the offset main beam **12** can be positioned by the surgical frame **300** to place the patient **P** in a prone position (e.g., FIGS. **27** and **28**), lateral positions (e.g., FIGS. **29** and **30**), and in a position 45° between the prone and lateral positions. The translating beam **302** can be positioned to accommodate the rotational position of the offset main beam **12** to aid in the stability of the surgical frame **300**. For example, when the patient **P** is in the prone position, the translating beam **302** can preferably be moved to the center of the surgical frame **300** underneath the patient **P**. Furthermore, when the patient **P** is in one of the lateral positions, the translating beam **302** can be moved toward one of the corresponding lateral sides L_1 and L_2 of the surgical frame **300** to position underneath the patient **P**. Such positioning of the translating beam **302** can serve to increase the stability of the surgical frame **300**.

A portion of a surgical frame **400** incorporating an upper leg support **402** in accordance with a first embodiment of the present disclosure is described hereinbelow. The surgical frame **400** can incorporate the features of the above-discussed surgical frames, and the upper leg support **402** can also be incorporated in the above-discussed surgical frames. As discussed below, the upper leg support **402** is reconfigurable such reconfiguration can be done via articulation using manual adjustment or controlled automation of the componentry thereof. In addition to the upper legs of the patient **P**, the upper leg support **402** can be used to at least partially support the pelvic area of the patient **P**, and to facilitate manipulation of the lumbar spine of the patient **P**.

Like the surgical frames **10** and **300**, the surgical frame **400** can serve as an exoskeleton to support the body of the patient **P** as the patient's body is manipulated thereby. In doing so, the surgical frame **400** serves to support the patient **P** such that the patient's spine does not experience unnecessary stress/torsion.

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Like the surgical frame 300, the surgical frame 400 can include a translating beam 302 and a support structure 304 having a support platform 306 incorporating the translating beam 302. Besides the support platform 306, the support structure 304 can include a first vertical support portion 308A and a second vertical support portion 308B. The first vertical support portion 308A and the second vertical support portion 308B are capable of expansion and contraction.

As depicted in FIGS. 31-33B, the surgical frame 400 also incorporates a main beam 404 having a first end (not shown) attached relative to the first support portion 308A and a second end (not shown) attached relative to the second support portion 308B. Like in the surgical frame 300, the main beam includes a first portion (not shown) at the first end, a second portion (not shown) at the second end, and a third portion 410 extending between the first portion and the second portion. The main beam 404 is similar to the offset main beam 12, and, as discussed below, the main beam 404 can incorporate features associated with the offset main beam 12. To illustrate, the offset main beam 404, like the main beam 12, is used in supporting the patient P on the surgical frame 400 and includes various components similar to those incorporated in the surgical frames 10 and 300. For example, in addition to the upper leg support 402, the main beam 404 can incorporate a head support (not shown), a chest support (not shown), arm supports (not shown), and a lower leg support (not shown). The upper leg support 402, the chest support, the arm supports, and the lower leg support can be attached to the third portion 410 of the main beam 404.

An operator such as a surgeon can control actuation of the various support components to manipulate the position of the patient's body. Soft straps (not shown) are used with these various support components to secure the patient P to the frame and to enable either manipulation or fixation of the patient P. Furthermore, reusable soft pads can be used on the load-bearing areas of the various support components. Additionally, the main beam 404 can be rotated a full 360° before, during, and even after surgery to facilitate various positions of the patient P to afford various surgical pathways to the patient's spine depending on the surgery to be performed. For example, the main beam 404 can be positioned by the surgical frame 400 to place the patient P in a prone position, lateral positions, and in a position 45° between the prone and lateral positions.

The surgical frame 400 can be used to facilitate access to different parts of the spine of the patient P. In particular, the surgical frame 400 can be used to facilitate access to portions of the patient's lumbar spine. To illustrate, the patient P is simultaneously supported by the chest support and the upper leg support 402 on the main beam 404, and uninterrupted access is provided to portions of the patient's lumbar spine by the positions of the chest support and the upper leg support 402. The upper leg support 402 can be used to support the patient P during rotation of the main beam 404, and articulation of the other componentry of the surgical frame 400. Furthermore, the upper leg support 402, as depicted in FIGS. 36 and 38, is actuatable to facilitate positioning and repositioning thereof before, during, and after surgery to manipulate the patient P about an adjustable center of rotation CR located in and/or along a portion of the spine, including but not limited to the lumbar spine. As discussed below, the adjustable center of rotation CR is both adjustable to accommodate patients having different body sizes, and adjustable to facilitate, for example, flexing of the lumbar spine with the center of rotation CR location located above (posterior), within, or below (anterior) the lumbar

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spine to afford surgical access thereto with or without distribution or compression of the lumbar spine or portions thereof.

The main beam 404 is moveably attached relative to the first vertical support post 308A and the second vertical support post 308B. Like those of the surgical frames 10 and 300, the first vertical support post 308A and the second vertical support post 308B of the surgical frame 400 each include a clevis (not shown) supporting componentry facilitating rotation of the main beam 404. In addition to the clevis, the first vertical support post 308A includes a support block portion (not shown), a pin portion (not shown) pivotally attaching the support block portion to the clevis, and an axle portion (not shown) rotatably supported by the support block and interconnected to the first portion at the first end of the main beam 404. The support block portion, via interaction of the pin portion with the clevis, is capable of pivotal movement relative to the clevis to accommodate different heights for the first vertical support post 308A and the second vertical support post 308B. And the main beam 404, via interaction of the axle portion with the support block portion, is capable of rotational movement relative to the support block portion to accommodate rotation of the patient P supported by the main beam 404.

Furthermore, in addition to the clevis, the second vertical support post 308B includes a coupler (not shown) and a pin portion (not shown) pivotally attaching the coupler to the clevis. The coupler includes a base portion (not shown) that is pinned to the clevis with the pin portion, a body portion (not shown) that includes a transmission (not shown), a motor (not shown) that drives the transmission in the body portion, and a head portion (not shown) that is rotatable with respect to the body portion and driven rotationally by the transmission via the motor. The head portion is interconnected with the second portion at the second end of the main beam 404, and the head portion (via the transmission and the motor) can rotate the main beam 404 a full 360° before, during, and even after surgery to facilitate various positions of the patient P.

As depicted in FIGS. 31-34B, the upper leg support 402 can be attached to and incorporated into portions of the third portion 410 of the main beam 404. The upper leg support 402, as depicted in FIGS. 32B-38, can include a first arm portion 450, a second arm portion 452, and a platform portion 454. The first arm portion 450 includes a first end portion 460 and a second end portion 462, and the second arm portion 452 includes a first end portion 464 and a second end portion 466. The first end portion 460 and the second end portion 462 of the first arm portion 450 are pivotally attached, respectively, to the main beam 404 and the first end portion 464 of the second arm portion 452, and the first end portion 464 and the second end portion 466 of the second arm portion 452 are pivotally attached, respectively, to the second end portion 466 of the first arm portion 450 and the main beam 404.

The first arm portion 450 is extendable, and includes a base portion 470 that includes the first end portion 460 and an extendable portion 472 that includes the second end portion 462. The extendable portion 472 is moveable inwardly and outwardly relative to the base portion 470, and such inward and outward movement serves to pivot the first arm portion 450 relative to the main beam 404, pivot the first arm portion 450 and the second arm portion 452 relative to one another, and pivot the second arm portion 452 relative to the main beam 404. As discussed below, such pivotal movement serves in facilitating positioning and repositioning of the platform portion 454 relative to the main beam

404. To illustrate, increasing the amount of extension of the extendable portion 472 relative to the base portion 470 moves the platform portion 454 away from the third portion 410 of the main beam 404, and toward the first end and away from the second end.

The third portion 410, as depicted in FIGS. 32B and 33B, includes an interior cavity 480 defined by a first sidewall portion 482, a second sidewall portion 484, and a connecting-wall portion 486 joining the first sidewall portion 482 and the second sidewall portion 484 to one another. Portions of the upper leg support 402 are received within the interior cavity 480. To illustrate, the first end portion 460 of the first arm portion 450 and the second end portion 466 of the second arm portion 452 are received with the cavity 480.

The first end portion 460 of the first arm portion 450, as depicted in FIG. 32B, includes an aperture 490 for receiving a pin 492 extending between the first sidewall portion 482 and the second sidewall portion 484 to facilitate pivotal attachment of the first arm portion 450 to the main beam 404, and the second end portion 466 of the second arm portion 452 includes an aperture 494 for receiving a pin 496 extending between the first sidewall portion 482 and the second sidewall portion 484 to facilitate pivotal attachment of the second arm portion 452 to the main beam 404. Furthermore, the second end portion 462 of the first arm portion 450 and the first end portion 464 of the second arm portion 452 can form a clevis-tang joint, wherein one of the second end portion 462 and the first end portion 464 is a clevis, and the other of the second end portion 462 and the first end portion 464 is a tang. As depicted in FIGS. 32B and 33B, the second end portion 462 is configured as a tang with an aperture 500 extending therethrough, and the first end portion 464 is configured as a clevis with apertures 502 extending therethrough. The aperture 500 and the apertures 502, as depicted in FIGS. 32B and 33B, are configured to receive a pin 504 to facilitate pivotal attachment of the first arm portion 450 and the second arm portion 452 to one another.

Additionally, the pin 504 is used in facilitating attachment of the platform portion 454 to the first arm portion 450 and the second arm portion 452. As depicted in FIGS. 32B and 33B, the platform portion 454 includes a base portion 510, a first upstanding portion 512, a second upstanding portion 514, and a third upstanding portion 516. Portions of the first upstanding portion 512 and the second upstanding portion 514 can be received in the interior cavity 480, and portions of the second end portion 462 of the first arm portion 450 and the first end portion 464 of the second arm portion 452 are received between the first upstanding portion 512 and the second upstanding portion 514. The first upstanding portion 512 includes an aperture 520, the second upstanding portion 514 includes an aperture 522, and each of the first aperture 520 and the second aperture 522 are configured to receive portions of the pin 504 to attach the second end portion 462 of the first arm portion 450 and the first end portion 464 of the second arm portion 452 to platform portion 454.

The extension of the extendable portion 472 relative to the base portion 470 serves in pivoting the first arm portion 450 and the second arm portion 452 relative to one another such that increasing the amount of extension decreases an angle between the first arm portion 450 and the second arm portion 452, and decreasing the amount of extension increases the angle between the first arm portion 450 and the second arm portion 452. Given that the platform portion 454 is attached to the second end portion 462 of the first arm portion 450 and the first end portion 464 of the second arm portion 452, increasing the amount of extension of the first arm portion

450 moves the platform portion 454 away from the third portion 410 of the main beam 404, and toward the first end and away from the second end, and decreasing the amount of extension of the first arm portion 450 moves the platform portion 454 toward the third portion 410 of the main beam 404, and away from the first end and toward the second end. Furthermore, when the extension of the first arm portion 450 is decreased, portions of the first upstanding portion 512 and the second upstanding portion 514 are drawn into the cavity 480. As discussed below, the movement of the platform portion 454 using the extension of the extendable portion 472 ultimately serves in adjusting the position of the patient's spine. Such adjustment can occur before, during, and/or after surgery using the surgical frame 400.

As depicted in FIGS. 32B and 33B, the upper leg support 402 also includes a telescoping shaft portion 530 that is connected between the second arm portion 452 and the platform portion 454. The telescoping shaft portion 530 is used to pivot the platform portion 454 relative to the first arm portion 450 and the second arm portion 452. The telescoping shaft portion 530 includes a first end portion 532, a second end portion 534, a base portion 536 including the first end portion 532, and an extendable portion 538 including the second end portion 534. As discussed below, the extendable portion 538 is moveable inwardly and outwardly relative to the base portion 536, and such inward and outward movement serves to pivot the platform portion 454.

To facilitate connection between the telescoping shaft portion 530 and the second arm portion 452, one of the first end portion 532 and the second arm portion 452 can form a clevis, and the other of the first end portion 532 and the second arm portion 452 can form a tang. Furthermore, to facilitate connection between the telescoping shaft portion 530 and the platform portion 454, one of the second end portion 534 and the platform portion 454 can form a clevis, and the other of the second end portion 534 and the platform portion 454 can form a tang. As depicted in FIGS. 32B and 33B, the second arm portion 452 includes a clevis 540 having apertures 542, the first end portion 532 is used as a tang having an aperture 544, and a pin 546 is received through the apertures 542 and 544 to join the telescoping shaft portion 530 to the second arm portion 452. Furthermore, as depicted in FIGS. 32B and 33B, the platform portion 454 includes a post 550, the second end portion 534 includes an aperture 552, and the post 550 is received in the aperture 552 to join the telescoping shaft portion 530 to the platform portion 454.

The extendable portion 538 is moveable inwardly and outwardly relative to the base portion 536, and such inward and outward movement serves to pivot the platform portion 454 relative to the first arm portion 450 and the second arm portion 452. Such pivotal movement serves in facilitating positioning and repositioning of the platform portion 454 relative to first arm portion 450 and the second arm portion 452. To illustrate, the base portion 510 of the platform portion 554 includes a first end 560 and a second end 562, and increasing the amount of extension of the extendable portion 538 relative to the base portion 536 moves the second end 562 away from the third portion 410 of the main beam 404. As discussed below, the movement of the platform portion 454 using the extension of the telescoping shaft portion 530 ultimately serves in adjusting the position of the patient's spine. Such adjustment can occur before, during, and/or after surgery using the surgical frame 400.

As depicted in FIGS. 34A and 34B, the upper leg support 402 also includes a linear movement assembly 570. The linear movement assembly 570 includes a track portion 572

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attached to the third upstanding portion 516, two trucks 574 moveable along the track portion 572, and a support bracket 576 attached to the two trucks 574. The third upstanding portion 516, as depicted in FIGS. 33A and 33B, includes a first end 580 and a second end 582, and is larger than the first upstanding portion 512 and the second upstanding portion 514. The third upstanding portion 516 is attached at and adjacent the first end 580 to the base portion 510, and extends from the base portion 510 toward the first portion of the main beam 404 to the second end 582. The third upstanding portion 516 supports the track portion 572, the two trucks 574, the support bracket 576, and additional components of the upper leg support 402.

The linear movement assembly 570, as depicted in FIGS. 34A and 34B, also includes a telescoping shaft portion 590 that is connected between the third upstanding portion 516 and the support bracket 576. The telescoping shaft portion 590 includes a first end portion 592, a second end portion 594, a base portion 596 including the first end portion 592, and an extendable portion 598 including the second end portion 594. To attach the telescoping shaft portion 590 to the third upstanding portion 516, the first end portion 592 of the telescoping shaft portion 590 can include an aperture 600, the third upstanding portion 516 can include an aperture 602, and a pin 604 can be received in the apertures 600 and 602. Furthermore, to attach the telescoping shaft portion 590 to the support bracket 576, the second end portion 594 of the telescoping shaft portion 590 can include an aperture 606, the support bracket 576 can include a projection 608 including an aperture 610, and a pin 612 can be received in the apertures 606 and 610.

The extendable portion 598 is moveable inwardly and outwardly relative to the base portion 596, and such inward and outward movement relative to the base portion 596 serves to move the support bracket 576 via movement of the two trucks 574 along the track portion 572 between at least a first position closer to the second end 582 of the main beam 404 to a second position closer to the first end 580 of the main beam 404. As discussed below, the movement of the support bracket 576 using the extension of the extendable portion 598 ultimately serves in adjusting the position of the patient's spine. Such adjustment can occur before, during, and/or after surgery using the surgical frame 400.

The upper leg support 402 also includes a support assembly 620 that is carried by the support bracket 576. The support assembly 620 includes a first support post 622, a second support post 624, and a connecting bracket 626 connecting the first support post 622 and the second support post 624 to one another. The support assembly 620 also includes a first support block 630, a second support block 632, a third support block 634, a fourth support block 636, a first support plate 640, a second support plate 642, and a third support plate 644. Each of the first support plate 640, the second support plate 642, and the third support plate 644 include upper surfaces 646A, 646B, and 646C, respectively, that can be used to contact the upper legs of the patient. The upper surfaces 646A and 646C can be covered with padding (not shown) for contacting portions of the patient's upper legs, and the padding can include pressure sensors (not shown) incorporated therein. The resulting pressure sensing padding can be used to determine if undue stress is placed on the patient P via articulation of the upper leg support 402.

The first support plate 640 and the second support plate 642, as discussed below, are moveable with respect to the support bracket 576 and the third support plate 644. As depicted in FIGS. 33A and 33B, the third support plate 644 is attached to the support bracket 576, and the second

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support plate 642 is positioned such that it can move underneath the third support plate 644. As such, during movement of the first support plate 640 and the second support plate 642, the area defined by the upper surfaces 646A, 646B, and 646C can be respectively decreased or increased as the second support plate 642 is moved under or out from under the third support plate 644.

As depicted in FIG. 34A, the first support plate 640 is attached to the first support block 630 and the second support block 632, and the first support block 630 is moveable along the first support post 622 and the second support block 632 is moveable along the second support post 624. Furthermore, the second support plate 642 is attached to the third support block 634 and the fourth support block 636, and the third support block 634 is moveable along the first support post 622 and the fourth support block 636 is moveable along the second support post 624.

The first and second support blocks 630 and 632 include apertures 650 and 652 for receiving the first and second support posts 622 and 624, respectively, and the third and fourth support blocks 634 and 636 include apertures 654 and 656 for receiving the first second support posts 622 and 624, respectively. The first support plate 640 and the second support plate 642 are moveable inwardly and outwardly relative to the support bracket 576 and the third support plate 644 with threads complementary to those of the threaded shaft via movement of the first and third support blocks 630 and 634 on the first support post 622 and via movement of the second and fourth support blocks 632 and 636 on the second support post 624.

The support assembly 620 also includes a threaded shaft 660 that is retained in position between the support bracket 576 and the connecting bracket 626. As discussed below, the threaded shaft 660 is used to constrain movement of the first support plate 640 and the second support plate 642 relative to the third support plate 644 and the main beam 404.

Furthermore, the first support plate 640 includes a first support collar 664 opposite from the upper surface 646A, and the second support plate 642 includes a second support collar 666 opposite from the upper surface 646B. The first support collar 664 includes a first aperture 670 that can include threads complementary to those of the threaded shaft 660, and the second support collar 666 includes a second aperture 672 that can include threads complementary to those of the threaded shaft 660. The threaded shaft 660 can be received in the first aperture 670 and the second aperture 672. The first support collar 664 and the second support collar 666 can include one or more latches (not shown) that can be engaged and disengaged from the threaded shaft 660 via actuation thereof. The one or more latches can be attached to the first support collar 664 and/or the second support collar 666, and a user can actuate the one or more latches to engage or disengage the threaded shaft 660 to correspondingly prevent movement or allow movement of the first support collar 664 and the second support collar 666 along the threaded shaft 660. When the one or more latches are engaged, the interactions of the one or more latches with the threaded shaft 660 prevent movement of the first support plate 640 and the second support plate 642 relative to the third support plate 644. When the one or more latches are disengaged, the first support plate 640 and the second support plate 642 can move relative to the third support plate 644. Rather than using the threaded shaft 660, a shaft with catches and/or teeth to which the one or more latches can be engaged and disengaged.

Alternatively, a motor/transmission/actuator (not shown) can be used to facilitate rotation of the threaded shaft 660,

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and rotation of the threaded shaft 660 and the interaction in the first aperture 670 and the second aperture 672 causes corresponding movement of the first support plate 640 and the second support plate 642. As such, rotation of the threaded shaft 660 via actuation of the motor/transmission/actuator is translated into movement of the first support plate 640 and the second support plate 642. To illustrate, the threads of the threaded shaft 660, the first aperture 670, and the second aperture 672 can be configured such that clockwise rotation of the threaded shaft 660 via actuation of the motor/transmission/actuator causes inward movement of the first support plate 640 and the second support plate 642, and counterclockwise rotation of the threaded shaft 660 via actuation of the motor/transmission/actuator causes outward movement of the first support plate 640 and the second support plate 642. The inward and outward movement of the first support plate 640 and the second support plate 642 is relative to the third support plate 644 and the main beam 404.

The movement of the first support plate 640 and the second support plate 642 ultimately serves in adjusting a total width of a combination of the first support plate 640, the second support plate 642, and the third support plate 644. Adjustment of the combined width of the first support plate 640, the second support plate 642, and the third support plate 644 affords the accommodation of differently sized patients on the upper leg support 402.

The movement of the componentry of the upper leg support 402 can be effectuated via manual adjustment and/or controlled automation. To illustrate, the first arm portion 450 includes the extendable portion 472 that is moveable with respect to the base portion 470 thereof, the telescoping shaft portion 530 includes the extendable portion 538 that is moveable with respect to the base portion 536 thereof, the telescoping shaft portion 590 includes the extendable portion 598 that is moveable with respect to the base portion 596 thereof, and the motor/transmission/actuator facilitates movement of the first support plate 640 and the second support plate 642 is relative to the third support plate 644 and the main beam 404.

Such reconfiguration of the upper leg support 402 can be actuated using the manual adjustment and/or the controlled automation, and as discussed below, the extension and retraction of the extendable portion 472, the extendable portion 538, and the extendable portion 598, as depicted in FIGS. 35 and 37, for example, via such actuation can be used to both adjust the adjustable center of rotation CR to accommodate patients having different body sizes, and to facilitate flexing of the lumbar spine to afford surgical access thereto via the manipulation of portions thereof. The extension/retraction of the extendable portion 472 serves to change the angle of the first arm portion 450 and the second arm portion 452 relative to one another, the extension/retraction of the extendable portion 538 serves to change the angle of the platform portion 454 relative to the second arm portion 452, and the extendable portion 598 serves to change position of the bracket 576 (which supports the first support plate 640, the second support plate 642, and the third support plate 644) relative to the platform portion 454.

Using the upper leg support 402, the position of the patient's upper legs can be altered, which correspondingly affects the flexure of the lumbar spine of the patient P, and care should be taken to prevent unwanted torsion thereof when manipulating the patient's spine. To illustrate, the amounts of extension/retraction of the extendable portion 472, the extendable portion 538, and the extendable portion 598 can be constrained with respect to one another to

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prevent unwanted torsion of the lumbar spine during articulation of the upper leg support 402. As such, the amounts of extension/retraction of the extendable portion 472, the extendable portion 538, and the extendable portion 598 can be contingent upon one another to facilitate such approximate preservation.

A controller (not shown) with a user interface (not shown) can be used to control the constrained/contingent extension and/or retraction of the extendable portion 472, the extendable portion 538, and the extendable portion 598 via the controlled automation. Furthermore, because patients' heights can vary, the amounts of extension/retraction of the extendable portion 472, the extendable portion 538, and the extendable portion 598 can be altered to accommodate these different heights while still being constrained/contingent upon one another to provide for the desired amount of distraction/compression of portions of the lumbar spine during articulation of the upper leg support 402.

The controller with input via the user interface can allow the user to select the desired center of rotation and the desired amount of manipulation or angulation of the segmental portions of the lumbar spine. To illustrate, the user interface can be used to display a graphical or actual representation of the patient's spine, and the user interface can permit the user to input the desired center of rotation and the desired amount of manipulation by, for example, highlighting a portion of the graphical or actual representation of the patient's spine on the user interface. The selection of the desired amount of manipulation can allow the user to select where the forces applied via the actuation of the extendable portion 472, the extendable portion 538, and the extendable portion 598 are applied during flexure of the patient's spine. In addition to or alternatively to use of the user interface, a navigation tool interconnected with the controller and/or the user interface can be positioned on or adjacent the patient's spine to facilitate inputting of the desired center of rotation and the desired amount of manipulation. The inputting of the desired center of rotation and the desired amount of manipulation can be done with the main beam 404 and the patient P supported on the main beam 404 in various rotational positions including, but not limited to, prone, lateral, and supine positions.

When the upper leg support 402 is articulated such that the lumbar spine of the patient P is in an unflexed neutral position, as depicted in FIG. 36, the controller can be used to extend/retract the extendable portion 472 (of the first arm portion 450), the extendable portion 538 (of the telescoping shaft portion 530), and the extendable portion 598 (of the telescoping shaft portion 590) such that the first arm portion 450, the telescoping shaft portion 530, and the telescoping shaft portion 590 have lengths of 24.862, 8.639, and 11.963 inches, respectively, that accommodate the height of the patient P. Furthermore, when the upper leg support 402 is articulated such that the lumbar spine of the patient P has a 30 degree flex, as depicted in FIG. 38, the controller can be used to extend/retract the extendable portion 472 (of the first arm portion 450), the extendable portion 538 (of the telescoping shaft portion 530), and the extendable portion 598 (of the telescoping shaft portion 590) such that the first arm portion 450, the telescoping shaft portion 530, and the telescoping shaft portion 590 have lengths of 40.375, 6.652, and 9.540 inches, respectively, that flex the lumbar spine of the patient P to afford surgical access thereto. Moreover, during the transition between the positions of FIGS. 36 and 38, the controller can serve to prevent unwanted torsion of the lumbar spine during articulation of the upper leg support 402 by properly adjusting the amounts of extension/retraction

tion of the extendable portion 472, the extendable portion 538, and the extendable portion 598.

FIG. 39, for example, includes a table that illustrates the relative amounts of increase/decrease of the lengths of the first arm portion 450 (Cylinder 1), the telescoping shaft portion 530 (Cylinder 2), and the telescoping shaft portion 590 (Cylinder 3) via extension/retraction of the extendable portion 472, the extendable portion 538, and the extendable portion 598 for a patient P having a height of 6'3". As such, the upper leg support 402 provides an adjustable center of rotation CR located in the lumbar spine to accommodate patients having different body sizes, and also to afford surgical access to the lumbar spine via manipulation of portions thereof.

Thus, using the user interface of the controller, the operator of the surgical frame 400 can input the height of the patient P, and input the desired degree of flexure of the lumbar spine, and the controller can actuate the first arm portion 450 (to extend or retract the extendable portion 472), the telescoping shaft portion 530 (to extend or retract the extendable portion 538), and the telescoping shaft portion 590 (to extend or retract the extendable portion 598) the appropriate amounts to provide such flexion, while also preventing unwanted torsion of the patient's spine. As discussed above, the extension/retraction of the extendable portion 472 serves to change the angle of the first arm portion 450 and the second arm portion 452 relative to one another, the extension/retraction of the extendable portion 538 serves to change the angle of the platform portion 454 relative to the second arm portion 452, and the extendable portion 598 serves to change position of the bracket 576 (which supports the first support plate 640, the second support plate 642, and the third support plate 644) relative to the platform portion 454. During such manipulation of the patient's spine using the upper leg support 402, the lengths of the first arm portion 450, the telescoping shaft portion 530, and the telescoping shaft portion 590 may each alternately increase/decrease or decrease/increase to provide for the desired adjustable center of rotation CR. The operator can use the controller to manipulate the upper leg support 402 to flex the lumbar spine of the patient P into position for surgery, while simultaneously inhibiting the unwanted torsion of the patient's spine that may be caused by reconfiguration of the upper leg support 402.

A portion of a surgical frame 1400 incorporating an upper leg support 1402 in accordance with a second embodiment of the present disclosure is described hereinbelow. The surgical frame 1400 can incorporate the features of the above-discussed surgical frames, and the upper leg support 1402 can also be incorporated in the above-discussed surgical frames. As discussed below, the upper leg support 1402 is reconfigurable such reconfiguration can be done via articulation using manual adjustment or controlled automation of the componentry thereof. In addition to the upper legs of the patient P, the upper leg support 1402 can be used to at least partially support the pelvic area of the patient P, and to facilitate manipulation of the lumbar spine of the patient P.

Like the surgical frames 10 and 300, the surgical frame 1400 can serve as an exoskeleton to support the body of the patient P as the patient's body is manipulated thereby. In doing so, the surgical frame 1400 serves to support the patient P such that the patient's spine does not experience unnecessary stress/torsion.

Like the surgical frame 300, the surgical frame 1400 can include a translating beam 302 and a support structure 304 having a support platform 306 incorporating the translating

beam 302. Besides the support platform 306, the support structure 304 can include a first vertical support portion 308A and a second vertical support portion 308B. The first vertical support portion 308A and the second vertical support portion 308B are capable of expansion and contraction.

As depicted in FIGS. 40-44, the surgical frame 1400 also incorporates a main beam 1404 having a first end 1405A attached relative to the first support portion 308A and a second end 1405B attached relative to the second support portion 308A. Like in the surgical frame 300, the main beam includes a first portion 1406 at the first end 1405A, a second portion (not shown) at the second end 1405B, and a third portion 1410 extending between the first portion 1406 and the second portion. The main beam 1404 is similar to the offset main beam 12, and, as discussed below, the main beam 1404 can incorporate features associated with the offset main beam 12. To illustrate, the offset main beam 1404, like the main beam 12, is used in supporting the patient P on the surgical frame 1400 and includes various components similar to those incorporated in the surgical frames 10 and 300. For example, in addition to the upper leg support 1402, the main beam 1404 can incorporate a head support (not shown), a chest support 1412, arm supports (not shown), and a lower leg support 1416. The upper leg support 1402, the chest support 1412, the arm supports, and the lower leg support 1416 can be attached to the third portion 1410 of the main beam 1404.

An operator such as a surgeon can control actuation of the various support components to manipulate the position of the patient's body. Soft straps (not shown) are used with these various support components to secure the patient P to the frame and to enable either manipulation or fixation of the patient P. Furthermore, reusable soft pads can be used on the load-bearing areas of the various support components. Additionally, the main beam 1404 can be rotated a full 360° before, during, and even after surgery to facilitate various positions of the patient P to afford various surgical pathways to the patient's spine depending on the surgery to be performed. For example, the main beam 1404 can be positioned by the surgical frame 1400 to place the patient P in a prone position, lateral positions, and in a position 45° between the prone and lateral positions.

The surgical frame 1400 can be used to facilitate access to different parts of the spine of the patient P. In particular, the surgical frame 1400 can be used to facilitate access to portions of the patient's lumbar spine. To illustrate, the patient P is simultaneously supported by the chest support 1412 and the upper leg support 1402 on the main beam 1404, and uninterrupted access is provided to portions of the patient's lumbar spine by the positions of the chest support 1412 and the upper leg support 1404. The upper leg support 402 can be used to support the patient P during rotation of the main beam 1404, and articulation of the other componentry of the surgical frame 1400. Furthermore, the upper leg support 1402 is actuatable to facilitate positioning and repositioning of the patient P before, during, and after surgery to manipulate the patient P about an adjustable center of rotation CR located in the lumbar spine. As discussed below, the adjustable center of rotation CR is both adjustable to accommodate patients having different body sizes, and adjustable to facilitate flexing of the lumbar spine to facilitate surgical access thereto via the manipulation of portions thereof.

The main beam 1404 is moveably attached relative to the first vertical support post 308A and the second vertical support post 308B. Like those of the surgical frames 10 and 300, the first vertical support post 308A and the second

vertical support post **308B** of the surgical frame **1400** each include a clevis (not shown) supporting componentry facilitating rotation of the main beam **1404**. In addition to the clevis, the first vertical support post **308A** includes a support block portion (not shown), a pin portion (not shown) pivotally attaching the support block portion to the clevis, and an axle portion (not shown) rotatably supported by the support block and interconnected to the first portion **1406** at the first end **1405A** of the main beam **1404**. The support block portion, via interaction of the pin portion with the clevis, is capable of pivotal movement relative to the clevis to accommodate different heights for the first vertical support post and the second vertical support post. And the main beam **1404**, via interaction of the axle portion with the support block portion, is capable of rotational movement relative to the support block portion to accommodate rotation of the patient **P** supported by the main beam **1404**.

Furthermore, in addition to the clevis, the second vertical support post **308B** includes a coupler (not shown) and a pin portion (not shown) pivotally attaching the coupler to the clevis. The coupler includes a base portion (not shown) that is pinned to the clevis with the pin portion, a body portion (not shown) that includes a transmission (not shown), a motor (not shown) that drives the transmission in the body portion, and a head portion (not shown) that is rotatable with respect to the body portion and driven rotationally by the transmission via the motor. The head portion is interconnected with the second portion at the second end **1405B** of the main beam **1404**, and the head portion (via the transmission and the motor) can rotate the main beam **1404** a full 360° before, during, and even after surgery to facilitate various positions of the patient **P**.

As depicted in FIGS. **40-46**, the upper leg support **1402** can be attached to and incorporated into portions of the third portion **1410** of the main beam **1404**. The upper leg support **1402**, as depicted in FIGS. **42-46**, can include a first arm portion **1450**, a second arm portion **1452**, and a platform portion **1454**. The first arm portion **1450** includes a first end portion **1460** and a second end portion **1462**, and the second arm portion **1452** includes a first end portion **1464** and a second end portion **1466**. The first end portion **1460** and the second end portion **1462** of the first arm portion **1450** are pivotally attached, respectively, to the main beam **1404** and the first end portion **1464** of the second arm portion **1452**, and the first end portion **1464** and the second end portion **1466** of the second arm portion **1452** are pivotally attached, respectively, to the second end portion **1466** of the first arm portion **1450** and the main beam **1404**.

The first arm portion **1450** is extendable, and includes a base portion **1470** that includes the first end portion **1460** and an extendable portion **1472** that includes the second end portion **1462**. The extendable portion **1472** is moveable inwardly and outwardly relative to the base portion **1470**, and such inward and outward movement serves to pivot the first arm portion **1450** relative to the main beam **1404**, pivot the first arm portion **1450** and the second arm portion **1452** relative to one another, and pivot the second arm portion **1452** relative to the main beam **1404**. As discussed below, such pivotal movement serves in facilitating positioning and repositioning of the platform portion **1454** relative to the main beam **1404**. To illustrate, increasing the amount of extension of the extendable portion **1472** relative to the base portion **1470** moves the platform portion **1454** away from the third portion **1410** of the main beam **1404**, and toward the first end **1405A** and away from the second end **1405B**.

The third portion **1410** includes an interior cavity **1890** defined by a first sidewall portion **1482**, a second sidewall

portion **1484**, and a connecting-wall portion **1486** joining the first sidewall portion **1482** and the second sidewall portion **1484** to one another. As depicted in FIGS. **42, 44, and 45**, portions of the upper leg support **1402** are received within the interior cavity **1480**. To illustrate, the first end portion **1460** of the first arm portion **1450** and the second end portion **1466** of the second arm portion **1452** are received with the cavity **1480**.

The first end portion **1460** of the first arm portion **1450** includes an aperture **1490** for receiving a pin **1492** extending between the first sidewall portion **1482** and the second sidewall portion **1484** to facilitate pivotal attachment of the first arm portion **1450** to the main beam **1404**, and the second end portion **1466** of the second arm portion **1452** includes an aperture **1494** for receiving a pin **1496** extending between the first sidewall portion **1482** and the second sidewall portion **1484** to facilitate pivotal attachment of the second arm portion **1452** to the main beam **1404**. Furthermore, the second end portion **1462** of the first arm portion **1450** and the first end portion **1464** of the second arm portion **1452** can form a clevis-tang joint, wherein one of the second end portion **1462** and the first end portion **1464** is a clevis, and the other of the second end portion **1462** and the first end portion **1464** is a tang. As depicted in FIG. **43**, the second end portion **1462** is configured as a tang with an aperture **1500** extending therethrough, and the first end portion **1464** is configured as a clevis with apertures **1502** extending therethrough. The aperture **1500** and the apertures **1502** are configured to receive a pin **1504**, as depicted in FIG. **42**, to facilitate pivotal attachment of the first arm portion **1450** and the second arm portion **1452** to one another.

Additionally, a pin **1506** is used in facilitating attachment of the platform portion **1454** to the second arm portion **1452**. As depicted in FIGS. **42 and 45**, the platform portion **1454** includes a base portion **1510**, a first upstanding portion **1512**, a second upstanding portion **1514**, and a third upstanding portion **1516**. Portions of the second end portion **1462** of the first arm portion **1450** and the first end portion **1464** of the second arm portion **1452** are received between the first upstanding portion **1512** and the second upstanding portion **1514**. The first upstanding portion **1512** includes an aperture **1520**, the second upstanding portion **1514** includes an aperture **1522**, and each of the first aperture **1520** and the second aperture **1522** are configured to receive portions of the pin **1506** to attach the first end portion **1464** of the second arm portion **1452** to platform portion **1454**.

The extension of the extendable portion **1472** relative to the base portion **1470** serves in pivoting the first arm portion **1450** and the second arm portion **1452** relative to one another such that increasing the amount of extension decreases an angle between the first arm portion **1450** and the second arm portion **1452**, and decreasing the amount of extension increases the angle between the first arm portion **1450** and the second arm portion **1452**. Given that the platform portion **1454** is attached to the first end portion **1464** of the second arm portion **1452**, increasing the amount of extension of the first arm portion **1450** moves the platform portion **1454** away from the third portion **1410** of the main beam **1404**, and toward the first end **1405A** and away from the second end **1405B**, and decreasing the amount of extension of the first arm portion **1450** moves the platform portion **1454** toward the third portion **1410** of the main beam **1404**, and away from the first end **1405A** and toward the second end **1405B**. As discussed below, the movement of the platform portion **1454** using the extension of the extendable portion **1472** ultimately serves in adjusting the position of

the patient's spine. Such adjustment can occur before, during, and/or after surgery using the surgical frame 1400.

As depicted in FIGS. 42-45, the upper leg support 1402 also includes a telescoping shaft portion 1530 that is connected between the second arm portion 1452 and the platform portion 1454. The telescoping shaft portion 1530 is used to pivot the platform portion 1454 relative to the second arm portion 1452. The telescoping shaft portion 1530 includes a first end portion 1532, a second end portion 1534, a base portion 1536 including the first end portion 1532, and an extendable portion 1538 including the second end portion 1534. As discussed below, the extendable portion 1538 is moveable inwardly and outwardly relative to the base portion 1536, and such inward and outward movement serves to pivot the platform portion 1454.

To facilitate connection between the telescoping shaft portion 1530 and the second arm portion 1452, one of the first end portion 1532 and the second arm portion 1452 can form a clevis, and the other of the first end portion 1532 and the second arm portion 1452 can form a tang. Furthermore, to facilitate connection between the telescoping shaft portion 1530 and the platform portion 1454, one of the second end portion 1534 and the platform portion 1454 can form a clevis, and the other of the second end portion 1534 and the platform portion 1454 can form a tang. As depicted in FIGS. 42 and 45, the second arm portion 1452 includes a clevis 1540 having apertures 1542, the first end portion 1532 is used as a tang having an aperture 1544, and a pin 1546 is received through the apertures 1542 and 1544 to join the telescoping shaft portion 1530 to the second arm portion 1542. Furthermore, as depicted in FIGS. 42 and 45, the platform portion 1454 includes a clevis 1550 having apertures 1552, the second end portion 1534 is used as a tang having an aperture 1554, and a pin 1556 is received through the apertures 1552 and 1554 to join the telescoping shaft portion 1530 to the platform portion 1454.

The extendable portion 1538 is moveable inwardly and outwardly relative to the base portion 1536, and such inward and outward movement serves to pivot the platform portion 1454 relative to the second arm portion 1452. Such pivotal movement serves in facilitating positioning and repositioning of the platform portion 1454 relative to the second arm portion 1452. To illustrate, the base portion 1510 of the platform portion 1454 includes a first end 1560 and a second end 1562, and increasing the amount of extension of the extendable portion 1538 relative to the base portion 1536 moves the second end 1562 away from the third portion 1410 of the main beam 1404. As discussed below, the movement of the platform portion 1454 using the extension of the telescoping shaft portion 1530 ultimately serves in adjusting the position of the patient's spine. Such adjustment can occur before, during, and/or after surgery using the surgical frame 1400.

As depicted in FIGS. 45 and 46, the upper leg support 1402 also includes a linear movement assembly 1570. The linear movement assembly 1570 includes a track portion 1572 attached to the third upstanding portion 1516, two trucks 1574 moveable along the track portion 1572, and a support bracket 1576 attached to the two trucks 1574. The third upstanding portion 1516, as depicted in FIG. 46, includes a first end 1580 and a second end 1582, and is larger than the first upstanding portion 1512 and the second upstanding portion 1514. The third upstanding portion 1516 is attached at and adjacent the first end 1580 to the base portion 1510, and extends from the base portion 510 toward the first portion 1406 of the main beam 1404 to the second end 1582. The third upstanding portion 1516 supports the

track portion 1572, the two trucks 1574, the support bracket 1576, and additional components of the upper leg support 1402.

The linear movement assembly 1570 also includes a telescoping shaft portion 1590 that is connected between the third upstanding portion 1516 and the support bracket 1576. The telescoping shaft portion 1590 includes a first end portion 1592, a second end portion 1594, a base portion 1596 including the first end portion 1592, and an extendable portion 1598 including the second end portion 1594. To attach the telescoping shaft portion 1590 to the third upstanding portion 1516, the first end portion 1592 of the base portion 1596 can include an aperture 1600, the third upstanding portion 1516 can include a clevis 1602 with apertures 1603, and a pin 1604 can be received in the apertures 1600 and 1603. Furthermore, to attach the telescoping shaft portion 1590 to the support bracket 1576, the second end portion 1594 of the extendable portion 1598 can include an aperture 1606, the support bracket 1576 include a clevis 1608 with apertures 1609, and a pin 1610 can be received in the apertures 1606 and 1609.

The extendable portion 1598 is moveable inwardly and outwardly relative to the base portion 1596, and such inward and outward movement relative to the base portion 1596 serves to move the support bracket 1576 via movement of the two trucks 1574 along the track portion 1572 between at least a first position closer to the second end 1582 of the third upstanding portion 1516 to a second position closer to the first end 1580 of the third upstanding portion 1516. As discussed below, the movement of the support bracket 1576 using the extension of the extendable portion 1598 ultimately serves in adjusting the position of the patient's spine. Such adjustment can occur before, during, and/or after surgery using the surgical frame 1400.

The upper leg support 1402 also includes a support assembly 1620 that is carried by the support bracket 1576. The support assembly 1620 includes a first support post 1622, a second support post 1624, and a connecting bracket 1626 connecting the first support post 1622 and the second support post 1624 to one another. The support assembly 1620 also includes a first support block 1630, a second support block 1632, a third support block 1634, a fourth support block 1636, a first support plate 1640, a second support plate 1642, and a third support plate 1644. Each of the first support plate 1640, the second support plate 1642, and the third support plate 1644 include upper surfaces 1646A, 1646B, and 1646C, respectively, that can be used to contact the upper legs of the patient. The upper surfaces 1646A and 1646C can be covered with padding (not shown) for contacting portions of the patient's upper legs, and the padding can include pressure sensors (not shown) incorporated therein. The resulting pressure sensing padding can be used to determine if undue stress is placed on the patient P via articulation of the upper leg support 1402.

The first support plate 1640 and the second support plate 1642, as discussed below, are moveable with respect to the support bracket 1576 and the third support plate 1644. As depicted in FIG. 46, the third support plate 1644 is attached to the support bracket 1576, and the second support plate 1642 is positioned such that it can move underneath the third support plate 1644. As such, during movement of the first support plate 1640 and the second support plate 1642, the area defined by the upper surfaces 1646A, 1646B, and 1646C can be respectively decreased or increased as the second support plate 1642 is moved under or out from under the third support plate 1644.

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As depicted in FIG. 45, the first support plate 1640 is attached to the first support block 1630 and the second support block 1632, and the first support block 1630 is moveable along the first support post 1622 and the second support block 1632 is moveable along the second support post 1624. Furthermore, the second support plate 1642 is attached to the third support block 1634 and the fourth support block 1636, and the third support block 1634 is moveable along the first support post 1622 and the fourth support block 1636 is moveable along the second support post 1624.

The first and second support blocks 1630 and 1632 include apertures 1650 and 1652 for receiving the first and second support posts 1622 and 1624, respectively, and the third and fourth support blocks 1634 and 1636 include apertures 1654 and 1656 for receiving the first and second support posts 1622 and 1624, respectively. The first support plate 1640 and the second support plate 1642 are moveable inwardly and outwardly relative to the support bracket 1576 and the third support plate 1644 via movement of the first and third support blocks 1630 and 1634 on the first support post 1622 and via movement of the second and fourth support blocks 1632 and 1636 on the second support post 1624.

The support assembly 1620 also includes a threaded shaft 1660 that is retained in position between the support bracket 1576 and the connecting bracket 1626. As discussed below, the threaded shaft 1660 is used to constrain movement of the first support plate 1640 and the second support plate 1642 relative to the third support plate 1644 and the main beam 1404.

Furthermore, the first support plate 1640 includes a first support collar 1664 opposite from the upper surface 1646A, and the second support plate 1642 includes a second support collar 1666 opposite from the upper surface 1646B. The first support collar 1664 includes a first aperture 1670 that can include threads complementary to those of the threaded shaft 1660, and the second support collar 1666 includes a second aperture 1672 that can include threads complementary to those of the threaded shaft 1660. The threaded shaft 1660 can be received in the first aperture 1670 and the second aperture 1672. The first support collar 1664 and the second support collar 1666 can include one or more latches (not shown) that can be engaged and disengaged from the threaded shaft 1660 via actuation thereof. The one or more latches can be attached to the first support collar 1664 and/or the second support collar 1666, and a user can actuate the one or more latches to engage or disengage the threaded shaft 1660 to correspondingly prevent movement or allow movement of the first support collar 1664 and the second support collar 1666 along the threaded shaft 1660. When the one or more latches are engaged, the interactions of the one or more latches with the threaded shaft 1660 prevent movement of the first support plate 1640 and the second support plate 1642 relative to the third support plate 1644. When the one or more latches are disengaged, the first support plate 1640 and the second support plate 1642 can move relative to the third support plate 1644. Rather than using the threaded shaft 1660, a shaft with catches and/or teeth to which the one or more latches can be engaged and disengaged.

Alternatively, a motor/transmission/actuator (not shown) can be used to facilitate rotation of the threaded shaft 1660, and rotation of the threaded shaft 1660 and the interaction in the first aperture 1670 and the second aperture 1672 causes corresponding movement of the first support plate 1640 and the second support plate 1642. As such, rotation of the threaded shaft 1660 via actuation of the motor/transmission/actuator is translated into movement of the first support plate

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1640 and the second support plate 1642. To illustrate, the threads of the threaded shaft 1660, the first aperture 1670, and the second aperture 1672 can be configured such that clockwise rotation of the threaded shaft 1660 via actuation of the motor/transmission/actuator causes inward movement of the first support plate 1640 and the second support plate 1642, and counterclockwise rotation of the threaded shaft 1660 via actuation of the motor/transmission/actuator causes outward movement of the first support plate 1640 and the second support plate 1642. The inward and outward movement of the first support plate 1640 and the second support plate 1642 is relative to the third support plate 1644 and the main beam 1404.

The movement of the first support plate 1640 and the second support plate 1642 ultimately serves in adjusting a total width of a combination of the first support plate 1640, the second support plate 1642, and the third support plate 1644. Adjustment of the combined width of the first support plate 1640, the second support plate 1642, and the third support plate 1644 affords the accommodation of differently sized patients on the upper leg support 1402.

The movement of the componentry of the upper leg support 1402 can be effectuated via manual adjustment and/or controlled automation. To illustrate, the first arm portion 1450 includes the extendable portion 1472 that is moveable with respect to the base portion 1470 thereof, the telescoping shaft portion 1530 includes the extendable portion 1538 that is moveable with respect to the base portion 1536 thereof, the telescoping shaft portion 1590 includes the extendable portion 1598 that is moveable with respect to the base portion 1596 thereof, and the motor/transmission/actuator facilitates movement of the first support plate 1640 and the second support plate 1642 is relative to the third support plate 1644 and the main beam 1404.

Such reconfiguration of the upper leg support 1402 can be actuated using the manual adjustment and/or the controlled automation, and as discussed below, the extension and retraction of the extendable portion 1472, the extendable portion 1538, and the extendable portion 1598 via such actuation can be used to both adjust the adjustable center of rotation CR to accommodate patients having different body sizes, and to facilitate flexing of the lumbar spine to afford surgical access thereto via manipulation of portions thereof. The extension/retraction of the extendable portion 1472 serves to change the angle of the first arm portion 1450 and the second arm portion 1452 relative to one another, the extension/retraction of the extendable portion 1538 serves to change the angle of the platform portion 1454 relative to the second arm portion 1452, and the extendable portion 1598 serves to change position of the bracket 1576 (which supports the first support plate 1640, the second support plate 1642, and the third support plate 1644) relative to the platform portion 1454.

Using the upper leg support 1402, the position of the patient's upper legs can be altered, which correspondingly affects the flexure of the lumbar spine of the patient P, and care should be taken to prevent unwanted torsion thereof when manipulating the patient's spine. To illustrate, the amounts of extension/retraction of the extendable portion 1472, the extendable portion 1538, and the extendable portion 1598 can be constrained with respect to one another to prevent unwanted torsion of the lumbar spine during articulation of the upper leg support 1402. As such, the amounts of extension/retraction of the extendable portion 1472, the extendable portion 1538, and the extendable portion 1598 can be contingent upon one another to facilitate such approximate preservation.

A controller (not shown) with a user interface (not shown) can be used to control the constrained/contingent extension and/or retraction of the extendable portion 1472, the extendable portion 1538, and the extendable portion 1598 via the controlled automation. Such extension and/or retraction, as depicted in FIGS. 48 and 49, affords positioning and repositioning of the support assembly 1620. Furthermore, because patients' heights can vary, the amounts of extension/retraction of the extendable portion 1472, the extendable portion 1538, and the extendable portion 1598 can be altered to accommodate these different heights while still being constrained/contingent upon one another to provide for the desired amount of manipulation of portions of the lumbar spine during articulation of the upper leg support 1402.

The controller with input via the user interface can allow the user to select the desired center of rotation and the desired amount of manipulation of the portions of the lumbar spine. To illustrate, the user interface can be used to display a graphical or actual representation of the patient's spine, and the user interface can permit the user to input the desired center of rotation and the desired amount of manipulation by, for example, highlighting a portion of the graphical or actual representation of the patient's spine on the user interface. The selection of the desired amount of manipulation can allow the user to select where the forces applied via the actuation of the extendable portion 1472, the extendable portion 1538, and the extendable portion 1598 are applied during flexure of the patient's spine. In addition to or alternatively to use of the user interface, a navigation tool interconnected with the controller and/or the user interface can be positioned on or adjacent the patient's spine to facilitate inputting of the desired center of rotation and the desired amount of manipulation. The inputting of the desired center of rotation and the desired amount of manipulation can be done with the main beam 404 and the patient P supported on the main beam 1404 in various rotational positions including, but not limited to, prone, lateral, and supine positions.

When the upper leg support 1402 is articulated such that the lumbar spine of the patient P is in an unflexed neutral position, as depicted in FIG. 48, the controller can be used to extend/retract the extendable portion 1472 (of the first arm portion 1450), the extendable portion 1538 (of the telescoping shaft portion 1540), and the extendable portion 1598 (of the telescoping shaft portion 1590) such that the first arm portion 1450, the telescoping shaft portion 1540, and the telescoping shaft portion 1590 have lengths that accommodate the height of the patient P. Furthermore, when the upper leg support 1402 is articulated such that the lumbar spine of the patient P has a 30 degree flex, as depicted in FIG. 49, the controller can be used to extend/retract the extendable portion 1472 (of the first arm portion 1450), the extendable portion 1538 (of the telescoping shaft portion 1530), and the extendable portion 1598 (of the telescoping shaft portion 1590) such that the first arm portion 1450, the telescoping shaft portion 1540, and the telescoping shaft portion 1590 have lengths that flex the lumbar spine of the patient P to afford surgical access thereto. Moreover, during the transition between the positions of FIGS. 48 and 49, the controller can serve to prevent unwanted torsion of the lumbar spine during articulation of the upper leg support 1402 by properly adjusting the amounts of extension/retraction of the extendable portion 1472, the extendable portion 1538, and the extendable portion 1598.

The relative amounts of extension/retraction can be provided for different patient heights and different degrees of flex of the lumbar spine, and can be included as presets in

the controller. Thus, using the user interface of the controller, the operator of the surgical frame 1400 can input the height of the patient P, and input the desired degree of flexure of the lumbar spine, and the controller can actuate the first arm portion 1450 (to extend or retract the extendable portion 1472), the telescoping shaft portion 1530 (to extend or retract the extendable portion 1538), and the telescoping shaft portion 1590 (to extend or retract the extendable portion 1598) the appropriate amounts to provide such flexion, while also preventing unwanted torsion of the patient's spine. As discussed above, the extension/retraction of the extendable portion 1472 serves to change the angle of the first arm portion 1450 and the second arm portion 1452 relative to one another, the extension/retraction of the extendable portion 1538 serves to change the angle of the platform portion 1454 relative to the second arm portion 1452, and the extendable portion 1598 serves to change position of the bracket 1576 (which supports the first support plate 1640, the second support plate 1642, and the third support plate 1644) relative to the platform portion 1454. During such manipulation of the patient's spine using the upper leg support 1402, the lengths of the first arm portion 1450, the telescoping shaft portion 1530, and the telescoping shaft portion 1590 may each be alternately increased/decreased or decreased/increased to provide for the desired adjustable center of rotation CR. As such, the operator can use the controller to manipulate the upper leg support 1402 to flex the lumbar spine of the patient P into position for surgery, while simultaneously inhibiting the unwanted torsion of the patient's spine caused by reconfiguration of the upper leg support 1402.

In addition to the upper leg support 1402, the surgical frame 1400, as depicted in FIGS. 42 and 47 includes the lower leg support 1416 that supports the lower legs of the patient P. The lower leg support includes a support plate portion 1680, a first arm portion 1682, a second arm portion 1684, a first plate portion 1686, a second plate portion 1688, and a connecting rib 1690. The first arm portion 1682 and the second arm portion 1684 include first end portions 1692 and second end portions (not shown). The first end portions 1692 can include apertures 1696 facilitating attachment thereof to the third portion 1410 via receipt of the pin 1492 therein. Furthermore, the second end portions can be attached to the first plate portion 1686, and the first plate portion 1686 can be attached to the first arm portion 1450. As such, portions of the lower leg support 1416 can move with the articulation of the first arm portion 1450. The first plate portion 1686 connects the second plate portion 1688 and the connecting rib 1690 to one another, and the connecting rib 1690 attaches the support plate portion 1680 to the first plate portion 1686. As depicted in FIG. 42, the connecting rib 1690 spaces the support plate portion 1680 from the first plate portion 1686. The support plate portion 1680 can be used to support the patient's lower legs thereon.

Further, other types of mechanism or actuators, such as servomotors, can be used or configuration to provide for the mechanical articulations and movements necessary to support the biomechanical manipulations of the spine described herein.

It should be understood that various aspects disclosed herein may be combined in different combinations than the combinations specifically presented in the description and the accompanying drawings. It should also be understood that, depending on the example, certain acts or events of any of the processes of methods described herein may be performed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not

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be necessary to carry out the techniques). In addition, while certain aspect of this disclosure are described as being performed by a single module or unit for purposes of clarity, it should be understood that the techniques of this disclosure may be performed by a combination of units or modules associated with, for example, a medical device.

We claim:

1. A method of adjusting a position of a patient supported on a surgical frame, the method comprising:

positioning the patient on the surgical frame by supporting upper legs of the patient on a support plate portion; extending a first arm portion relative to a second arm portion to adjust a position of a platform portion supporting the support plate portion relative to a portion of the surgical frame, the first arm portion including a first end attached relative to the portion of the surgical frame and a second end attached relative to the platform portion, and the second arm portion including a first end attached relative to the platform portion and a second end attached relative to the portion of the surgical frame;

extending a telescoping shaft to adjust the position of the platform portion and the support plate portion supported thereby relative to at least one of the first arm portion and the second arm portion, the telescoping shaft including a first end attached relative to the second arm portion and a second end attached relative to the platform portion; and

adjusting a position of a lumbar portion of a spine of the patient by coordinating extension of the first arm portion and the telescoping shaft to adjust the position of the platform and the support plate portion supported thereby downwardly and rotatably relative to the portion of the surgical frame to correspondingly adjust the upper legs of the patient supported by the support plate portion.

2. The method of claim 1, wherein the second end of the first arm portion and the first end of the second arm portion are pivotally attached to one another and to the platform portion.

3. The method of claim 2, wherein during extension of the first arm portion, the first arm portion and the second arm portion are pivoted with respect to one another.

4. The method of claim 3, wherein during extension of the telescoping shaft, the platform portion is rotated relative to the at least one of the first arm portion and the second arm portion.

5. The method of claim 1, wherein the first arm portion and the second arm portion are pivoted with respect to the portion of the surgical frame and one another during extension of the first arm portion, and the platform portion is rotated relative to the at least one of the first arm portion and the second arm portion during extension of the telescoping shaft.

6. The method of claim 1, further comprising rotating the portion of the surgical frame to position and reposition the portion of the surgical frame relative to remaining portions of the surgical frame.

7. The method of claim 1, wherein the support plate portion is attached to the platform portion via a support bracket, and further comprising moving the support bracket along portions of the platform portion, and moving the support plate portion inwardly and outwardly relative to the support bracket, the support bracket being moveable along a track attached to the platform portion, and the support plate portion being moveable along at least one support post extending outwardly from the support bracket.

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8. A method of adjusting a position of a patient supported on a surgical frame, the method comprising:

supporting upper legs of the patient on a support plate portion of the surgical frame;

supporting the support plate portion relative to a portion of the surgical frame by a platform portion, a first arm portion, and a second arm portion, the first arm portion and the second portion being pivotal with respect to one another via pivotal attachment thereof to the platform portion;

extending the first arm portion to adjust a position of the platform portion and the support plate portion supported thereby relative to the portion of the surgical frame, the first arm portion including a first end attached relative to the portion of the surgical frame and a second end attached to the platform portion, and the second arm portion including a first end attached to the platform portion and a second end attached relative to the portion of the surgical frame;

extending a telescoping shaft to adjust the position of the platform portion and the support plate portion supported thereby relative to the first arm portion and the second arm portion, the telescoping shaft including a first end attached to second arm portion and a second end attached to the platform portion; and

adjusting a position of a lumbar portion of a spine of the patient by coordinating extension of the first arm portion and the telescoping shaft to adjust the position of the platform portion and the support plate portion supported thereby downwardly and rotatably relative to the portion of the surgical frame to correspondingly adjust the upper legs of the patient supported by the support plate portion.

9. The method of claim 8, wherein during extension of the first arm portion, the first arm portion and the second arm portion are pivoted with respect to one another.

10. The method of claim 9, further comprising retracting the first arm portion to further adjust the position of the platform portion relative to the surgical frame.

11. The method of claim 8, wherein during extension of the telescoping shaft, a rotational position of the platform portion is adjusted.

12. The method of claim 8, further comprising rotating the portion of the surgical frame to position and reposition the portion of the surgical frame relative to remaining portions of the surgical frame.

13. The method of claim 8, wherein the support plate portion is attached to the platform portion via a support bracket; and further comprising moving the support bracket along portions of the platform portion, and moving the support plate portion inwardly and outwardly relative to the support bracket, the support bracket being moveable along a track attached to the platform portion; and the support plate portion being moveable along at least one support post extending outwardly from the support bracket.

14. A method of adjusting a position of a patient supported on a surgical frame, the method comprising:

pivotaly attaching each of a first arm portion and a second arm portion to a portion of the surgical frame and to a platform portion;

positioning the patient on the surgical frame by supporting upper legs of the patient on a support plate portion supported by the platform portion;

pivoting and extending the first arm portion relative to the surgical frame and pivoting the second arm portion

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relative to the surgical frame to adjust the position of the platform portion relative to the portion of the surgical frame; and

extending a telescoping shaft to adjust the position of the platform portion and the support plate portion supported thereby relative to at least one of the first arm portion and the second arm portion, the telescoping shaft including a first end attached relative to the second arm portion and a second end attached relative to the portion of the platform portion; and

adjusting a position of a lumbar portion of a spine of the patient by coordinating extension of the first arm portion and the telescoping shaft to adjust the position of the platform portion and the support plate portion supported thereby downwardly and rotatably relative to the portion of the surgical frame to correspondingly adjust the upper legs of the patient supported by the support plate portion.

15. The method of claim 14, wherein the pivoting of the first arm portion is afforded by extension of the first arm portion.

16. The method of claim 14, wherein during extension of the telescoping shaft, the platform portion is rotated relative to the first arm portion.

17. The method of claim 14, wherein a first end the first arm portion is pivotally attached to the portion of the surgical frame and a second end of the first arm portion is

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attached to the platform portion, and a first end of the second arm portion is pivotally attached to the portion of the surgical frame and a second end of the second arm portion is attached to the platform portion.

18. The method of claim 17, wherein the support plate portion is attached to the platform portion via a support bracket; and further comprising moving the support bracket along portions of the platform portion, and moving the support plate portion inwardly and outwardly relative to the support bracket, the support bracket being moveable along a track attached to the platform portion, and the support plate portion being moveable along at least one support post extending outwardly from the support bracket.

19. The method of claim 14, further comprising rotating the portion of the surgical frame to position and reposition the portion of the surgical frame relative to remaining portions of the surgical frame.

20. The method of claim 14, wherein the support plate portion is attached to the platform portion via a support bracket; and further comprising moving the support bracket along portions of the platform portion, and moving the support plate portion inwardly and outwardly relative to the support bracket, the support bracket being moveable along a track attached to the platform portion, and the support plate portion being moveable along at least one support post extending outwardly from the support bracket.

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